Fake–Real gaps in observed spread and detectability differ between health misinformation datasets built in different ways: a comparison of CoAID and FakeHealth

Ji Zhao[1], Jesús Valdiviezo[2]

[1]New England Innovation Academy

[2]Department of Computer Science, Princeton University

ORCID (Ji Zhao): https://orcid.org/0009-0001-5218-0192

ORCID (Jesús Valdiviezo): https://orcid.org/0000-0002-8542-1247

Abstract

Health misinformation datasets don't reach a common ground about what misinformation is. FakeHealth labels mainstream health news by expert-rated reporting quality, while CoAID compares fact-checked COVID-19 claims with material from health agencies and health information publishers. We compared model-standardized Fake/Real ratios of observed social media posts, including items with no observed posts, between the CoAID News and FakeHealth HealthStory subsets, and evaluated leakage-controlled text classifiers within, across and beyond the two datasets. Fake items drew fewer observed posts than Real items in both datasets, but the ratio was 0.253 in CoAID and 0.840 in FakeHealth (ratio-of-ratios 0.301; 95% interval 0.154–0.645). The CoAID gap lay entirely in whether an item was posted at all and was concentrated among Fake items with no identifiable publisher, which points to limited findability as well as limited spread. In two FakeNewsNet collections outside health, Fake items drew more observed posts in GossipCop (ratio 2.144), and PolitiFact's interval included parity. Text showed a parallel split: Fake and Real were highly separable within CoAID (macro F1 0.926) and modestly within FakeHealth (0.675), transfer was poor in both directions, and word–label associations were uncorrelated across datasets. Fake items were shorter in both datasets, and length alone ranked transferred items and independent COVID rumors better than the word-based models. The distance between Fake and Real items, in language and in observed engagement, differed between datasets that define and source misinformation differently; these are comparisons between dataset contexts, not estimates of a pandemic effect. Engagement studies and detection benchmarks should report how labels, sources and collection were defined, check for source–label entanglement, shared-post dependence and simple length baselines, and test beyond a single dataset.

1. Introduction

1.1 Why the definition of misinformation matters

Health misinformation can affect health intentions. In a randomized exposure experiment, COVID-19 vaccine misinformation reduced the share of respondents who said they would definitely accept vaccination by 6.2 percentage points in the UK and 6.4 points in the USA, relative to factual information (Loomba et al., 2021). Research on health misinformation follows two lines: measuring how it circulates on social media and building classifiers to detect it. Both of them depend on labeled datasets, and the datasets do not have a common definition of misinformation.

In FakeHealth, a story is labeled Fake when expert reviewers rate its reporting as poor. For example, an item is considered Fake because it fails to quantify benefits or harms, ignores costs, or misjudges the quality of the evidence. The stories come from major US media outlets (Dai et al., 2020), and many publishers carry both labels (Supplementary Table S15). In CoAID, Fake items are claims flagged by fact-checking sites, while Real items come from health agencies and health-information publishers such as the CDC, WHO, NIH, Healthline and WebMD (Cui & Lee, 2020). The first construct contrasts weaker and stronger reporting from the same kind of outlet, while the second contrasts fact-checked claims with institutional health information. If Fake–Real differences in engagement or language depend on which construct a dataset encodes, results from a single benchmark cannot generalize well beyond that construct.

1.2 Engagement: what is known and what is missing

The best known diffusion result is that false news on Twitter traveled faster and reached more people than true news, in fact-checked rumor cascades from 2006 to 2017 (Vosoughi et al., 2018). That analysis began from content that already formed cascades. Counting every labeled item, including those with no observed posts, asks a prior question: is an item taken up at all? Juul and Ugander (2021) later showed that the reported structural differences between false and true cascades largely disappear once cascade size is matched, because false news cascades were generally larger. For COVID-19, Cinelli et al. (2020) found platform differences in the volume and amplification of questionable sources, but no different spreading patterns for reliable and questionable sources within their data. To our knowledge, no study has compared the Fake–Real engagement gap across datasets that encode different misinformation constructs on a common zero inclusive scale.

Measuring that gap requires two additional steps. Repeated captures of an updated web page must be resolved into content items before engagement is counted, and items that share the same posts must be treated as dependent when estimating uncertainty.

1.3 Detection: what is known about portability

A classifier can learn a dataset's labels without learning anything that holds elsewhere. Torabi Asr and Taboada (2018) found that classifiers trained on labels derived from source reputation generalized poorly to article level veracity labels and performed unevenly across topics, though their predictions showed some correspondence with article veracity. Murayama et al. (2021) showed that proper names specific to the period in which a fake-news dataset was created can become predictive. They examined how this bias affects robustness on datasets that differ in domain and creation period. Leakage between training and test material can also inflate performance estimates (Kapoor & Narayanan, 2023). Within-dataset scores are hence informative about portability only when they are reported alongside transfer to other datasets, evaluated on the same target items.

1.4 This study

We compare CoAID News, a COVID-focused collection, with FakeHealth HealthStory, a non-COVID health news collection. Both provide health news content, Fake/Real labels and item-linked social posts, which lets us study engagement and text within the same pair of datasets. We ask three questions:

- **RQ1:** How does the model-standardized Fake to Real ratio of expected unique observed posts differ between CoAID News and FakeHealth HealthStory?
- **RQ2:** How well do learned Fake/Real textual associations generalize within and across the 2 datasets?
- **RQ3:** Do CoAID-trained and FakeHealth-trained classifiers differ in portability to the same independent COVID-Rumor NEWS target?

RQ1 contributes a zero-inclusive engagement comparison on a common standardized scale, with content versions resolved and shared-post dependence reflected in the uncertainty. RQ2 contributes a leakage-controlled measure of how far each dataset's textual signal travels, set against how strongly source and label are entangled in each dataset and against a length-only baseline. RQ3 tests whether the source ordering seen between the two datasets holds on a third collection that neither model was trained on. Two non-health collections from FakeNewsNet provide an out-of-domain check on both branches.

We treat each dataset as an information context, defined jointly by what it labels as misinformation and how it was assembled, and describe how the distance between its Fake and Real items varies across contexts. The study is a descriptive comparison between dataset contexts. Engagement is measured as captured posts instead of exposure or complete cascades, and the text models predict labels without modeling why items were shared.

2. Methods

2.1 Design overview

The study has two branches with separate populations and outcomes (Figure M1). The engagement branch models observed post counts per content item. The text branch predicts each dataset's Fake/Real labels from title and body text. Engagement measures never enter the text models. The primary protocols were CoAID reconstruction A for engagement and natural combined-text logistic regression for text; all other analyses are sensitivity or robustness checks. The primary estimand, model-selection rules, data splits and transfer design were set before engagement outcomes were modeled or holdout labels were scored. Analyses specified after the primary results were known are identified as such in Sections 2.9 and 2.11 and in Supplementary Methods S8.

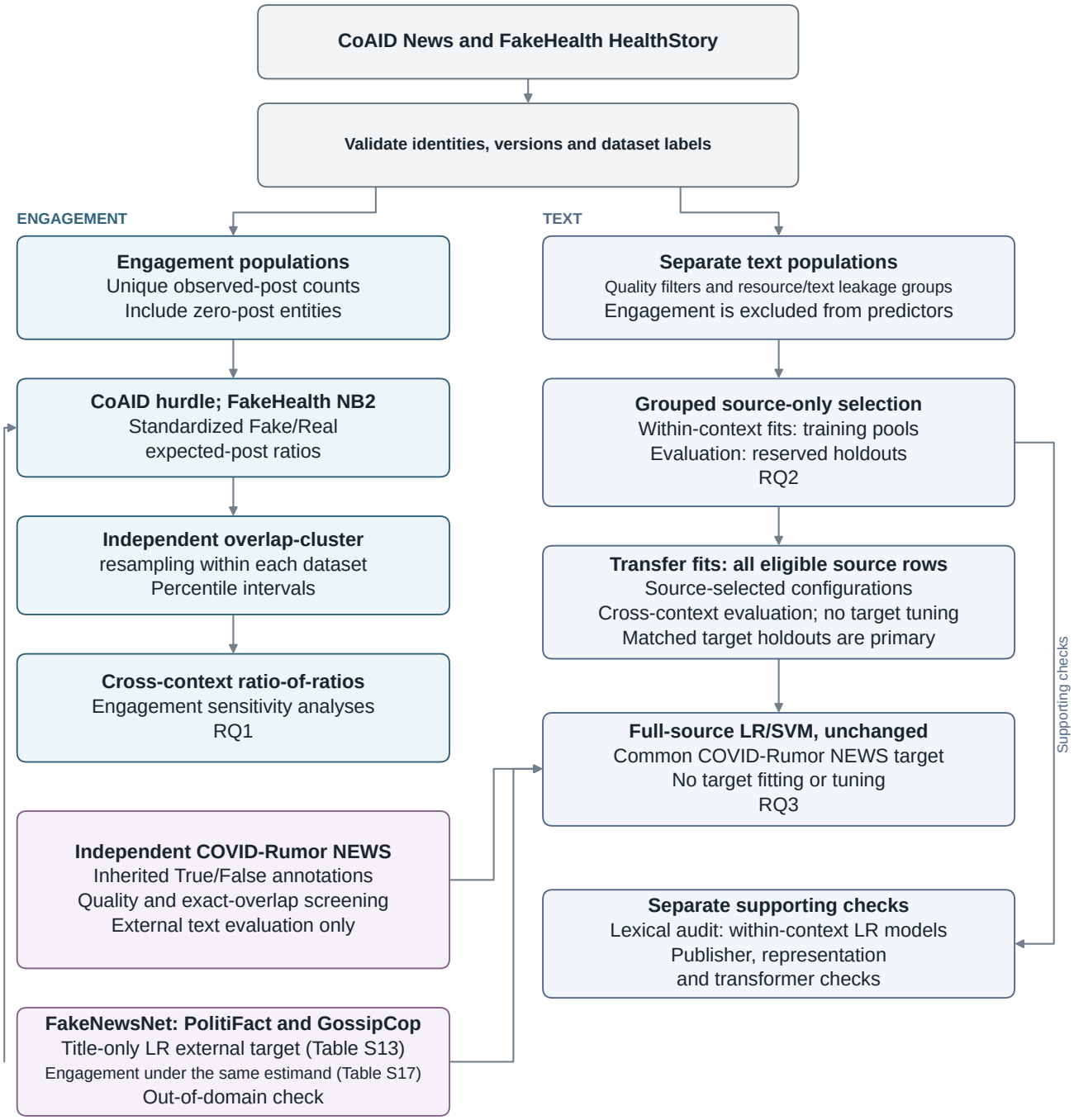


Figure M1. Analytical workflow. The engagement branch uses zero-inclusive observed-post populations; the text branch uses separately filtered text populations. Model selection only uses source data; target data are never used for fitting or tuning. COVID-Rumor only serves as an external text target. The FakeNewsNet extension (title-only text target and engagement under the same estimand) is an out-of-domain check described in Supplementary Methods S8.

2.2 Datasets and labels

CoAID News. We used the News subset of the four official releases (`05-01-2020`, `07-01-2020`, `09-01-2020`, `11-01-2020`), with Fake/Real labels from the original News filenames. CoAID's Fake items were drawn from fact-checking sites and its Real items from nine health agencies and health information publishers (Cui & Lee, 2020; cuilimeng, n.d.). The primary engagement population (reconstruction A) contained 5,337 items: 922 Fake and 4,415 Real. Content records were linked to post mappings through the content ID and to reply mappings through the news ID, within release and file category, because release-local IDs are not globally unique.

FakeHealth HealthStory. Stories with an expert-review score below 3 were labeled Fake and those at 3 or above were labeled Real (Dai et al., 2020; EnyanDai, n.d.). Reviews, article content and engagement mappings were joined by news ID. The engagement population contained 1,690 items: 472 Fake and 1,218 Real. HealthRelease was not used. FakeHealth's retweet mappings were completely excluded from the common outcome because CoAID has no equivalent.

COVID-Rumor NEWS. For external text evaluation, we used the NEWS subset, mapping True to Real and False to Fake and excluding Unverified items and the Twitter subset (Cheng et al., 2021; MickeysClubhouse, n.d.). Annotations were inherited rather than independently re-verified.

The analytical labels share an orientation (Fake = 1) but do not share a construct. FakeHealth labels reporting quality, while CoAID and COVID-Rumor labels derive from veracity judgments. This difference is the organizing variable of the study.

2.3 CoAID content identity

CoAID re-captures many of the same web pages across its four releases, sometimes after updates, so item identity had to be resolved before engagement could be counted. Identity relied on the specific URL or resource and on audited source lineage; shared domains, generic titles, branding and shared social posts never established identity. An early representation that merged on such cues produced one component joining 110 source records, which motivated this rule.

The primary reconstruction (A) merged captures with strong evidence of identical content or only minor updates, split material updates into separate versions, and kept captures with insufficient evidence as separate items. Reconstruction B additionally assumed stability for insufficient evidence captures of the same validated resource. Reconstruction C excluded uncertain version and unresolved identity captures. Version partitions used complete-link checks so that chains of pairwise matches could not join incompatible captures. Decision rules and audit records are in Supplementary Methods S1.

2.4 Engagement outcome and dependence

The primary outcome was the number of unique observed posts linked to each content item, including items with none. CoAID counts were the union of post IDs across an item's member captures; FakeHealth counts were unique tweet IDs. These are counts of captured posts. They do not measure exposure, complete cascades or network structure. Reply counts and occurrence summaries were secondary.

Distinct items can share the same posts, so items linked by shared post or reply IDs were resampled together. For each dataset and reconstruction we built a graph with items as nodes and shared IDs as edges. Its connected components, including singletons, were the resampling clusters. CoAID A had 5,056 clusters, the largest containing 187 items. FakeHealth had 1,662, the largest containing four. An influence analysis, planned with the primary protocol, refitted CoAID without its largest cluster.

2.5 Engagement models and estimand

CoAID counts were modeled with a hurdle model. It contained a logistic component for any observed post and a zero-truncated negative binomial (NB2) component for positive counts, each with the Fake indicator and first-recorded-release fixed effects. Release describes capture provenance rather than time of publication or exposure. The hurdle

was chosen by a rule set before fitting (AIC improvement over NB2 greater than 2, or NB2 instability). FakeHealth counts were modeled with NB2 and the Fake indicator. No exposure offset was used. Model families were fixed before resampling.

Because the two model families have incomparable coefficients, we compared standardized expected counts. Let m_d(a, X_i) be the fitted expected count for item i in dataset d with its Fake indicator set to a and other covariates unchanged. For each dataset, the standardized ratio is given by equation (1),

$$R_d = \frac{N_d^{-1} \sum_{i=1}^{N_d} m_d(1, X_i)}{N_d^{-1} \sum_{i=1}^{N_d} m_d(0, X_i)}. \quad (1)$$

For the CoAID hurdle, the row-level expectation is equation (2),

$$m_d(a, X_i) = P(Y_i > 0 \mid a, X_i)\, E(Y_i \mid Y_i > 0, a, X_i), \quad (2)$$

with the row level products of (2) averaged before forming the ratio in (1). Each dataset kept its own covariate distribution, including zero-count items. For CoAID we also report the two hurdle components, the standardized probability of any observed post and the standardized expected count given at least one post, with intervals from the same bootstrap draws. Because the estimand (1) averages row level products, the two component ratios do not multiply exactly to the overall ratio. FakeHealth has no equivalent split as it was modeled without an occurrence component. The cross-context estimand was C = R_CoAID / R_FakeHealth. Values below 1 indicate a larger relative Fake–Real gap in CoAID. The estimand compares expected posts per item, which is a mean. Where counts are heavily skewed, a few highly shared items can set its direction, and the typical Fake and Real items can differ in the opposite direction. We therefore also report medians and the probability that a randomly chosen Fake item has more observed posts than a randomly chosen Real one, counting ties as one half (Supplementary Table S21).

Uncertainty came from an unstratified pairs bootstrap of resampling clusters. Each draw sampled clusters with replacement, refitted the model and re-standardized, targeting 1,000 successful draws per dataset and reconstruction (seed 42). The cross-context interval combined all 1,000 × 1,000 pairs of independent CoAID and FakeHealth draws. These million values share marginal draws and are not independent replicates. Intervals are 95% percentile intervals. Where one cluster dominates the outcome, the interval also reflects how often that cluster is drawn (Supplementary Table S20, Figure S1). Intervals assume independence between datasets and do not include uncertainty from entity resolution, missing links or model family choice, which the B/C reconstructions and the cluster exclusion address separately.

2.6 Text cohorts and leakage control

Text was the title and body joined into one field, retaining case, punctuation and URLs. Empty and artifact-dominated records were excluded by label-blind rules. Predictors excluded IDs, ratings, source and review metadata, engagement and split fields. The modeling cohorts contained 1,647 FakeHealth items (459 Fake, 1,188 Real) plus 3,577 CoAID items (825 Fake, 2,752 Real).

Leakage groups joined items that share a resource, source lineage, or exact or near-exact text, so that versions of the same resource never crossed a partition. A grouped stratified split reserved about 20% of each dataset as a holdout, chosen on class counts rather than performance (FakeHealth 1,317 training / 330 holdout; CoAID 2,875 / 702), and 5 grouped cross-validation folds were formed within the training pool. A masked text variant replaced known explicit source names by using a label-blind dictionary. Because that dictionary drew on evaluation metadata too, masking is a sensitivity check rather than a training-only procedure. Details are in Supplementary Methods S3.

2.7 Classical models and within-dataset evaluation

The primary classifier was word TF-IDF with L2-regularized logistic regression (LR). A linear support vector machine (SVM) tested robustness to classifier choice. Vocabularies and IDF weights were only learned within each training fold. Hyperparameters (n-gram range, minimum document frequency, regularization strength, class weighting; 24 LR and 32 SVM configurations) were chosen by mean grouped cross-validation macro F1 on the source training pool and

fixed before any holdout prediction. LR classified Fake at probability ≥ 0.5 and SVM at decision score ≥ 0. We report macro F1 (the unweighted mean of class-wise F1), ROC-AUC and average precision with Fake as the positive class. Grids, tie rules and selected settings are in Supplementary Methods S4.

2.8 Cross-dataset transfer and lexical alignment

Transfer asks whether what a model learns in one dataset holds in another. Each source model was refitted on all eligible source items with its selected configuration and applied to the target without fitting, calibration or threshold changes, in both directions. Transfer and within-target models were compared on the identical target holdout items. Predictions were saved before target labels were joined.

To compare what the 2 models learned, we intersected the unigram vocabularies of the within-dataset LR models (4,174 shared words) and summarized the agreement of their signed coefficients by Pearson and Spearman correlation and the share of sign reversals, overall and among the top-ranked words. Coefficients are conditional associations within differently fitted models, so the comparison concerns direction instead of magnitude.

2.9 Robustness analyses

- **Title-only text.** LR tuned and evaluated on titles alone, with transfer compared on identical title-eligible target rows.
- **Unseen publishers (FakeHealth).** 3-fold outer evaluation holding out whole publisher families (49 families), with nested 3-fold tuning inside each outer training set. CoAID could not support an equivalent design because its publishers are almost entirely single-label and unresolved publishers are especially concentrated among Fake items.
- **Transformer.** DistilRoBERTa at a pinned revision (Sanh et al., n.d.), with architecture and distillation attributions to Liu et al. (2019) and Sanh et al. (2019). Title plus body prefix up to 512 tokens, with a 256-token sensitivity; learning rate and epochs chosen on source cross-validation. At 512 tokens, 71.22% of FakeHealth texts and no CoAID texts were truncated. A 3-seed follow-up (42, 123, 2026) compared the two token limits for FakeHealth-to-CoAID transfer.

Full protocols are in Supplementary Methods S5 and S6.

Diagnostics specified after the primary results. Once the primary results were known, we wrote a protocol for a further set of diagnostics before computing any of them. All reused the selected configurations and saved splits (Supplementary Methods S8, Tables S18–S23). For CoAID, we broke down any observed post by whether an item's publisher could be resolved and by first-recorded release. To test whether document length acts as a shared class cue, we fitted a logistic regression on log token count alone on each source training pool and scored it on every target. We summarized the saved bootstrap draws by how often the dominant cluster entered each draw, and computed medians and the probability that a random Fake item outdraws a random Real one. We refitted the FakeHealth LR on the text before and after DistilRoBERTa's 512-token cut and correlated LR coefficients across the five training folds within each dataset. After the length-baseline results were seen, we added one re-scoring check: the existing LR and SVM scores were evaluated, without refitting, within quintiles of document length, reporting ROC-AUC within each quintile and pooled over Fake–Real pairs from the same quintile.

2.10 External COVID-Rumor evaluation

Of 3,700 NEWS items with True/False labels, 33 overlapping exactly with source texts or resources were excluded, leaving 3,667 (3,015 Fake, 652 Real; 82.22% Fake). The 4 full-source LR and SVM models were applied unchanged. Quality and overlap-screening rules are in Supplementary Methods S6.

2.11 Out-of-domain checks with FakeNewsNet

To see if the patterns held outside health, we used FakeNewsNet's PolitiFact and GossipCop subsets (Shu et al., 2020). Their protocols were written after the primary results were known and before any FakeNewsNet text or engagement estimate was computed (Supplementary Methods S8). Labels were inherited. PolitiFact labels are fact-checker

conclusions. In GossipCop, stories scored below 5 of 10 by the site were labeled fake, and real items are documented as articles from E! Online. The URLs in our local copy, however, suggest a broad set of entertainment and news domains rather than to E! Online (Supplementary Table S17). Two PolitiFact items listed under both labels were excluded.

For engagement, the outcome was distinct tweet IDs per item, including zeros (PolitiFact 1,052 items: 430 Fake, 622 Real; GossipCop 22,140: 5,323 Fake, 16,817 Real). Tweets were collected by headline search, and no reply or retweet mapping exists. Model selection, estimand and cluster bootstrap followed Sections 2.4–2.5, with the Fake indicator as the only covariate. The selection rule chose the hurdle model in both collections. As for CoAID, we also refitted each collection without its largest cluster of items linked by shared tweets. For text, only titles were available locally, so items passing the COVID-Rumor quality and overlap rules (PolitiFact 429 Fake, 596 Real; GossipCop 5,315 Fake, 16,799 Real) were scored with the existing title-only LR models; no title-only SVM exists.

2.12 Statistical reporting and reproducibility

Engagement uncertainty is reported as bootstrap percentile intervals. Fold and seed standard deviations describe variability and are not confidence intervals. Text comparisons are descriptive. No significance tests were used to declare one model or source superior. Analyses used Python 3.14 with scikit-learn (Pedregosa et al., 2011) and statsmodels (Seabold & Perktold, 2010), and Python 3.12 with PyTorch (Paszke et al., 2019) and Transformers (Wolf et al., 2020) for the transformer; versions, seeds and provenance records are given in Supplementary Methods S7.

Code and data availability are described at the end of the paper.

3. Results

3.1 Fake items drew far fewer observed posts than Real items in CoAID, and nearly as many in FakeHealth

Fake items drew fewer observed posts than Real items in both datasets, but the gap was far larger in CoAID. The standardized Fake/Real ratio was 0.253 in CoAID (95% interval 0.132–0.522): a Fake item was expected to draw about a quarter as many observed posts as a Real one. In FakeHealth the ratio was 0.840 (0.714–0.997), close to parity. The ratio of the two ratios was 0.301 (0.154–0.645; Figure 1).

The direction held under each alternative reconstruction of CoAID items (ratio-of-ratios 0.286 for B and 0.343 for C; Table 1). Its magnitude depends on one cluster. Excluding CoAID's largest group of items linked by shared posts moved the ratio-of-ratios to 0.489 (0.310–0.710), weaker but in the same direction and inside the primary interval. How often this cluster is drawn also shapes the width of the primary CoAID interval (Supplementary Table S20, Figure S1).

Absolute levels differed by an order of magnitude. Standardized expected counts were 199.9 (Fake) and 237.9 (Real) posts per FakeHealth item, against 8.8 and 34.9 per CoAID item, which is why the comparison is made between within-dataset ratios.

The CoAID gap falls entirely in whether an item was posted at all. Standardized over the same population, a Fake item had an expected 11.7% chance of any observed post against 61.7% for a Real item (ratio 0.190; 95% interval 0.164–0.218). Among items with at least one post, Fake items drew somewhat more posts than Real ones (expected 51.4 vs 39.5; ratio 1.30, 0.73–2.55), an interval that includes parity (Supplementary Table S11). Nearly all FakeHealth items had observed posts, so its ratio reflects posting volume alone.

The occurrence gap partly reflects which items the collection could find. Because CoAID's Fake items were selected by fact-checkers, most had already circulated somewhere. Fake items whose publisher could be resolved had any observed post 33.0% of the time, against 7.5% for those hosted on platforms without an identifiable account or lacking source metadata, which make up two-thirds of CoAID's Fake items. Resolved Real items were posted 57.6% of the time, so a gap remains among items that could be traced. Collection timing mattered as much as class. Real items first captured in CoAID's May and July 2020 releases had observed posts 91.3% and 90.2% of the time, but those first

captured in September and November only 7.6% and 5.3%, close to the Fake shares in those releases (7.1% and 3.2%) (Supplementary Table S18). Release enters the model as a fixed effect, so the standardized contrast compares items within the same release, and it rests mainly on the two earlier releases, where posts were collected for most Real items.

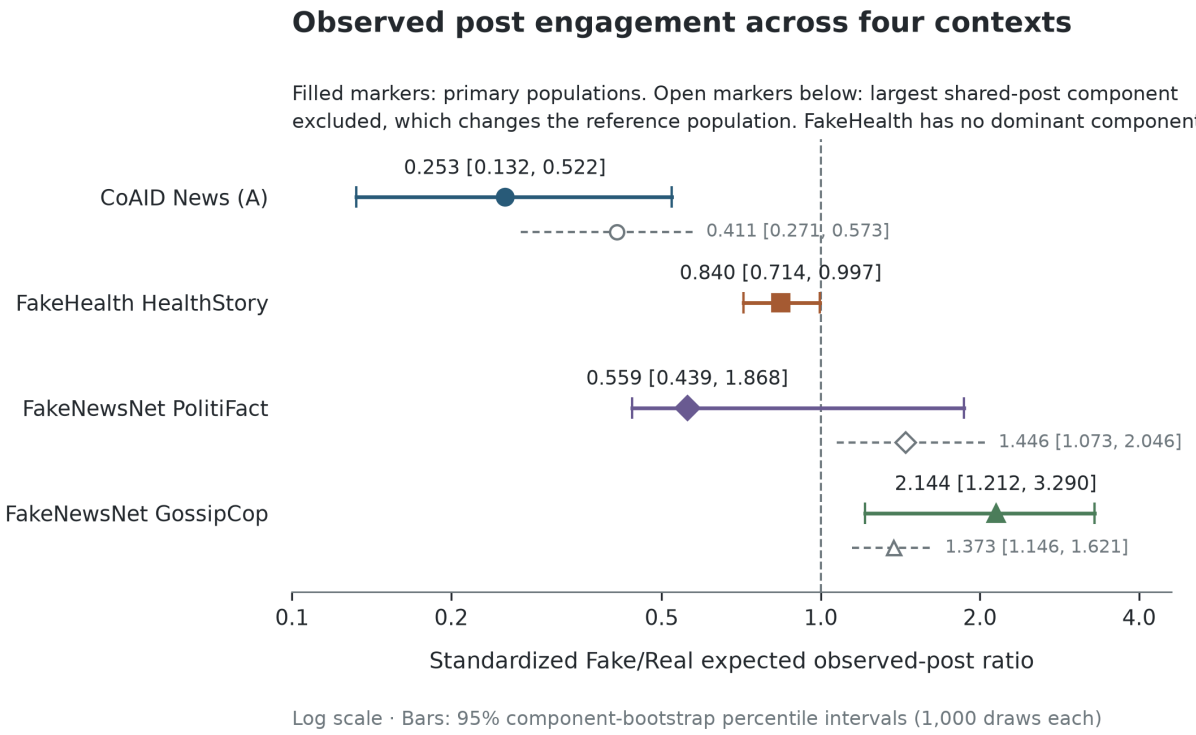


Figure 1. Fake/Real observed-post ratios in 4 contexts. Filled markers are model-standardized ratios of expected unique observed posts, including items with no observed posts, for CoAID News (A), FakeHealth HealthStory and the supplementary FakeNewsNet PolitiFact and GossipCop collections. Bars are 95% component-bootstrap percentile intervals. Open markers below each primary value show the ratio after excluding the largest shared-post component, which changes the reference population; FakeHealth has no dominant component. Log scale; the dashed line marks parity. CoAID-to-FakeHealth ratio-of-ratios 0.301 (0.154–0.645).

3.2 Outside health, the size, direction and source of the gap all changed

Two non-health collections show that neither the size nor the direction of the gap is fixed (Table 2; Supplementary Table S17). Under the same estimand, the Fake/Real ratio was 0.559 (0.439–1.868) for PolitiFact and 2.144 (1.212–3.290) for GossipCop.

Where the gap arose was also different. In CoAID, Fake items were far less likely than Real ones to have any observed post. In PolitiFact, which like CoAID labels items by fact-checker results, this reversed: Fake items were more likely to have any observed post (0.907 vs 0.654). PolitiFact's interval includes parity. One cluster of 84 items, 79 of them Real, holds two-thirds of the collection's post links. Its items are mostly generically titled pages and transcripts, such as web-portal pages and debate transcripts, linked by shared tweets rather than paraphrases of one story (Supplementary Table S20). Excluding the cluster gives 1.446 (1.073–2.046), and posted Fake and Real items then drew similar numbers of posts (ratio 0.96). The same exclusion weakened CoAID's contrast but reversed PolitiFact's.

In GossipCop, Fake items drew more expected posts than Real ones in both components. The ratio stayed above 1 without its largest cluster (1.373, 1.146–1.621) and when items with repeated titles were removed (1.767). Its ratio-of-ratios against FakeHealth was 2.552 (1.357–4.087), or 1.635 (1.277–2.057) without the cluster. The excess rests on a heavy Fake tail. The median item had 12 posts if Fake and 45 if Real, and a random Fake item had more posts than a random Real one only 37% of the time (Table 2). PolitiFact shows the mirror image: its mean ratio is below 1, but the median Fake item had 77.5 posts against 7.5 for the median Real item, and a random Fake item outdrew a random Real one 63% of the time. In both non-health collections the mean and the typical item point in opposite directions. In the health datasets they agree (0.29 in CoAID, 0.46 in FakeHealth).

Table 2. Fake/Real observed-post ratios and their components in four collections.

Collection	Fake label	Fake/Real ratio (95% interval)	Any observed post, Fake / Real	Posts given any, Fake/Real ratio (95% interval)	Median observed posts, Fake / Real	P(Fake > Real)
CoAID News	Fact-checked claims (Real: health agencies and publishers)	0.253 (0.132–0.522)	0.117 / 0.617	1.30 (0.73–2.55)	0 / 3	0.29
FakeHealth HealthStory	Expert-rated poor reporting	0.840 (0.714–0.997)	0.998 / 0.995 (observed)	Not separated (single-part model)	113 / 129	0.46
FakeNewsNet PolitiFact	Fact-checker verdicts	0.559 (0.439–1.868)	0.907 / 0.654	0.404 (0.336–1.255)	77.5 / 7.5	0.63
PolitiFact, largest cluster excluded		1.446 (1.073–2.046)	0.906 / 0.604	0.964 (0.726–1.351)	—	—
FakeNewsNet GossipCop	Site score below 5 of 10	2.144 (1.212–3.290)	0.965 / 0.937	2.083 (1.185–3.179)	12 / 45	0.37
GossipCop, largest cluster excluded		1.373 (1.146–1.621)	0.960 / 0.937	1.340 (1.118–1.577)	—	—

Standardized expected unique observed posts, including items with none. CoAID and FakeHealth are the primary contexts; the FakeNewsNet rows are an out-of-domain check with title-search tweet collection and no release covariate. Probabilities are standardized (CoAID), fitted class values (FakeNewsNet) or observed shares of items with at least one post (FakeHealth, single-part model). Cluster-excluded rows change the reference population and are influence checks. Component ratios do not multiply exactly to the overall ratio because the estimand averages row-level products. Medians and P(Fake > Real) are observed rather than model-standardized, and count items with no posts. P(Fake > Real) is the probability that a randomly chosen Fake item has more observed posts than a randomly chosen Real item, with ties counted as one half (Supplementary Table S21). Sources: Supplementary Tables S11 and S17.

3.3 Fake and Real were highly separable by text in CoAID and modestly separable in FakeHealth

On reserved holdouts, logistic regression reached macro F1 0.926 and ROC-AUC 0.983 in CoAID, against 0.675 and 0.770 in FakeHealth (Figure 2). The linear SVM gave nearly identical results (CoAID 0.913 and 0.979; FakeHealth 0.670 and 0.770).

Separability did not depend on explicit source names, on particular publishers or on the formatting markers we could identify. Masking source names barely changed either dataset (CoAID 0.926 to 0.928; FakeHealth 0.675 to 0.673). In FakeHealth, most of the signal survived on publishers the model had never seen: with whole publisher families held out, macro F1 averaged 0.620 across folds (SD 0.021; range 0.603–0.643; mean ROC-AUC 0.700), against 0.660 under resource-grouped cross-validation, a descriptive comparison between designs that differ in grouping and tuning. CoAID's classes differ sharply in format (Supplementary Table S16): character-encoding errors appeared in 25.9% of CoAID Fake texts and no Real texts, and year tokens and photo credits appeared in 49.0% and 24.8% of Real texts against 8.4% and 0.1% of Fake texts. In FakeHealth the same markers were spread evenly across classes (years 58.8% Fake vs 62.5% Real; credits 23.1% vs 17.9%). Repairing the encoding errors left CoAID holdout macro F1 at 0.928, and removing years, month names and credits as well gave 0.919. These checks address explicit names, publisher identity and identifiable markers; the broadest difference between CoAID's sources is genre, and its clearest trace, length, is examined next.

Document length accounts for much of the separability in both datasets (Supplementary Table S19). A logistic regression on log token count alone reached ROC-AUC 0.882 on the CoAID holdout, against 0.983 for the word model, and 0.729 on the FakeHealth holdout, against 0.770. In CoAID the shortest fifth of texts (4–36 tokens) was 93% Fake, and the other fifths were 2–10% Fake. The word models' scores fell with length within both datasets (Spearman ρ −0.54 in CoAID, −0.63 in FakeHealth). FakeHealth's Fake items also tend to be shorter, though the classes overlap more. Words add to length. Within length quintiles, the LR still separated Fake from Real with ROC-AUC 0.878 on the CoAID holdout (SVM 0.858) and 0.672 on the FakeHealth holdout, against 0.983 and 0.770 unstratified (Supplementary Table S19). CoAID's macro F1 of 0.926 therefore reflects both how easily short fact-checked excerpts are told apart from long institutional articles and words that separate the classes at any length. The word signal was weakest in CoAID's shortest fifth, where 131 Fake items faced 10 Real ones (0.584).

3.4 What each model learned did not carry over to the other dataset

Transfer cut performance sharply in both directions, leaving ranking only modestly above chance (Figure 2). On the CoAID holdout, the FakeHealth-trained LR reached macro F1 0.429 and ROC-AUC 0.571, a macro-F1 drop of 0.497 from the CoAID-trained model on the same items. On the FakeHealth holdout, the CoAID-trained LR reached 0.451 and 0.619, a drop of 0.224. SVM reproduced both results (0.424 and 0.583; 0.449 and 0.631). The larger drop on CoAID reflects its higher within-dataset baseline.

The errors ran in opposite directions as well. FakeHealth-trained models labeled too many CoAID items Fake, while CoAID-trained models labeled too few FakeHealth items Fake.

The two datasets' word–label associations were essentially unrelated (Figure 3). Across the 4,174 words shared by the two LR models, coefficients correlated at r = −0.015 (Spearman ρ = −0.003), and 48.4% reversed direction between datasets. Among the 100 most strongly weighted shared words, 48.0% reversed, so the strongest cues were no more consistent than the rest. A reversal share near one half is what independent associations would produce, so the two datasets' cues are unrelated rather than opposed. Each dataset's own coefficients were stable: refitted on different training folds, they agreed closely with themselves (mean Spearman ρ 0.78 in FakeHealth and 0.82 in CoAID), against −0.003 between datasets (Supplementary Table S23).

The most heavily weighted words differed in kind as well as direction (Supplementary Table S12). CoAID's strongest Real-leaning word was "and", and its Real-leaning words also included dates ("2020", "april", "march") and image-credit fragments ("getty", "images", "medscape"); its Fake-leaning words were mostly topical and political terms ("coronavirus", "trump", "facebook", "stimulus", "president") along with encoding artifacts. FakeHealth's weights were much smaller and centered on health-reporting vocabulary, with press-release terms ("release") leaning Fake and words such as "costs" and "says" leaning Real. Between 9 and 14 of each dataset's 25 most Fake- or Real-leaning words leaned the other way in the other model.

One cue did carry across. Length alone ranked the transfer targets better than either word model. On all eligible items, the length baseline reached ROC-AUC 0.852 on CoAID when fitted on FakeHealth, against 0.537 for the FakeHealth word model, and 0.649 on FakeHealth when fitted on CoAID, against 0.605 (Supplementary Table S19). Both datasets encode short as Fake, but the word models did not pass this on equally. CoAID-trained scores fell with length on FakeHealth (ρ −0.69), so long FakeHealth articles were scored as Real. FakeHealth-trained scores barely tracked length on CoAID (ρ −0.12), because its vocabulary weights outweighed it. After stratifying by length, the transferred word models ranked both targets close to chance (ROC-AUC 0.50–0.52 within length quintiles), so the ranking that did transfer between the datasets came mainly through length.

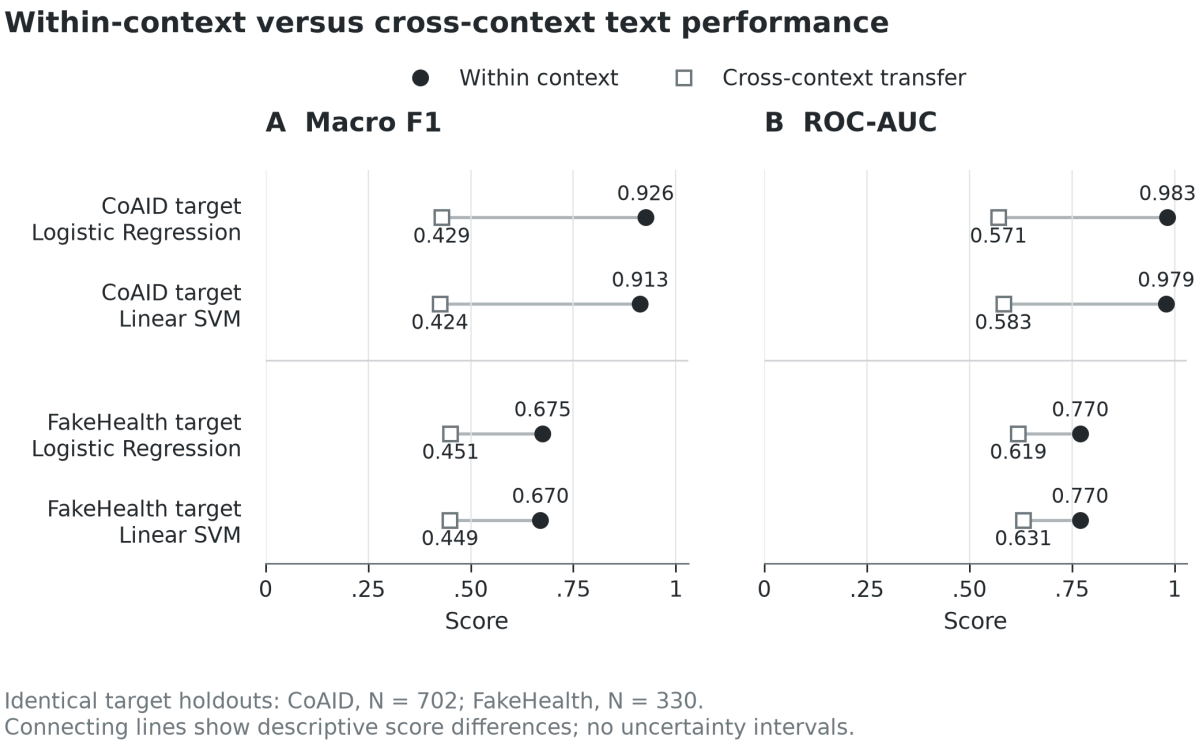


Figure 2. Within-dataset and transferred text performance. Macro F1 (A) and ROC-AUC (B) for TF-IDF logistic regression and linear SVM. Filled circles: models trained in the target dataset; open squares: models transferred from the other dataset. Both are scored on the same target holdout items (CoAID N = 702; FakeHealth N = 330).

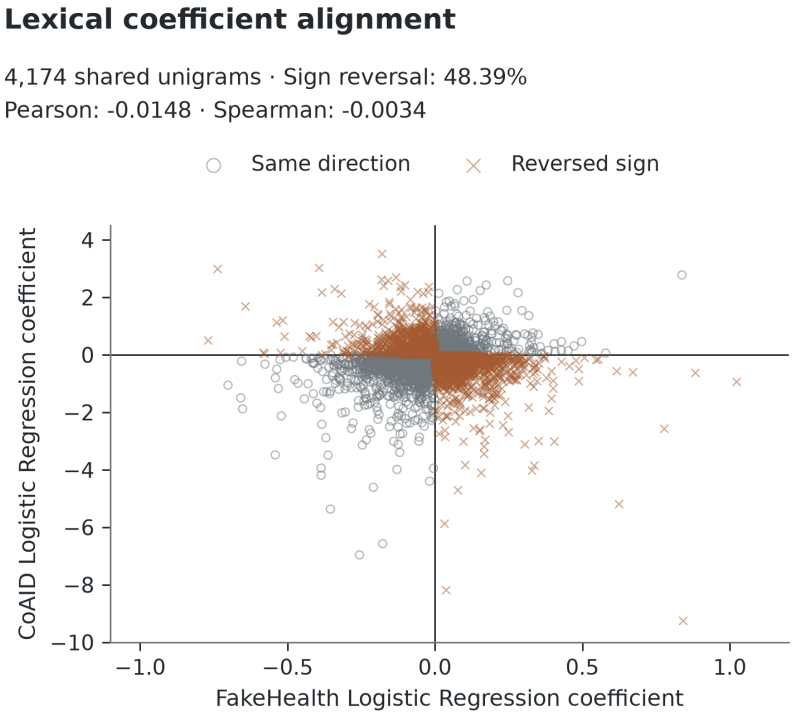


Figure 3. Word–label associations in the two datasets. Each point is one of 4,174 unigrams shared by the within-dataset logistic regression models; positive values point toward Fake. Crosses mark words whose association reverses sign between datasets (48.4%). Pearson r = −0.015; Spearman ρ = −0.003. Coefficient scales are model-specific, so the figure shows direction, not magnitude.

3.5 CoAID-trained models ranked independent COVID rumors better, largely through length

On the 3,667 COVID-Rumor NEWS items, CoAID-trained models ranked items considerably better than FakeHealth-trained models (Figure 4). ROC-AUC was 0.700 versus 0.522 for LR and 0.682 versus 0.525 for SVM; FakeHealth-trained models were close to chance. Macro F1 showed the same ordering with smaller differences (LR 0.558 vs 0.497; SVM 0.531 vs 0.497). The ordering held when approximate overlaps were also excluded and when all eligible items were retained.

Length explains most of this ordering. COVID-Rumor items are short statements, and a length-only model fitted on either source pool reached ROC-AUC 0.716 on them, slightly above the CoAID word model (Supplementary Table S19). The CoAID-trained models' scores fell with length on this target (ρ −0.36 for LR), whereas the FakeHealth-trained models' did not (ρ 0.04). What CoAID's models carried to COVID-Rumor was mostly that its Fake items, like CoAID's, are short.

The portable part of the signal was ranking. With fixed source thresholds, all four models labeled 89.6–94.4% of items Fake against a true rate of 82.2%, and balanced accuracy stayed between 0.50 and 0.55. Both source vocabularies covered most target words (89.5% of target word occurrences for CoAID, 81.9% for FakeHealth), and no target item lacked features.

A non-COVID target showed a weaker version of the same ordering. Scored with title-only LR, the CoAID-trained model ranked FakeNewsNet's fact-checked PolitiFact items better than the FakeHealth-trained model (ROC-AUC 0.608 vs 0.503) and GossipCop's entertainment items slightly better (0.532 vs 0.477). Both models were near chance on thresholded metrics, and the FakeHealth title model carried little signal even within FakeHealth (macro F1 0.513) (Supplementary Table S13). Length alone, fitted on CoAID, ranked PolitiFact titles better than the CoAID title model (ROC-AUC 0.739 vs 0.608), so this ordering also follows the length of fact-checked claims rather than their vocabulary.

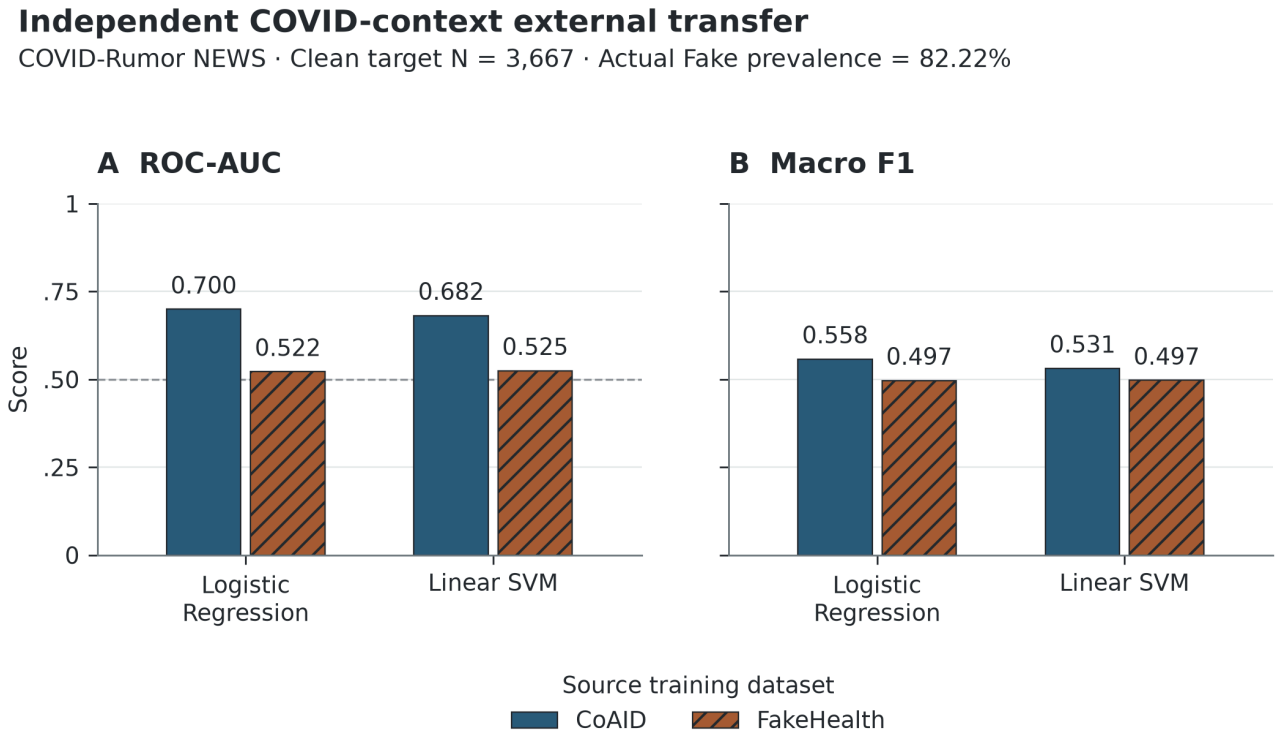


Figure 4. Transfer to independent COVID rumors. ROC-AUC (A) and macro F1 (B) on the same 3,667 COVID-Rumor NEWS items (82.22% Fake) for models trained on CoAID (solid) and FakeHealth (hatched). Source models and thresholds were applied unchanged. Dashed line in A marks ROC-AUC = 0.5.

3.6 Robustness

The main patterns held across the robustness checks (Table 1). Title-only models gave small macro-F1 gains on transfer but lower ROC-AUC in both directions, so shortening the text did not restore portability. DistilRoBERTa matched the classical within-dataset pattern (macro F1 0.939 in CoAID, 0.621 in FakeHealth) but did not improve transfer: FakeHealth-to-CoAID ROC-AUC was 0.442 and CoAID-to-FakeHealth 0.437. Truncation explains part of DistilRoBERTa's shortfall in FakeHealth (macro F1 0.621 against 0.675 for LR), but not all of it. Refitting the LR on only the text before the 512-token cut gave 0.651, and the text after the cut alone gave 0.653 with a higher ROC-AUC (0.766 vs 0.718). The FakeHealth signal is spread through each article, and 40.9% of the occurrences of the LR's 50 strongest terms fall after the cut (Supplementary Table S22). Its transfer scores were also unstable. Across three seeds with identical settings, FakeHealth-to-CoAID macro F1 on the matched CoAID holdout at 512 tokens ranged from 0.287 to 0.510. Shortening inputs to 256 tokens raised macro F1 in all three seeds (mean change +0.101, SD 0.079) but improved ranking in only one.

Table 1. Robustness and sensitivity checks.

Analysis	Result	Reading
Engagement, reconstructions B and C	Ratio-of-ratios 0.286 (B), 0.343 (C)	Direction unchanged
Engagement, largest CoAID cluster excluded	Ratio-of-ratios 0.489 (0.310–0.710) vs 0.301	Weaker, same direction (influence check)
Linear SVM	Within macro F1: CoAID 0.913, FakeHealth 0.670; transfer near chance	Replicates LR
Source-name masking	LR holdout macro F1: CoAID 0.926 → 0.928; FakeHealth 0.675 → 0.673	Names not needed
CoAID encoding repair; plus years, months, credits removed	LR holdout macro F1 0.928; 0.919 (vs 0.926)	These markers not needed
FakeHealth unseen publishers	Mean macro F1 0.620 (SD 0.021) vs 0.660 resource-grouped	Most signal survives
Title-only transfer	Small macro-F1 gains; lower ROC-AUC in both directions	Does not restore transfer
DistilRoBERTa, 512 tokens	Transfer ROC-AUC 0.442 and 0.437	No transfer gain
DistilRoBERTa, 256 vs 512 tokens, 3 seeds	Macro F1 +0.101 (SD 0.079); ranking better in 1 of 3	Seed-sensitive
Length-only baseline (log token count)	ROC-AUC within CoAID 0.882 (word LR 0.983), FakeHealth 0.729 (0.770); beats word LR on both transfer targets and COVID-Rumor; word LR within length quintiles 0.878 (CoAID), 0.672 (FakeHealth)	Length carries much of the within-dataset signal and words add to it; on transfer, word models are near chance within length quintiles (0.50–0.52)
FakeHealth LR on text before 512-token cut	Macro F1 0.651 (vs 0.675); text after cut alone 0.653	Truncation explains part of transformer shortfall
Fold-to-fold LR coefficient stability	Mean Spearman ρ 0.78 (FakeHealth), 0.82 (CoAID) vs −0.003 across	Cross-dataset non-alignment is not noise

SD = standard deviation across folds or seeds, not a confidence interval.

4. Discussion

4.1 One pattern across both branches

The engagement and text branches ordered the two health datasets the same way. In FakeHealth, Fake and Real items come from overlapping publishers, text separates them modestly (macro F1 0.675, most of which survives evaluation on unseen publishers and much of which is length), and Fake items draw nearly as many observed posts as Real ones

(ratio 0.840). In CoAID, source and label are nearly the same variable (Cramér's V 0.971 against 0.352 in FakeHealth, in the text-modeling cohorts; Supplementary Table S14), Fake and Real are highly separable by text (0.926), through both length and words, and Fake items draw about a quarter of the observed posts of Real ones (0.253). FakeHealth's Fake label marks weaker reporting from mainstream outlets, many of which also supply its Real items. CoAID's marks fact-checked claims, set against health agencies and health-information publishers. In both branches, the dataset whose Fake and Real items differ more in construct and source showed the larger Fake–Real distance. With two primary datasets, this is an association between contexts.

Two further results fit this association. First, word–label associations were unrelated between the datasets, as expected if the two Fake labels encode different constructs, and each dataset's coefficients were stable across training folds, so the two vocabularies differ reliably. Second, length was the only cue we found that carried between the datasets. Short items lean Fake in both, a length-only model transferred better than either word model, and CoAID-trained scores fell with length on FakeHealth (ρ −0.69), consistent with those models labeling too few FakeHealth items, all full articles, as Fake. The words each model weighted most also differed in kind. In FakeHealth, "costs" leaned Real, and coverage of costs is one of the review criteria; "release" leaned Fake, which may reflect the criterion on whether a story relies solely or largely on a news release (Dai et al., 2020), although the word alone does not show such reliance. CoAID's model leaned toward Real on dates, image-credit fragments and the word "and". These follow the formatting differences between its classes: 49.0% of CoAID's Real texts carry a year and 24.8% a photo credit, against 8.4% and 0.1% of its Fake texts.

Across all four collections, the Fake/Real ratio ranged from about a quarter in CoAID to about two in GossipCop, and the component behind the gap varied: in CoAID it lay in whether items were posted at all, while in GossipCop posted Fake items drew more posts.

4.2 Weighing the explanations

Three explanations could produce this pattern, and the evidence for them differs. The first is label construct and source ecology: the datasets define Fake differently and draw Fake and Real items from differently separated sources. It is consistent with the text results if genre is counted as part of source ecology: fact-checked claims are short statements in CoAID and COVID-Rumor, PolitiFact's titles behaved the same way, and length is the cue that carried between them, while the word models carried almost nothing between CoAID and FakeHealth beyond it. For engagement, the label construct alone does not determine the gap, because PolitiFact shares CoAID's construct but not its engagement pattern; the two collections also differ in what they count as Real.

The second explanation is how posts were collected and counted. Both CoAID and FakeHealth gathered tweets by searching for item titles, and FakeHealth also searched by keywords and URLs (Cui & Lee, 2020; Dai et al., 2020). The procedures also differ in how queries were formed, when crawls ran and, in CoAID, how posts were pooled across repeated captures. These differences bear on the engagement counts, not on the text results. The whole CoAID gap lies in whether an item has any observed post, which is where differences in capture would appear. PolitiFact, also collected by headline search, showed no Fake occurrence deficit, so headline search does not produce one by itself. The findability breakdown supports this explanation for part of the occurrence gap. CoAID's Fake items had any observed post far less often when no publisher could be resolved (7.5%) than when one could (33.0%), and Real items first captured in CoAID's later releases rarely had any observed post either. If fact-checked claims had generally circulated before being checked, these zeros reflect what the collection could find and when it looked. Resolved Fake items were still posted less often than resolved Real ones (33.0% against 57.6%), so publisher identifiability alone does not account for the gap.

The third explanation is the COVID context itself: a crisis may shift attention toward official sources and change the language of both true and false claims. CoAID-trained models did rank COVID rumors better, but length alone matched that advantage and did the same for non-COVID PolitiFact titles. The ordering therefore tracks the short-claim form of fact-checked items, and the text results offer little evidence of a crisis effect on language.

In sum, the first explanation is consistent with both branches. The second is supported for part of CoAID's occurrence gap but cannot explain the text results. The third cannot be assessed with these data.

4.3 Engagement in context

Our estimates and those of Vosoughi et al. (2018) measure different quantities. Their unit was the fact-checked rumor cascade, so every item had been shared at least once. Ours is the labeled item, including items with no observed posts, and the outcome is captured posts rather than cascade depth, speed or reach. The hurdle components separate the two quantities. In CoAID, fact-checked Fake items were about one-fifth as likely as Real items to have any observed post, but those that were posted drew a comparable number of posts (ratio 1.30, with an interval that includes parity). A design that begins from cascades observes only the second quantity; the zero-inclusive design adds the first, which the findability results tie partly to what the collection could find. Juul and Ugander (2021) likewise showed that comparisons of false and true news depend on what is held fixed: structural differences largely disappeared once cascade size was matched. The split did not recur in PolitiFact, where Fake items were more often posted and, without the dominant cluster, posted Fake and Real items drew similar numbers of posts (ratio 0.96).

In FakeHealth, stories that expert reviewers judged poorly reported drew 84% of the observed posts of well-reported stories. In this collection of mainstream health news, reporting quality was only weakly related to observed engagement.

The size of the CoAID contrast depends on one cluster of items linked by shared posts (ratio-of-ratios 0.489 without it, against 0.301); its direction held in every check. The ratios describe expected posts per item, and in both non-health collections the typical item pointed the other way: a random Fake item outdrew a random Real one only 37% of the time in GossipCop and 63% of the time in PolitiFact (Table 2).

4.4 What this means for detection

High within-dataset accuracy is weak evidence of a general detector. In CoAID, macro F1 of 0.926 within the dataset coexisted with transfer only modestly above chance and with word associations unrelated to FakeHealth's. This parallels evidence that classifiers trained on source-reputation labels generalize poorly to article-level veracity, with performance that depends on topic (Torabi Asr & Taboada, 2018), and that period-specific proper names can become spurious cues (Murayama et al., 2021).

Threshold and ranking metrics need to be read together. Title-only models and shorter transformer inputs raised macro F1 without consistently improving ranking, and on the external target the signal that transferred was in ranking: balanced accuracy stayed between 0.50 and 0.55. A threshold fixed on one dataset's class balance need not suit another's.

DistilRoBERTa did not improve transfer either (ROC-AUC 0.442 and 0.437). It matched the classical models within CoAID but fell short in FakeHealth (macro F1 0.621 against 0.675), partly because most FakeHealth articles were truncated, and its transfer scores varied widely across seeds. With one transformer and three seeds, this does not rank model families; it shows that this pretrained model did not close the transfer gap. Joint learning across misinformation tasks (Lee et al., 2021) would be most informative if evaluated against fixed external targets with explicit label constructs.

4.5 What the results ask of related work

Several results concern how misinformation datasets are built and analyzed, and each suggests a check for related work.

Misinformation research. Studies should say what both classes consist of: CoAID and PolitiFact share a Fake construct but differ in their Real class and in their engagement patterns. They should also say whether the outcome counts every item or only items that circulated. In CoAID the whole gap lay in whether items were posted at all, which a design that starts from cascades cannot see. Comparisons across datasets need a common estimand, such as standardized expected counts, and uncertainty that respects shared posts, because a single dependent cluster changed the magnitude of CoAID's contrast and the direction of PolitiFact's point estimate. Studies should also say whether they compare means or typical items, since heavy tails made the two disagree in both non-health collections.

Machine-learning work. A high within-dataset score can reflect how a dataset was assembled. In CoAID, source and label were nearly the same variable (Cramér's V 0.971), Fake texts were short excerpts while Real texts were long institutional articles, and length alone ranked the holdout at ROC-AUC 0.882. Real texts also carried years and photo credits that Fake texts rarely did, and encoding errors appeared only in Fake texts. Removing source names, encoding errors, or dates and credits barely changed CoAID's 0.926 macro F1, so none of these markers was necessary on its own. Six practical checks follow:

- examine how source and label co-vary before training;
- compare every model with a length-only baseline, within and across datasets, and report its scores within length strata;
- scan for formatting and encoding markers that differ by class;
- group data splits by resource, version lineage and near-duplicate text rather than by row;
- report transfer on the same target items alongside within-dataset scores;
- state the text representation, because titles, excerpts and full articles give a model different information.

Data processing. Several decisions that affected these results were made before any model ran. In an early representation of CoAID, generic branding and domain-level links merged distinct pages into one item joining 110 source records, so we counted engagement only after resolving items by specific resource and version lineage. Items that share posts are not independent rows. Treating them as clusters set the width of every engagement interval here, and in three collections whether one cluster was drawn largely determined where a bootstrap draw landed. Engagement scope also has to match across datasets. We excluded FakeHealth's retweets because CoAID has no equivalent, and pooled CoAID posts across its four releases. Both choices change the counts. Real items first captured in CoAID's two later releases rarely had any observed post, against about nine in ten in the earlier two, which suggests uneven collection across releases. Dataset releases that document these steps would let users judge which findings reflect collection.

4.6 Limitations

Label construct, source composition, COVID timing and collection procedure vary together between CoAID and FakeHealth. The study therefore describes how the Fake–Real distance differs between these dataset contexts; it cannot attribute the difference to any one of these features, and it does not estimate an effect of the pandemic. The COVID-Rumor target and the FakeNewsNet collections widen the comparison but do not remove this confounding, because each brings its own construct, period and collection procedure. Accordingly, the cross-branch pattern in Section 4.1 is an association across contexts, and the explanations in Section 4.2 are weighed by their consistency with the evidence, not identified.

Engagement is measured as captured posts. Deleted, private and uncaptured activity cannot be recovered, and if capture differed by class or dataset it could bias the ratios. CoAID's post coverage differed sharply by release, and part of its zero-inflation may reflect collection rather than circulation. The cluster bootstrap reflects dependence through shared posts only, assumes independence between datasets, and does not include uncertainty from entity resolution. The influence of one CoAID cluster shows that magnitude estimates are sensitive to a few highly shared items.

Text comparisons are bounded by the information each model received. CoAID stores short excerpts and FakeHealth full articles, most FakeHealth texts were truncated for the transformer, and the transformer follow-up covered three seeds in one direction. Publisher-held-out evaluation was possible only for FakeHealth. CoAID's Fake and Real texts also differ in format, including encoding errors confined to Fake texts. Removing these markers barely changed separability, but other collection traces remain, the largest being length: Fake texts are shorter than Real ones in both datasets. The text models were not adjusted for it. The within-quintile check, added after the length results were seen (Supplementary Table S19), shows that the word signal survives within both datasets, but it rests on small Fake counts in CoAID's longer quintiles and reports no intervals.

The external target consists of short statements with 82% Fake prevalence. Its results describe ranking portability to that genre and class balance, and conservative overlap screening cannot exclude every equivalent item. The FakeNewsNet checks are weaker still: text was limited to titles, engagement came from a different collection

procedure and period, PolitiFact's interval includes parity and its shape depends on one cluster of generic pages, and the local GossipCop Real items do not match their documented source.

4.7 Future work

Restricting or matching the existing collections would not yield a cleaner comparison: few publisher families carry both labels in both datasets, publisher identity is unresolved for most CoAID Fake items, and CoAID records no topic field (Supplementary Table S15). Separating crisis context from label construct therefore requires new data: a fixed publisher panel with adequate Fake and Real support in both crisis and comparison periods, a common item-level veracity protocol with reporting quality recorded as a separate construct, one platform outcome collected under identical query rules and a follow-up horizon fixed in advance, and retained timestamps and version lineage.

Explaining why sharing differs would also need exposure and behavioral measures. People discriminate true from false COVID-19 headlines less well when deciding what to share than when judging accuracy (Pennycook et al., 2020), although a preregistered replication found a smaller effect of an accuracy prompt (Roozenbeek et al., 2021), and a crisis may draw unusual attention to official sources such as CoAID's Real publishers. For detection, event-held-out and chronological evaluations with consistent label definitions would test whether the portability pattern recurs, and benchmarks built from such data should match Fake and Real items on length and format, or report results within length strata, so that portability can be judged on content rather than genre.

5. Conclusion

Fake health items drew fewer observed posts than Real ones in both health datasets, but the size of the gap differed: about a quarter as many in COVID-focused CoAID, where the shortfall came from Fake items rarely being posted at all, largely among items the collection could not trace to a publisher, and nearly as many in FakeHealth. Outside health, the gap was uncertain in PolitiFact and reversed in GossipCop. Text separability also differed: high within CoAID, modest within FakeHealth and only modestly above chance across them. Neither dataset's word cues carried over to the other; the cue that did was that Fake items were shorter. Across these collections, the distance between Fake and Real items, in language and in observed engagement, was larger where a dataset's classes differed more in construct and source. Which of the features that distinguish the datasets produces this difference, and whether crisis periods change it, is a question for data that hold platform, labeling and observation windows fixed. Until such data exist, engagement findings and detection scores should be reported alongside the definition of misinformation that produced them, the sources each class was drawn from, and the processing steps (item resolution, dependence and collection scope) that shaped the counts.

Author contributions

J.Z. designed the study, carried out the analyses and wrote the manuscript. J.V. introduced the evaluation metrics (ROC-AUC and macro F1), taught and guided the use of pandas, suggested the models and related literature to consult, and provided overall direction on the machine-learning methodology and data processing.

Use of AI tools

We used Anthropic’s Claude (including Claude Code) as an assistant in this work. It helped write and debug analysis and formatting scripts, run and audit the analysis pipeline, check reported numbers against saved outputs, verify references, and suggest edits to the manuscript’s wording and structure. The study design, interpretation of results and all final text were decided and reviewed by the authors, who take full responsibility for the content.

Data and code availability

Code, data splits, model locks, classical model bundles, aggregate tables, bootstrap draws and item-level predictions are available at https://github.com/jayzh312-rgb/health-misinfo-context (release v1.0.0). Code is released under the MIT license and derived tables under CC BY 4.0. The source datasets (CoAID, FakeHealth, the COVID-19 Rumor

Dataset and FakeNewsNet; Cui & Lee, 2020; Dai et al., 2020; Cheng et al., 2021; Shu et al., 2020) are available from their providers and are not redistributed. Processed tables are shared without article text or post identifiers, so rebuilding the text features requires the providers' data. Transformer checkpoints are not included; the pinned base model, locked settings, seeds and all predictions are.

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Supplementary Material

These notes specify implementation details supporting Methods Sections 2.1–2.12. All result tables are displayed below, with exact aggregate CSVs included in `data/`. Internal filenames identify archived provenance and are not dependencies for reading this package. They describe the completed protocols and do not define additional analyses.

S1. Content identity, lineage, and engagement linkage

The authoritative CoAID populations are `A_primary_strict`, `B_resource_sensitivity`, and `C_high_certainty`, with 5,337, 5,205, and 5,082 entities, respectively. Their construction is recorded in `outputs/reports/coaid_version_aware_rebuild.md`, `outputs/tables/coaid_version_aware_entity_lineage.csv`, and `outputs/tables/coaid_version_aware_rebuild_metadata.json`. Specification A is the main population; B/C remain sensitivities.

Version partitions are the audited partitions, not clusters regenerated from a single similarity threshold. Supporting evidence includes normalized equality, token overlap, edit similarity, and body/abstract similarity, interpreted jointly with informative titles and metadata. Complete-link checks enforce compatible within-version pairs for material updates. Missing or insufficient excerpts do not establish identity, and an identical short introduction does not verify an entire historical webpage. Item-specific resource normalization remains offline; unverified redirects and cross-URL aliases are not inferred. The content-version evidence and final qualitative refinements are documented in `outputs/reports/coaid_content_version_stability.md` and its linked evidence artifacts.

Within a final entity, the representative record is selected by descending stored body length, then descending title length, then release and original UID in ascending order. All original member captures remain in provenance. Entity UIDs incorporate the specific resource, aggregation mode, and sorted source UIDs. Post and reply IDs are set-unioned within the entity; legacy aggregate counts are not added together. These rules are implemented in `materialize` in `src/rebuild_coaid_version_aware_samples.py`.

CoAID content IDs originate in `Unnamed: 0`; post files join through `index` and supply `tweet_id`, while reply files join through `news_id` and supply `reply_id`, within the release/file category. FakeHealth joins review/content/engagement records using `news_id` and retains unique IDs from the engagement dictionary's `tweets` and `replies` lists. Its separate `retweets` list is not included in the common primary count. Linkage implementations are retained in `src/build_coaid_content_table.py` and `src/build_fakehealth_content_table.py`.

Source-title changes or dates alone did not establish material change; a material-change decision required corroborating evidence across the body, title, abstract and metadata fields.

S2. Engagement standardization and uncertainty

`src/run_version_aware_engagement_analysis.py` uses the validated count-fitting and standardization helpers in `src/run_primary_engagement_models.py` and `src/audit_engagement_estimand_robustness.py`. CoAID designs include an intercept, the Fake indicator, and first-recorded-release indicators with May as reference. FakeHealth uses the intercept and Fake indicator. For the hurdle, standardization averages each row's logistic probability multiplied by its conditional positive-count mean. Changing the reference indicator applies consistently to both model components.

Bootstrap components are sampled with replacement without class stratification; each selected component contributes all its rows. Attempt seeds depend on the sample and attempt index rather than worker scheduling. Failed warm-start fits receive the documented multistart retry on the same draw. Failure, convergence, and successful-draw records are retained; inference targets 1,000 successful post refits. Each draw uses its own sampled reference distribution. The interval excludes uncertainty about family selection, unobserved social dependence, and content-identity decisions.

`src/run_cross_context_engagement_contrast.py` combines all marginal draw pairs and uses NumPy linear empirical quantiles. The dominant-component exclusion has its own 1,000-successful-refit CoAID bootstrap, combined with the saved FakeHealth draws. This exclusion distribution is distinct from conditioning the original bootstrap on draws omitting the component. Top-five and top-1% outcome exclusions are secondary point-estimate diagnostics; their separately retained HC0 intervals are not overlap-cluster confirmation. Reply NB fits remain provisional secondary summaries rather than a replacement primary outcome.

The engagement populations were 5,337 CoAID Specification A entities (922 Fake, 4,415 Real) and 1,690 HealthStory items (472 Fake, 1,218 Real), both including entities with zero observed posts (Table S3).

FakeHealth retweets excluded from the primary outcome were also excluded from the overlap graph, so graph edges reflect only counted post and reply IDs. A connected component was a dependence unit for resampling, not a claim that its members represented identical content. The posts-only occurrence part used by the hurdle (any_post) was distinguished from the broader any-post-or-reply indicator reported as engagement_present; the two are tabulated separately in `version_aware_occurrence_results.csv`. The cross-context estimand was C = R_CoAID / R_FakeHealth, equivalently theta = log(R_CoAID) − log(R_FakeHealth); values of C below 1 indicate a smaller Fake/Real observed-post ratio in CoAID. The Cartesian bootstrap distribution was formed on these log differences and exponentiated.

S3. Text grouping, quality, and source masking

Exact title/body comparisons use the minimally normalized case-preserved text. Near-duplicate screening uses case-preserving word shingles, ignoring punctuation for the comparison only: title bigram Jaccard >=0.95 for strings with at least 8 words, and body five-gram Jaccard >=0.90 with at least 30 words. Candidate filtering is followed by exact Jaccard verification, without learned TF-IDF or embedding features. Source lineage and specific-resource barriers are combined with text links into evaluation components. Cross-dataset flags cover directly matched rows and affected within-dataset groups; this screen does not claim to identify all paraphrases.

The final quality audit's `primary_modeling_eligible` flags supersede initial text-availability eligibility for modeling counts. All original records and decisions remain stored. Mixed artifact/substantive fields are reviewed without silently rewriting language; high-risk review-label disclosure is an exclusion rule. Primary split assignments are preserved when quality exclusions are applied. The applicable reports are `text_analysis_cohort_preparation.md` and `text_quality_and_source_cue_audit.md`; final sizes are corroborated by tfidf_linear_svm_cohort_summary.csv.

The saved `CoAID grouped-CV`, `FakeHealth grouped-CV`, and corresponding `CoAID holdout` and `FakeHealth holdout` files define original UID assignments. `cv_fold=-1` is reserved holdout; training for fold f requires `cv_fold>=0` and `cv_fold!=f`, while validation uses `cv_fold==f`. Original ineligible rows have no usable training assignment. The candidate outer fold was chosen by minimizing summed absolute deviations from 20% of total, Fake, and Real counts, breaking ties by fold index. Quality filtering does not regenerate these partitions.

The source dictionary uses prepared publisher names, article hosts/URLs, and fixed metadata-activated aliases, without labels or engagement. Overlapping identifier spans are merged and replaced with `[SOURCE]`. Explicit URL hosts are masked while paths remain; selected ambiguous short names use case-sensitive matching. Evaluation metadata contributed to dictionary preparation, so masking is a transductive sensitivity. Masking can create identical strings across previously distinct groups through deliberate information

removal; existing resource barriers remain unchanged.

Text cohorts were derived from HealthStory items and CoAID Specification A entities (`A_primary_strict`), not from the B or C populations. Natural `combined_text` joined the usable title and body with a blank line; a title-only or body-only item remained eligible when the other field was unavailable. A substantive headline with a poor body could remain eligible for combined-text modeling while being excluded from body-only analyses. Exact generic-template matches could also join evaluation groups. Publisher identity or broad topic alone did not define the primary groups. Rows were sorted by stable UID before partitioning, and candidate outer partitions were generated by seed-42 StratifiedGroupKFold over leakage groups; the holdout fold was then chosen by the count rule above. For transfer evaluation, existing overlap flags and group safeguards were retained; target cohorts were not redefined in response to performance.

S4. Classical tuning and selected settings

Supplementary Table S1. Classical model grids and solver settings. The settings are reproduced from the validated supplementary methods; the word n-gram notation gives inclusive bounds.

Setting	LR	Linear SVM
Word n-grams	(1,1), (1,2)	(1,1), (1,2)
min_df	2, 5	2, 5
C	0.1, 1, 10	0.01, 0.1, 1, 10
Class weighting	None, balanced	None, balanced
Sublinear TF; smoothed IDF; normalization	True; True; L2	True; True; L2
max_df; data type	1.0; float64	1.0; float64
Estimator	LogisticRegression, L2/liblinear	LinearSVC, L2/squared-hinge, dual="auto"
Iteration limit; tolerance	5,000; 1e-4	20,000; 1e-4
Fake threshold; ranking score	probability >=0.5; Fake probability	decision score >=0; decision_function

Both vectorizers use the word-token pattern `(?u)\b\w\w+\b`, lowercase conversion, and no accent stripping. Intercepts are fitted with intercept scaling 1. In the pinned scikit-learn version, LR's pure L2 penalty is specified by `l1_ratio=0.0`. Balanced class weights depend on the training labels supplied to each fit. The mean-F1 and sample-SD tie tolerances are 0.001 for both classifiers; LR then uses unigram, smaller C, None weighting, larger min_df, and configuration ID. SVM omits the larger-min_df priority. The transformer tie policy differs: after the mean-F1 tolerance, its SD ordering has no additional tolerance.

Supplementary Table S2. Selected natural-text configurations. Combined-text and title-only settings retain their separate source selections.

Selected natural configuration	n-grams	min_df	C	Class weighting
Combined LR, FakeHealth	(1,2)	5	1	balanced
Combined LR, CoAID	(1,1)	2	10	balanced
Combined SVM, FakeHealth	(1,2)	5	0.1	balanced
Combined SVM, CoAID	(1,1)	2	0.1	balanced
Title LR, FakeHealth	(1,1)	5	10	balanced
Title LR, CoAID	(1,2)	2	10	None

Title-only eligibility gives 1,645 FakeHealth items and 3,571 CoAID items; their holdouts contain 330 and 701 items. Comparisons with combined text use those same target rows. Natural-title settings are selected independently from combined-text settings, and masked-title fits reuse natural-title settings. Final selection tables and the corresponding baseline, SVM, and title-sensitivity reports retain these locks.

For lexical alignment, absolute ranks use minimum rank for ties, with lexical order resolving top-N selection. The above-median subset strictly exceeds either model's absolute-coefficient median among shared terms. Near-zero coefficients satisfy absolute value <0.05; the descriptive meaningful-in-both category requires both absolute values >=0.05. These audit categories do not represent statistical significance or standardized causal effect sizes.

LogisticRegression and LinearSVC were fitted with random_state 42 (seed 42 for the solvers), in addition to the seed-42 partitions. Training-only most-frequent and seed-42 stratified dummy classifiers provided baseline references for the LR evaluation; they were not selection candidates. Reverse-direction transfer results did not modify the opposite source model; each direction's source fit was saved before the other direction was scored. Within-context comparators remained the saved training-pool-to-holdout models; only evaluation rows, not training populations or sample sizes, were matched between within-context and transfer scores. The title-only design does not isolate text length alone, because title eligibility and the separately selected title configurations differ from the combined-text protocol.

S5. Publisher-family evaluation

Names undergo Unicode NFKC normalization, case folding, and whitespace normalization. Hosts are lowercased and stripped of leading www; subdomains collapse only to an observed parent or audited identity host. Fixed aliases and unambiguous name-host links may join identities. Multiple attribution names on a host do not bridge unrelated hosts; conflicting audited identities and hosted tenants remain separate. Explicit Twitter/X account-status and YouTube channel/account URLs can define account proxies; generic platform hosts and unresolved routes do not.

The final FakeHealth outer and inner partitions use publisher/resource components, shuffled StratifiedGroupKFold, and seed 42. All outer test partitions meet the minimum support of 20 items per class, and all final inner partitions use three folds. The source map and component construction were fixed before model fitting. Selected outer-fold natural configurations are: fold 0, (1,2)/min_df=5/C=1/balanced; fold 1, (1,1)/5/1/balanced; fold 2, (1,1)/2/1/balanced. The corresponding masked fits reuse these natural settings.

Feasibility concerns in CoAID combine class-exclusive source families, sparse mixed-family data, and unresolved identities whose removal disproportionately excludes Fake support. The design therefore does not interpret a numerically feasible restricted partition as a clean language-generalization experiment. The full rationale and normalization rules are recorded in `publisher_heldout_feasibility.md`; the final experiment is `fakehealth_publisher_heldout.md`.

S6. Transformer and external-target details

The official fast tokenizer and checkpoint revision define pair special tokens; packing is checked against that template. All titles fit the available title budget in the evaluated cohorts. Training dynamically pads each microbatch and accumulates weighted gradients to the effective batch, retaining the partial final batch. The constant scheduler allows the epoch-2 candidate checkpoint to represent the same schedule as a two-epoch refit. Completed checkpoints retain model, optimizer, and random-state provenance; execution safeguards do not change the selected optimizer or predictor protocol.

The primary experiment and token-budget follow-up are recorded in `distilroberta_text_experiment.md` and `distilroberta_budget_seed_stability.md`. The selected source settings are learning rate 2e-5 and three epochs. The targeted follow-up uses seeds 42, 123, and 2026 for both caps, reusing original seed-42 artifacts. Source class weights are recomputed for each fit's source-training population only; no target-label statistic enters training.

For COVID-Rumor NEWS, quality screening strips URLs only when checking alphabetic tokens. Text normalization otherwise preserves case and punctuation. Exact statements are compared with source title/body/combined fields, including ineligible source records. Exact specific resources use the prior offline URL normalization, remove tracking parameters, retain path identity, and exclude homepage-only matches. Missing website files are retained. High-similarity flags use title bigram Jaccard >=0.95 with >=8 words, or body/combined five-gram Jaccard >=0.90 with >=30 words. Approximate matches are retained in the primary clean cohort unless they also meet an exact exclusion rule.

Token occurrence coverage counts default-analyzer unigrams represented in the fitted source vocabulary; zero-feature diagnostics inspect transformed TF-IDF rows separately. Natural class prevalence and fixed decision thresholds are retained. The complete source-overlap evidence, quality rules, prediction-freeze checks, and evaluation scopes are recorded in `covid_rumor_external_transfer.md` and its companion validation JSON.

The following implementation details were previously stated in the main-text Methods and are consolidated here. The checkpoint was `distilbert/distilroberta-base` at immutable revision `fb53ab8802853c8e4fbdbcd0529f21fc6f459b2b`, instantiated as AutoModelForSequenceClassification with two labels. The revision-specific model card identifies RoBERTa as the teacher and the DistilBERT distillation procedure (Sanh et al., 2019; Liu et al., 2019). Training used FP32 on Apple Silicon MPS in the separate Python 3.12 environment. Microbatch size was 2, with gradient accumulation of 8 and an effective batch size of 16. AdamW used weight decay 0.01 except for bias and LayerNorm weights, betas (0.9, 0.999), epsilon 1e-8 and gradient-norm clipping at 1.0, with a constant learning-rate schedule and zero warmup. Candidate learning rates were 2e-5 and 5e-5 and candidate epoch counts were 2 and 3; each learning-rate trajectory ran for three epochs with validation at epochs 2 and 3. No early stopping, mixed precision, oversampling or target threshold tuning was used. Within-context final models trained on the source training pool; transfer models started from the pinned pretrained checkpoint and trained on all eligible source rows, including the former source holdout, with probability >= 0.5 defining Fake on the transform-only target. In the budget follow-up, CoAID's unpadded target token sequences were identical at both caps; only the available FakeHealth source body prefix changed.

The headerless NEWS item table was read using its documented rumor-ID, label, statement-content and sentiment columns. IDs, sentiment, stance, dates, engagement and source metadata were excluded as separate predictors. The quality screen required at least two alphabetic analyzer tokens after URL stripping and at least 90% Latin letters among alphabetic characters, in addition to the missing-text and artifact-only exclusions. The repository's English designation and this script check do not establish complete language identification; mixed-language statements may remain.

S7. Reproducibility records

Saved protocol metadata provide software versions and input/output SHA-256 fingerprints. Classical versions are recorded in `engagement_model_run_metadata.json` and the classical model reports' `signature.versions` fields. Transformer versions and checkpoint identity are recorded in `transformer_feasibility.md`, `transformer_environment_audit.csv`, `requirements-transformer.txt`, and the final experiment metadata. The final experiment's saved settings take precedence over prospective feasibility proposals.

The Methods protocol checklist links each main-text and supplementary block to its report, table, or exact implementation helper. Reproducibility checks cover cohort/UID identity, group separation, training-only fitting, fixed configurations, prediction timing, convergence or finite-loss checks, and artifact integrity. The engagement contrast retains the documented limitation that the original within-engagement run lacked historical output-file hashes; later stages fingerprinted the saved outputs and checked their internal consistency without claiming a nonexistent earlier checksum.

Supplementary Table S7a. Recorded software versions, checkpoint identity and seed policy. Values are the recorded environment metadata cited in Methods; they are reproduced here so the main text can refer to them by environment name.

Environment or item	Component	Recorded value
`.venv` (classical and engagement)	Python	3.14.2
`.venv`	NumPy	2.5.1
`.venv`	pandas	3.0.5
`.venv`	scikit-learn	1.9.0
`.venv`	SciPy	1.18.0
`.venv`	statsmodels	0.15.0
`.venv-transformer`	Python	3.12.7
`.venv-transformer`	PyTorch	2.14.0
`.venv-transformer`	Transformers	5.17.0
Checkpoint	`distilbert/distilroberta-base` revision	`fb53ab8802853c8e4fbdbcd0529f21fc6f459b2b`
Seed policy	Primary seed	42 (partitions, solvers, bootstrap resampling)
Seed policy	Additional seeds	123 and 2026, used only for the targeted token-budget/seed stability check
Seed policy	MPS runs	Seeded; bitwise reproducibility across software or devices is not asserted

S8. Supplementary analyses and their timing

The primary analyses (Methods 2.2–2.8 and 2.10) followed protocols that fixed the populations, estimand, model-family rule, data splits, hyperparameter grids and transfer design before engagement outcomes were modeled or holdout labels were scored. The analyses below were specified later, after the primary results were known. Each has a written protocol under `outputs/extension_v1/protocols/`, written before the quantities it governs were computed, and each reuses the selected configurations, saved splits, saved fits or saved bootstrap draws. None reselects a hyperparameter or replaces a primary estimate. They fall into three groups.

Reporting of saved primary quantities (Tables S11–S12). For the CoAID Specification A hurdle, occurrence probabilities and conditional positive means by class, their ratios and the product of the two component ratios were read from the saved `version_aware_hurdle_components.csv` and the 1,000 saved successful draws in `version_aware_bootstrap_refits.csv`; percentile intervals recomputed from those draws reproduce the saved intervals exactly. The product of component ratios (0.2471) differs from the row-level-product estimand (0.2528), as expected. Table S12 lists the 25 most Fake-leaning and 25 most Real-leaning unigrams of each within-context natural LR model with coefficient, training-pool document frequency (obtained with the fitted vectorizer, without refitting), shared-vocabulary membership, the other model's sign, and whether the source-mask dictionary would remove the term as a full identifier.

Extensions specified after the primary results (Tables S13, S16, S17). `protocols/fakenewsnet_external_protocol.md` fixed the FakeNewsNet title-only cohort, quality and overlap rules (identical to the COVID-Rumor rules), the label-conflict gate, the title-only LR transfer fits, Fake = 1, threshold 0.5 and the metrics before scoring. Only titles exist in the local copy, so the combined-text models were not used and no title-only SVM exists; the comparison is LR-only and title-only. `protocols/fakenewsnet_engagement_protocol.md` mirrored Methods 2.4–2.5: distinct tweet IDs per item including zeros, a Fake-indicator-only design, the primary family-selection rule, the row-level-product estimand, a shared-tweet overlap graph with connected components as resampling units, 1,000 successful component-bootstrap refits with seed 42, and ratio-of-ratios against the saved CoAID A and FakeHealth draws by the same Cartesian-product rule. Descriptive feasibility medians had already been observed when this protocol was written; the protocol records this and leaves no rule discretionary. `protocols/coaid_encoding_sensitivity.md` fixed label-blind mojibake, year, month and credit patterns, a hand-written repair mapping (ftfy is not installed), and a refit of the selected CoAID LR configuration on the saved training pool with repaired text, scored on the saved holdout; the natural-text refit reproduces the primary holdout metrics exactly.

Diagnostics (Tables S18–S23, Figure S1). `protocols/round4_reviewer_checks.md` specified the following before any label- or outcome-dependent computation: CoAID any-post shares by label-blind publisher-resolution status, first-recorded release and fact-check site (S18); score–length correlations for every model and target and a length-only logistic regression on log token count, a diagnostic of a shared cue and not a candidate classifier (S19.1–S19.3); the saved bootstrap draws summarized by how often the dominant component entered each draw, with a label-blind title audit of PolitiFact's component (S20, Figure S1); zero-inclusive medians, percentiles and the probability that a random Fake item outdraws a random Real one (S21); refits of the selected FakeHealth LR on the text before and after the transformer's 512-token cut (S22); and fold-to-fold coefficient correlations under the selected configuration (S23). One further re-scoring check was added after the length results had been seen: the ROC-AUC of the saved LR and SVM scores within length quintiles and pooled over same-quintile Fake–Real pairs (S19.4), without refitting.

Reading the aggregate tables

Tables S1–S2 are embedded in Section S4. Tables S3–S10 below display every scientific row and field from the included aggregate CSVs. Wide files are split into panels with repeated source-row and identity columns; scope and condition are stated for each group. Row IDs are one-based CSV data-row positions, not new observations.

Numbers are displayed to four decimal places where necessary; integers remain exact, and nonzero magnitudes below 0.0001 use scientific notation to avoid presenting them as zero. Percent fields retain their original scale. **— means a blank or non-finite value in the saved source, not zero or an imputed result.** Empty interval fields remain absent; fold/seed sample SDs are not confidence intervals. Exact source strings remain in the CSV companions. SHA-256 columns are retained only in companion data and internal mapping records. No rows are averaged or scopes combined.

Table S3. Engagement analytical populations

Engagement populations are distinct from text cohorts. Posts and replies are captured ID totals.

S3.1 — version_aware_sample_confirmation.csv

Exact source CSV.

Panel S3.1.1.1.1

All saved rows. Fields: `N`, `fake_N`, `real_N`, `posts`.

CSV row	specification	N	fake N	real N	posts
1	A_primary_strict	5337	922	4415	152223
2	B_resource_sensitivity	5205	921	4284	152144
3	C_high_certainty	5082	920	4162	124743
4	FakeHealth_HealthStory	1690	472	1218	384073

Panel S3.1.1.2.1

All saved rows. Fields: `replies`.

CSV row	specification	replies
1	A_primary_strict	120604
2	B_resource_sensitivity	120604
3	C_high_certainty	86165
4	FakeHealth_HealthStory	27601

Table S4. Engagement specification and influence contrasts

Dataset reference populations vary by specification/exclusion. Uncertainty-method notes are repeated below. Cartesian pair counts reuse marginal draws and are not independent bootstrap samples. Tail-exclusion point estimates have no valid saved overlap-bootstrap intervals.

S4.1 — cross_context_engagement_contrast.csv

Exact source CSV.

Panel S4.1.1.1.1

All saved rows. Fields: `fakehealth_ratio`, `coaid_ratio`, `ratio_of_ratios`, `log_ratio_of_ratios`.

CSV row	coaid specification	fakehealth ratio	coaid ratio	ratio of ratios	log ratio of ratios
1	A_primary_strict	0.8403	0.2528	0.3009	-1.2011
2	B_resource_sensitivity	0.8403	0.2401	0.2857	-1.2528
3	C_high_certainty	0.8403	0.2885	0.3433	-1.0692

Panel S4.1.1.2.1

All saved rows. Fields: `CI_lower_ratio`, `CI_upper_ratio`, `CI_lower_log`, `CI_upper_log`.

CSV row	coaid specification	CI lower ratio	CI upper ratio	CI lower log	CI upper log
1	A_primary_strict	0.1543	0.6452	-1.8689	-0.4382
2	B_resource_sensitivity	0.1541	0.6051	-1.8700	-0.5024
3	C_high_certainty	0.1817	0.6202	-1.7053	-0.4778

Panel S4.1.1.3.1

All saved rows. Fields: `bootstrap_median`, `bootstrap_median_log`, `skewness`, `excess_kurtosis`.

CSV row	coaid specification	bootstrap median	bootstrap median log	skewness	excess kurtosis
1	A_primary_strict	0.3163	-1.1512	0.7383	0.0160
2	B_resource_sensitivity	0.2967	-1.2150	0.8198	0.2221
3	C_high_certainty	0.3535	-1.0398	0.5984	0.2879

Panel S4.1.1.4.1

All saved rows. Fields: `skewness_log`, `excess_kurtosis_log`, `ratio_interval_contains_1`, `log_interval_contains_0`.

CSV row	coaid specification	skewness log	excess kurtosis log	ratio interval contains 1	log interval contains 0
1	A_primary_strict	-0.0411	-0.5507	False	False
2	B_resource_sensitivity	0.0754	-0.6289	False	False
3	C_high_certainty	-0.2302	-0.1370	False	False

Panel S4.1.1.5.1

All saved rows. Fields: `coaid_successful_draws`, `fakehealth_successful_draws`, `cartesian_combinations`, `percentile_025_ratio`.

CSV row	coaid specification	coaid successful draws	fakehealth successful draws	cartesian combinations	percentile 025 ratio
1	A_primary_strict	1000	1000	1000000	0.1543
2	B_resource_sensitivity	1000	1000	1000000	0.1541
3	C_high_certainty	1000	1000	1000000	0.1817

Panel S4.1.1.6.1

All saved rows. Fields: `percentile_025_log`, `percentile_25_ratio`, `percentile_25_log`, `percentile_50_ratio`.

CSV row	coaid specification	percentile 025 log	percentile 25 ratio	percentile 25 log	percentile 50 ratio
1	A_primary_strict	-1.8689	0.2444	-1.4091	0.3163
2	B_resource_sensitivity	-1.8700	0.2318	-1.4618	0.2967
3	C_high_certainty	-1.7053	0.2843	-1.2578	0.3535

Panel S4.1.1.7.1

All saved rows. Fields: `percentile_50_log`, `percentile_75_ratio`, `percentile_75_log`, `percentile_975_ratio`.

CSV row	coaid specification	percentile 50 log	percentile 75 ratio	percentile 75 log	percentile 975 ratio
1	A_primary_strict	-1.1512	0.4394	-0.8223	0.6452
2	B_resource_sensitivity	-1.2150	0.4033	-0.9082	0.6051

CSV row	coaid specification	percentile 50 log	percentile 75 ratio	percentile 75 log	percentile 975 ratio
3	C_high_certainty	-1.0398	0.4416	-0.8174	0.6202

Panel S4.1.1.8.1

All saved rows. Fields: `percentile_975_log`.

CSV row	coaid specification	percentile 975 log
1	A_primary_strict	-0.4382
2	B_resource_sensitivity	-0.5024
3	C_high_certainty	-0.4778

analysis role; CSV rows 1: primary observational

analysis role; CSV rows 2, 3: entity/version sensitivity

uncertainty method; CSV rows 1, 2, 3: Full Cartesian product of independent overlap-cluster bootstrap ratios; 1000 x 1000 empirical pairs; percentile 95% interval

interpretation note; CSV rows 1, 2, 3: Observational dataset-context contrast; distinct standardization populations; not a causal COVID effect; dependence graph and dominant-component limitations remain

S4.2 — cross_context_sensitivity_results.csv

Exact source CSV.

Panel S4.2.1.1.1

All saved rows. Fields: `coaid_specification`, `fakehealth_ratio`, `coaid_ratio`, `ratio_of_ratios`.

CSV row	sensitivity	coaid specification	fakehealth ratio	coaid ratio	ratio of ratios
1	all_retained	A_primary_strict	0.8403	0.2528	0.3009
2	exclude_top_1pct	A_primary_strict	0.8403	0.2865	0.3410
3	exclude_top_5_items	A_primary_strict	0.8403	0.2611	0.3108
4	exclude_dominant_component	A_exclude_dominant_component	0.8403	0.4113	0.4895

Panel S4.2.1.2.1

All saved rows. Fields: `log_ratio_of_ratios`, `CI_lower_ratio`, `CI_upper_ratio`, `change_from_primary`.

CSV row	sensitivity	log ratio of ratios	CI lower ratio	CI upper ratio	change from primary
1	all_retained	-1.2011	0.1543	0.6452	0
2	exclude_top_1pct	-1.0758	—	—	0.0402
3	exclude_top_5_items	-1.1687	—	—	0.0099
4	exclude_dominant_component	-0.7145	0.3103	0.7101	0.1886

Panel S4.2.1.3.1

All saved rows. Fields: `relative_change_percent`, `CI_lower_log`, `CI_upper_log`, `bootstrap_median`.

CSV row	sensitivity	relative change percent	CI lower log	CI upper log	bootstrap median
1	all_retained	0	—	—	—
2	exclude_top_1pct	13.3459	—	—	—
3	exclude_top_5_items	3.2919	—	—	—
4	exclude_dominant_component	62.6842	-1.1701	-0.3424	0.4836

Panel S4.2.1.4.1

All saved rows. Fields: `bootstrap_median_log`, `skewness`, `excess_kurtosis`, `skewness_log`.

CSV row	sensitivity	bootstrap median log	skewness	excess kurtosis	skewness log
1	all_retained	—	—	—	—
2	exclude_top_1pct	—	—	—	—
3	exclude_top_5_items	—	—	—	—
4	exclude_dominant_component	-0.7265	0.4097	0.3085	-0.2362

Panel S4.2.1.5.1

All saved rows. Fields: `excess_kurtosis_log`, `ratio_interval_contains_1`, `log_interval_contains_0`, `coaid_successful_draws`.

CSV row	sensitivity	excess kurtosis log	ratio interval contains 1	log interval contains 0	coaid successful draws
1	all_retained	—	—	—	—
2	exclude_top_1pct	—	—	—	—
3	exclude_top_5_items	—	—	—	—
4	exclude_dominant_component	0.1799	False	False	1000

Panel S4.2.1.6.1

All saved rows. Fields: `fakehealth_successful_draws`, `cartesian_combinations`, `percentile_025_ratio`, `percentile_025_log`.

CSV row	sensitivity	fakehealth successful draws	cartesian combinations	percentile 025 ratio	percentile 025 log
1	all_retained	—	—	—	—
2	exclude_top_1pct	—	—	—	—
3	exclude_top_5_items	—	—	—	—
4	exclude_dominant_component	1000	1000000	0.3103	-1.1701

Panel S4.2.1.7.1

All saved rows. Fields: `percentile_25_ratio`, `percentile_25_log`, `percentile_50_ratio`, `percentile_50_log`.

CSV row	sensitivity	percentile 25 ratio	percentile 25 log	percentile 50 ratio	percentile 50 log
1	all_retained	—	—	—	—
2	exclude_top_1pct	—	—	—	—
3	exclude_top_5_items	—	—	—	—
4	exclude_dominant_component	0.4198	-0.8680	0.4836	-0.7265

Panel S4.2.1.8.1

All saved rows. Fields: `percentile_75_ratio`, `percentile_75_log`, `percentile_975_ratio`, `percentile_975_log`.

CSV row	sensitivity	percentile 75 ratio	percentile 75 log	percentile 975 ratio	percentile 975 log
1	all_retained	—	—	—	—
2	exclude_top_1pct	—	—	—	—
3	exclude_top_5_items	—	—	—	—
4	exclude_dominant_component	0.5526	-0.5931	0.7101	-0.3424

uncertainty method; CSV rows 1, 4: Full Cartesian product of independent overlap-cluster bootstrap ratios; 1000 x 1000 empirical pairs; percentile 95% interval

uncertainty method; CSV rows 2, 3: Point only; no valid saved overlap-bootstrap distribution; no mixed HC0/cluster CI

analysis role; CSV rows 1, 2, 3: — (not recorded in this saved field)

analysis role; CSV rows 4: dominant-component influence sensitivity

interpretation note; CSV rows 1, 2, 3: — (not recorded in this saved field)

interpretation note; CSV rows 4: Observational dataset-context contrast; distinct standardization populations; not a causal COVID effect; dependence graph and dominant-component limitations remain

Table S5. Final text cohort and partition counts

Partition labels distinguish all eligible rows, training, reserved holdout and grouped CV membership. Partitions are not additive independent populations.

S5.1 — tfidf_linear_svm_cohort_summary.csv

Exact source CSV.

Companion-only audit fields: `uid_sha256`.

Panel S5.1.1.1.1

dataset: FakeHealth. Fields: `N`, `fake_N`, `real_N`, `leakage_groups`.

CSV row	partition	N	fake N	real N	leakage groups
1	all	1647	459	1188	1634
2	training_pool	1317	367	950	1305
3	holdout	330	92	238	329
4	cv0_train	1053	293	760	1042
5	cv0_validation	264	74	190	263
6	cv1_train	1055	295	760	1044
7	cv1_validation	262	72	190	261
8	cv2_train	1053	293	760	1051
9	cv2_validation	264	74	190	254
10	cv3_train	1053	293	760	1041

Panel S5.1.1.1.2

dataset: FakeHealth. Fields: `N`, `fake_N`, `real_N`, `leakage_groups`.

CSV row	partition	N	fake N	real N	leakage groups
11	cv3_validation	264	74	190	264
12	cv4_train	1054	294	760	1042
13	cv4_validation	263	73	190	263

Panel S5.1.2.1.1

dataset: CoAID. Fields: `N`, `fake_N`, `real_N`, `leakage_groups`.

CSV row	partition	N	fake N	real N	leakage groups
14	all	3577	825	2752	2828
15	training_pool	2875	659	2216	2163

CSV row	partition	N	fake N	real N	leakage groups
16	holdout	702	166	536	665
17	cv0_train	2143	526	1617	2037
18	cv0_validation	732	133	599	126
19	cv1_train	2337	525	1812	1654
20	cv1_validation	538	134	404	509
21	cv2_train	2342	530	1812	1650
22	cv2_validation	533	129	404	513
23	cv3_train	2339	528	1811	1650

Panel S5.1.2.1.2

dataset: CoAID. Fields: `N`, `fake_N`, `real_N`, `leakage_groups`.

CSV row	partition	N	fake N	real N	leakage groups
24	cv3_validation	536	131	405	513
25	cv4_train	2339	527	1812	1661
26	cv4_validation	536	132	404	502

Table S6. Title-only and combined-text comparisons

Matched comparisons use title-eligible target rows. The original combined-text and unpaired columns are separately labeled and must not be substituted for matched contrasts. CA = CoAID; FH = FakeHealth. All scopes and conditions remain separate.

S6.1 — tfidf_logistic_title_vs_combined.csv

Exact source CSV.

Companion-only audit fields: `evaluation_uid_sha256`.

Panel S6.1.1.1.1

evaluation: transfer; condition: natural; scope: all_eligible. Fields: `title_N`, `combined_original_N`, `matched_target_N`, `title`.

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
1	CA_to_FH	FakeHealth	macro_f1	1645	1647	1645	0.4733
2	CA_to_FH	FakeHealth	balanced_accuracy	1645	1647	1645	0.4960
3	CA_to_FH	FakeHealth	fake_f1	1645	1647	1645	0.1524
4	CA_to_FH	FakeHealth	roc_auc	1645	1647	1645	0.4808
5	CA_to_FH	FakeHealth	average_precision	1645	1647	1645	0.2663
21	FH_to_CA	CoAID	macro_f1	3571	3577	3571	0.4148
22	FH_to_CA	CoAID	balanced_accuracy	3571	3577	3571	0.4222
23	FH_to_CA	CoAID	fake_f1	3571	3577	3571	0.2162
24	FH_to_CA	CoAID	roc_auc	3571	3577	3571	0.4367
25	FH_to_CA	CoAID	average_precision	3571	3577	3571	0.2019

Panel S6.1.1.2.1

evaluation: transfer; condition: natural; scope: all_eligible. Fields: `combined_same_rows`, `title_minus_combined`, `combined_original`, `unpaired_title_minus_combined_original`.

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
1	CA_to_FH	FakeHealth	macro_f1	0.4396	0.0336	0.4429	0.0304
2	CA_to_FH	FakeHealth	balanced_accuracy	0.4857	0.0103	0.4878	0.0082
3	CA_to_FH	FakeHealth	fake_f1	0.0710	0.0814	0.0776	0.0748
4	CA_to_FH	FakeHealth	roc_auc	0.6032	-0.1224	0.6047	-0.1239
5	CA_to_FH	FakeHealth	average_precision	0.3223	-0.0560	0.3246	-0.0583
21	FH_to_CA	CoAID	macro_f1	0.3746	0.0402	0.3757	0.0391
22	FH_to_CA	CoAID	balanced_accuracy	0.5210	-0.0988	0.5217	-0.0995

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
23	FH_to_CA	CoAID	fake_f1	0.3672	-0.1510	0.3694	-0.1532
24	FH_to_CA	CoAID	roc_auc	0.5370	-0.1003	0.5367	-0.1001
25	FH_to_CA	CoAID	average_precision	0.2560	-0.0541	0.2569	-0.0550

Panel S6.1.1.3.1

evaluation: transfer; condition: natural; scope: all_eligible. Fields: `exact_same_target_rows`, `original_cohorts_identical`, `source_training_cohorts_identical`.

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
1	CA_to_FH	FakeHealth	macro_f1	True	False	False
2	CA_to_FH	FakeHealth	balanced_accuracy	True	False	False
3	CA_to_FH	FakeHealth	fake_f1	True	False	False
4	CA_to_FH	FakeHealth	roc_auc	True	False	False
5	CA_to_FH	FakeHealth	average_precision	True	False	False
21	FH_to_CA	CoAID	macro_f1	True	False	False
22	FH_to_CA	CoAID	balanced_accuracy	True	False	False
23	FH_to_CA	CoAID	fake_f1	True	False	False
24	FH_to_CA	CoAID	roc_auc	True	False	False
25	FH_to_CA	CoAID	average_precision	True	False	False

Panel S6.1.2.1.1

evaluation: transfer; condition: natural; scope: same_target_holdout. Fields: `title_N`, `combined_original_N`, `matched_target_N`, `title`.

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
6	CA_to_FH	FakeHealth	macro_f1	330	330	330	0.4636
7	CA_to_FH	FakeHealth	balanced_accuracy	330	330	330	0.4880

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
8	CA_to_FH	FakeHealth	fake_f1	330	330	330	0.1385
9	CA_to_FH	FakeHealth	roc_auc	330	330	330	0.4265
10	CA_to_FH	FakeHealth	average_precision	330	330	330	0.2568
26	FH_to_CA	CoAID	macro_f1	701	702	701	0.4286
27	FH_to_CA	CoAID	balanced_accuracy	701	702	701	0.4288
28	FH_to_CA	CoAID	fake_f1	701	702	701	0.2028
29	FH_to_CA	CoAID	roc_auc	701	702	701	0.4742
30	FH_to_CA	CoAID	average_precision	701	702	701	0.2220

Panel S6.1.2.2.1

evaluation: transfer; condition: natural; scope: same_target_holdout. Fields: `combined_same_rows`, `title_minus_combined`, `combined_original`, `unpaired_title_minus_combined_original`.

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
6	CA_to_FH	FakeHealth	macro_f1	0.4507	0.0129	0.4507	0.0129
7	CA_to_FH	FakeHealth	balanced_accuracy	0.5028	-0.0148	0.5028	-0.0148
8	CA_to_FH	FakeHealth	fake_f1	0.0762	0.0623	0.0762	0.0623
9	CA_to_FH	FakeHealth	roc_auc	0.6191	-0.1925	0.6191	-0.1925
10	CA_to_FH	FakeHealth	average_precision	0.3409	-0.0842	0.3409	-0.0842
26	FH_to_CA	CoAID	macro_f1	0.4281	0.0005	0.4290	-0.0004
27	FH_to_CA	CoAID	balanced_accuracy	0.5618	-0.1330	0.5624	-0.1336
28	FH_to_CA	CoAID	fake_f1	0.4012	-0.1984	0.4030	-0.2002
29	FH_to_CA	CoAID	roc_auc	0.5707	-0.0965	0.5705	-0.0964
30	FH_to_CA	CoAID	average_precision	0.2670	-0.0449	0.2677	-0.0457

Panel S6.1.2.3.1

evaluation: transfer; condition: natural; scope: same_target_holdout. Fields: `exact_same_target_rows`, `original_cohorts_identical`, `source_training_cohorts_identical`.

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
6	CA_to_FH	FakeHealth	macro_f1	True	True	False
7	CA_to_FH	FakeHealth	balanced_accuracy	True	True	False
8	CA_to_FH	FakeHealth	fake_f1	True	True	False
9	CA_to_FH	FakeHealth	roc_auc	True	True	False
10	CA_to_FH	FakeHealth	average_precision	True	True	False
26	FH_to_CA	CoAID	macro_f1	True	False	False
27	FH_to_CA	CoAID	balanced_accuracy	True	False	False
28	FH_to_CA	CoAID	fake_f1	True	False	False
29	FH_to_CA	CoAID	roc_auc	True	False	False
30	FH_to_CA	CoAID	average_precision	True	False	False

Panel S6.1.3.1.1

evaluation: transfer; condition: source_masked; scope: all_eligible. Fields: `title_N`, `combined_original_N`, `matched_target_N`, `title`.

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
11	CA_to_FH	FakeHealth	macro_f1	1645	1647	1645	0.4719
12	CA_to_FH	FakeHealth	balanced_accuracy	1645	1647	1645	0.4943
13	CA_to_FH	FakeHealth	fake_f1	1645	1647	1645	0.1515
14	CA_to_FH	FakeHealth	roc_auc	1645	1647	1645	0.4819
15	CA_to_FH	FakeHealth	average_precision	1645	1647	1645	0.2668
31	FH_to_CA	CoAID	macro_f1	3571	3577	3571	0.4165
32	FH_to_CA	CoAID	balanced_accuracy	3571	3577	3571	0.4238
33	FH_to_CA	CoAID	fake_f1	3571	3577	3571	0.2170

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
34	FH_to_CA	CoAID	roc_auc	3571	3577	3571	0.4378
35	FH_to_CA	CoAID	average_precision	3571	3577	3571	0.2023

Panel S6.1.3.2.1

evaluation: transfer; condition: source_masked; scope: all_eligible. Fields: `combined_same_rows`, `title_minus_combined`, `combined_original`, `unpaired_title_minus_combined_original`.

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
11	CA_to_FH	FakeHealth	macro_f1	0.4399	0.0320	0.4432	0.0287
12	CA_to_FH	FakeHealth	balanced_accuracy	0.4861	0.0082	0.4882	0.0061
13	CA_to_FH	FakeHealth	fake_f1	0.0712	0.0803	0.0777	0.0737
14	CA_to_FH	FakeHealth	roc_auc	0.6059	-0.1239	0.6073	-0.1254
15	CA_to_FH	FakeHealth	average_precision	0.3227	-0.0560	0.3252	-0.0584
31	FH_to_CA	CoAID	macro_f1	0.3886	0.0279	0.3897	0.0269
32	FH_to_CA	CoAID	balanced_accuracy	0.5558	-0.1319	0.5563	-0.1324
33	FH_to_CA	CoAID	fake_f1	0.3934	-0.1764	0.3955	-0.1784
34	FH_to_CA	CoAID	roc_auc	0.5613	-0.1235	0.5608	-0.1230
35	FH_to_CA	CoAID	average_precision	0.2609	-0.0586	0.2617	-0.0594

Panel S6.1.3.3.1

evaluation: transfer; condition: source_masked; scope: all_eligible. Fields: `exact_same_target_rows`, `original_cohorts_identical`, `source_training_cohorts_identical`.

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
11	CA_to_FH	FakeHealth	macro_f1	True	False	False
12	CA_to_FH	FakeHealth	balanced_accuracy	True	False	False

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
13	CA_to_FH	FakeHealth	fake_f1	True	False	False
14	CA_to_FH	FakeHealth	roc_auc	True	False	False
15	CA_to_FH	FakeHealth	average_precision	True	False	False
31	FH_to_CA	CoAID	macro_f1	True	False	False
32	FH_to_CA	CoAID	balanced_accuracy	True	False	False
33	FH_to_CA	CoAID	fake_f1	True	False	False
34	FH_to_CA	CoAID	roc_auc	True	False	False
35	FH_to_CA	CoAID	average_precision	True	False	False

Panel S6.1.4.1.1

evaluation: transfer; condition: source_masked; scope: same_target_holdout. Fields: `title_N`, `combined_original_N`, `matched_target_N`, `title`.

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
16	CA_to_FH	FakeHealth	macro_f1	330	330	330	0.4540
17	CA_to_FH	FakeHealth	balanced_accuracy	330	330	330	0.4805
18	CA_to_FH	FakeHealth	fake_f1	330	330	330	0.1231
19	CA_to_FH	FakeHealth	roc_auc	330	330	330	0.4267
20	CA_to_FH	FakeHealth	average_precision	330	330	330	0.2561
36	FH_to_CA	CoAID	macro_f1	701	702	701	0.4323
37	FH_to_CA	CoAID	balanced_accuracy	701	702	701	0.4325
38	FH_to_CA	CoAID	fake_f1	701	702	701	0.2048
39	FH_to_CA	CoAID	roc_auc	701	702	701	0.4789
40	FH_to_CA	CoAID	average_precision	701	702	701	0.2238

Panel S6.1.4.2.1

evaluation: transfer; condition: source_masked; scope: same_target_holdout. Fields: `combined_same_rows`, `title_minus_combined`, `combined_original`, `unpaired_title_minus_combined_original`.

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
16	CA_to_FH	FakeHealth	macro_f1	0.4507	0.0033	0.4507	0.0033
17	CA_to_FH	FakeHealth	balanced_accuracy	0.5028	-0.0224	0.5028	-0.0224
18	CA_to_FH	FakeHealth	fake_f1	0.0762	0.0469	0.0762	0.0469
19	CA_to_FH	FakeHealth	roc_auc	0.6229	-0.1962	0.6229	-0.1962
20	CA_to_FH	FakeHealth	average_precision	0.3418	-0.0856	0.3418	-0.0856
36	FH_to_CA	CoAID	macro_f1	0.4420	-0.0097	0.4428	-0.0105
37	FH_to_CA	CoAID	balanced_accuracy	0.5996	-0.1671	0.5999	-0.1674
38	FH_to_CA	CoAID	fake_f1	0.4309	-0.2261	0.4325	-0.2277
39	FH_to_CA	CoAID	roc_auc	0.5989	-0.1199	0.5985	-0.1195
40	FH_to_CA	CoAID	average_precision	0.2776	-0.0537	0.2782	-0.0544

Panel S6.1.4.3.1

evaluation: transfer; condition: source_masked; scope: same_target_holdout. Fields: `exact_same_target_rows`, `original_cohorts_identical`, `source_training_cohorts_identical`.

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
16	CA_to_FH	FakeHealth	macro_f1	True	True	False
17	CA_to_FH	FakeHealth	balanced_accuracy	True	True	False
18	CA_to_FH	FakeHealth	fake_f1	True	True	False
19	CA_to_FH	FakeHealth	roc_auc	True	True	False
20	CA_to_FH	FakeHealth	average_precision	True	True	False
36	FH_to_CA	CoAID	macro_f1	True	False	False
37	FH_to_CA	CoAID	balanced_accuracy	True	False	False

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
38	FH_to_CA	CoAID	fake_f1	True	False	False
39	FH_to_CA	CoAID	roc_auc	True	False	False
40	FH_to_CA	CoAID	average_precision	True	False	False

Panel S6.1.5.1.1

evaluation: within; condition: natural; scope: same_target_holdout. Fields: `title_N`, `combined_original_N`, `matched_target_N`, `title`.

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
41	within_coaid	CoAID	macro_f1	701	702	701	0.8579
42	within_coaid	CoAID	balanced_accuracy	701	702	701	0.8594
43	within_coaid	CoAID	fake_f1	701	702	701	0.7831
44	within_coaid	CoAID	roc_auc	701	702	701	0.9444
45	within_coaid	CoAID	average_precision	701	702	701	0.8514
51	within_fakehealth	FakeHealth	macro_f1	330	330	330	0.5127
52	within_fakehealth	FakeHealth	balanced_accuracy	330	330	330	0.5155
53	within_fakehealth	FakeHealth	fake_f1	330	330	330	0.3251
54	within_fakehealth	FakeHealth	roc_auc	330	330	330	0.5398
55	within_fakehealth	FakeHealth	average_precision	330	330	330	0.3387

Panel S6.1.5.2.1

evaluation: within; condition: natural; scope: same_target_holdout. Fields: `combined_same_rows`, `title_minus_combined`, `combined_original`, `unpaired_title_minus_combined_original`.

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
41	within_coaid	CoAID	macro_f1	0.9256	-0.0677	0.9260	-0.0680

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
42	within_coaid	CoAID	balanced_accuracy	0.9310	-0.0716	0.9313	-0.0719
43	within_coaid	CoAID	fake_f1	0.8869	-0.1038	0.8876	-0.1044
44	within_coaid	CoAID	roc_auc	0.9826	-0.0382	0.9827	-0.0383
45	within_coaid	CoAID	average_precision	0.9267	-0.0754	0.9280	-0.0766
51	within_fakehealth	FakeHealth	macro_f1	0.6752	-0.1625	0.6752	-0.1625
52	within_fakehealth	FakeHealth	balanced_accuracy	0.6677	-0.1522	0.6677	-0.1522
53	within_fakehealth	FakeHealth	fake_f1	0.5176	-0.1925	0.5176	-0.1925
54	within_fakehealth	FakeHealth	roc_auc	0.7701	-0.2303	0.7701	-0.2303
55	within_fakehealth	FakeHealth	average_precision	0.5707	-0.2320	0.5707	-0.2320

Panel S6.1.5.3.1

evaluation: within; condition: natural; scope: same_target_holdout. Fields: `exact_same_target_rows`, `original_cohorts_identical`, `source_training_cohorts_identical`.

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
41	within_coaid	CoAID	macro_f1	True	False	False
42	within_coaid	CoAID	balanced_accuracy	True	False	False
43	within_coaid	CoAID	fake_f1	True	False	False
44	within_coaid	CoAID	roc_auc	True	False	False
45	within_coaid	CoAID	average_precision	True	False	False
51	within_fakehealth	FakeHealth	macro_f1	True	True	False
52	within_fakehealth	FakeHealth	balanced_accuracy	True	True	False
53	within_fakehealth	FakeHealth	fake_f1	True	True	False
54	within_fakehealth	FakeHealth	roc_auc	True	True	False
55	within_fakehealth	FakeHealth	average_precision	True	True	False

Panel S6.1.6.1.1

evaluation: within; condition: source_masked; scope: same_target_holdout. Fields: `title_N`, `combined_original_N`, `matched_target_N`, `title`.

CSV row	direction	target dataset	metric	title N	combined original N	matched target N	title
46	within_coaid	CoAID	macro_f1	701	702	701	0.8573
47	within_coaid	CoAID	balanced_accuracy	701	702	701	0.8573
48	within_coaid	CoAID	fake_f1	701	702	701	0.7818
49	within_coaid	CoAID	roc_auc	701	702	701	0.9400
50	within_coaid	CoAID	average_precision	701	702	701	0.8458
56	within_fakehealth	FakeHealth	macro_f1	330	330	330	0.5127
57	within_fakehealth	FakeHealth	balanced_accuracy	330	330	330	0.5155
58	within_fakehealth	FakeHealth	fake_f1	330	330	330	0.3251
59	within_fakehealth	FakeHealth	roc_auc	330	330	330	0.5398
60	within_fakehealth	FakeHealth	average_precision	330	330	330	0.3387

Panel S6.1.6.2.1

evaluation: within; condition: source_masked; scope: same_target_holdout. Fields: `combined_same_rows`, `title_minus_combined`, `combined_original`, `unpaired_title_minus_combined_original`.

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
46	within_coaid	CoAID	macro_f1	0.9274	-0.0701	0.9278	-0.0704
47	within_coaid	CoAID	balanced_accuracy	0.9319	-0.0746	0.9322	-0.0749
48	within_coaid	CoAID	fake_f1	0.8896	-0.1077	0.8902	-0.1084
49	within_coaid	CoAID	roc_auc	0.9817	-0.0417	0.9818	-0.0418
50	within_coaid	CoAID	average_precision	0.9257	-0.0799	0.9269	-0.0811
56	within_fakehealth	FakeHealth	macro_f1	0.6726	-0.1599	0.6726	-0.1599
57	within_fakehealth	FakeHealth	balanced_accuracy	0.6644	-0.1489	0.6644	-0.1489

CSV row	direction	target dataset	metric	combined same rows	title minus combined	combined original	unpaired title minus combined original
58	within_fakehealth	FakeHealth	fake_f1	0.5119	-0.1868	0.5119	-0.1868
59	within_fakehealth	FakeHealth	roc_auc	0.7648	-0.2250	0.7648	-0.2250
60	within_fakehealth	FakeHealth	average_precision	0.5643	-0.2256	0.5643	-0.2256

Panel S6.1.6.3.1

evaluation: within; condition: source_masked; scope: same_target_holdout. Fields: `exact_same_target_rows`, `original_cohorts_identical`, `source_training_cohorts_identical`.

CSV row	direction	target dataset	metric	exact same target rows	original cohorts identical	source training cohorts identical
46	within_coaid	CoAID	macro_f1	True	False	False
47	within_coaid	CoAID	balanced_accuracy	True	False	False
48	within_coaid	CoAID	fake_f1	True	False	False
49	within_coaid	CoAID	roc_auc	True	False	False
50	within_coaid	CoAID	average_precision	True	False	False
56	within_fakehealth	FakeHealth	macro_f1	True	True	False
57	within_fakehealth	FakeHealth	balanced_accuracy	True	True	False
58	within_fakehealth	FakeHealth	fake_f1	True	True	False
59	within_fakehealth	FakeHealth	roc_auc	True	True	False
60	within_fakehealth	FakeHealth	average_precision	True	True	False

Table S7. Publisher-held-out evaluation

Natural and masked results retain separate fold and pooled scopes. Outer mean/sample SD and pooled metrics are distinct. The manuscript summarizes ROC-AUC within folds; the saved pooled ranking column is shown for completeness, not substituted for that estimand.

S7.1 — fakehealth_publisher_heldout_results.csv

Exact source CSV.

Companion-only audit fields: `evaluation_uid_sha256`.

Panel S7.1.1.1.1

condition: natural; scope: outer_fold. Fields: `threshold`, `N`, `fake_N`, `real_N`.

CSV row	outer fold	threshold	N	fake N	real N
1	0	0.5000	549	153	396
2	1	0.5000	549	152	397
3	2	0.5000	549	154	395

Panel S7.1.1.2.1

condition: natural; scope: outer_fold. Fields: `accuracy`, `balanced_accuracy`, `macro_f1`, `fake_precision`.

CSV row	outer fold	accuracy	balanced accuracy	macro f1	fake precision
1	0	0.6685	0.6078	0.6030	0.4162
2	1	0.7086	0.6462	0.6432	0.4753
3	2	0.7177	0.6057	0.6125	0.4954

Panel S7.1.1.3.1

condition: natural; scope: outer_fold. Fields: `fake_recall`, `fake_f1`, `real_precision`, `real_recall`.

CSV row	outer fold	fake recall	fake f1	real precision	real recall
1	0	0.4706	0.4417	0.7846	0.7449
2	1	0.5066	0.4904	0.8062	0.7859
3	2	0.3506	0.4106	0.7727	0.8608

Panel S7.1.1.4.1

condition: natural; scope: outer_fold. Fields: `real_f1`, `roc_auc`, `average_precision`, `tn`.

CSV row	outer fold	real f1	roc auc	average precision	tn
1	0	0.7642	0.6772	0.4010	295
2	1	0.7959	0.7134	0.5006	312

CSV row	outer fold	real f1	roc auc	average precision	tn
3	2	0.8144	0.7078	0.4599	340

Panel S7.1.1.5.1

condition: natural; scope: outer_fold. Fields: `fp`, `fn`, `tp`, `actual_fake_prevalence`.

CSV row	outer fold	fp	fn	tp	actual fake prevalence
1	0	101	81	72	0.2787
2	1	85	75	77	0.2769
3	2	55	100	54	0.2805

Panel S7.1.1.6.1

condition: natural; scope: outer_fold. Fields: `predicted_fake_prevalence`, `mean_fake_probability`.

CSV row	outer fold	predicted fake prevalence	mean fake probability
1	0	0.3151	0.4517
2	1	0.2951	0.4426
3	2	0.1985	0.4213

Panel S7.1.2.1.1

condition: natural; scope: pooled. Fields: `threshold`, `N`, `fake_N`, `real_N`.

CSV row	outer fold	threshold	N	fake N	real N
4	-1	0.5000	1647	459	1188

Panel S7.1.2.2.1

condition: natural; scope: pooled. Fields: `accuracy`, `balanced_accuracy`, `macro_f1`, `fake_precision`.

CSV row	outer fold	accuracy	balanced accuracy	macro f1	fake precision
4	-1	0.6982	0.6197	0.6209	0.4572

Panel S7.1.2.3.1

condition: natural; scope: pooled. Fields: `fake_recall`, `fake_f1`, `real_precision`, `real_recall`.

CSV row	outer fold	fake recall	fake f1	real precision	real recall
4	-1	0.4423	0.4496	0.7872	0.7971

Panel S7.1.2.4.1

condition: natural; scope: pooled. Fields: `real_f1`, `roc_auc`, `average_precision`, `tn`.

CSV row	outer fold	real f1	roc auc	average precision	tn
4	-1	0.7921	0.6976	0.4467	947

Panel S7.1.2.5.1

condition: natural; scope: pooled. Fields: `fp`, `fn`, `tp`, `actual_fake_prevalence`.

CSV row	outer fold	fp	fn	tp	actual fake prevalence
4	-1	241	256	203	0.2787

Panel S7.1.2.6.1

condition: natural; scope: pooled. Fields: `predicted_fake_prevalence`, `mean_fake_probability`.

CSV row	outer fold	predicted fake prevalence	mean fake probability
4	-1	0.2696	0.4385

Panel S7.1.3.1.1

condition: source_masked; scope: outer_fold. Fields: `threshold`, `N`, `fake_N`, `real_N`.

CSV row	outer fold	threshold	N	fake N	real N
5	0	0.5000	549	153	396
6	1	0.5000	549	152	397
7	2	0.5000	549	154	395

Panel S7.1.3.2.1

condition: source_masked; scope: outer_fold. Fields: `accuracy`, `balanced_accuracy`, `macro_f1`, `fake_precision`.

CSV row	outer fold	accuracy	balanced accuracy	macro f1	fake precision
5	0	0.6794	0.6194	0.6147	0.4327
6	1	0.7013	0.6412	0.6370	0.4639
7	2	0.7086	0.6014	0.6071	0.4741

Panel S7.1.3.3.1

condition: source_masked; scope: outer_fold. Fields: `fake_recall`, `fake_f1`, `real_precision`, `real_recall`.

CSV row	outer fold	fake recall	fake f1	real precision	real recall
5	0	0.4837	0.4568	0.7910	0.7551
6	1	0.5066	0.4843	0.8042	0.7758
7	2	0.3571	0.4074	0.7714	0.8456

Panel S7.1.3.4.1

condition: source_masked; scope: outer_fold. Fields: `real_f1`, `roc_auc`, `average_precision`, `tn`.

CSV row	outer fold	real f1	roc auc	average precision	tn
5	0	0.7726	0.6781	0.3975	299
6	1	0.7897	0.7092	0.4982	308
7	2	0.8068	0.7072	0.4554	334

Panel S7.1.3.5.1

condition: source_masked; scope: outer_fold. Fields: `fp`, `fn`, `tp`, `actual_fake_prevalence`.

CSV row	outer fold	fp	fn	tp	actual fake prevalence
5	0	97	79	74	0.2787
6	1	89	75	77	0.2769
7	2	61	99	55	0.2805

Panel S7.1.3.6.1

condition: source_masked; scope: outer_fold. Fields: `predicted_fake_prevalence`, `mean_fake_probability`.

CSV row	outer fold	predicted fake prevalence	mean fake probability
5	0	0.3115	0.4495
6	1	0.3024	0.4458
7	2	0.2113	0.4233

Panel S7.1.4.1.1

condition: source_masked; scope: pooled. Fields: `threshold`, `N`, `fake_N`, `real_N`.

CSV row	outer fold	threshold	N	fake N	real N
8	-1	0.5000	1647	459	1188

Panel S7.1.4.2.1

condition: source_masked; scope: pooled. Fields: `accuracy`, `balanced_accuracy`, `macro_f1`, `fake_precision`.

CSV row	outer fold	accuracy	balanced accuracy	macro f1	fake precision
8	-1	0.6964	0.6204	0.6209	0.4547

Panel S7.1.4.3.1

condition: source_masked; scope: pooled. Fields: `fake_recall`, `fake_f1`, `real_precision`, `real_recall`.

CSV row	outer fold	fake recall	fake f1	real precision	real recall
8	-1	0.4488	0.4518	0.7881	0.7921

Panel S7.1.4.4.1

condition: source_masked; scope: pooled. Fields: `real_f1`, `roc_auc`, `average_precision`, `tn`.

CSV row	outer fold	real f1	roc auc	average precision	tn
8	-1	0.7901	0.6966	0.4438	941

Panel S7.1.4.5.1

condition: source_masked; scope: pooled. Fields: `fp`, `fn`, `tp`, `actual_fake_prevalence`.

CSV row	outer fold	fp	fn	tp	actual fake prevalence
8	-1	247	253	206	0.2787

Panel S7.1.4.6.1

condition: source_masked; scope: pooled. Fields: `predicted_fake_prevalence`, `mean_fake_probability`.

CSV row	outer fold	predicted fake prevalence	mean fake probability
8	-1	0.2750	0.4395

S7.2 — fakehealth_publisher_heldout_summary.csv

Exact source CSV.

Panel S7.2.1.1.1

condition: natural. Fields: `outer_mean`, `outer_sample_sd`, `outer_min`, `outer_max`.

CSV row	metric	outer mean	outer sample sd	outer min	outer max
1	macro_f1	0.6196	0.0210	0.6030	0.6432
2	balanced_accuracy	0.6199	0.0228	0.6057	0.6462
3	fake_f1	0.4476	0.0402	0.4106	0.4904
4	roc_auc	0.6995	0.0195	0.6772	0.7134
5	average_precision	0.4538	0.0500	0.4010	0.5006

Panel S7.2.1.2.1

condition: natural. Fields: `pooled`.

CSV row	metric	pooled
1	macro_f1	0.6209
2	balanced_accuracy	0.6197

CSV row	metric	pooled
3	fake_f1	0.4496
4	roc_auc	0.6976
5	average_precision	0.4467

Panel S7.2.2.1.1

condition: source_masked. Fields: `outer_mean`, `outer_sample_sd`, `outer_min`, `outer_max`.

CSV row	metric	outer mean	outer sample sd	outer min	outer max
6	macro_f1	0.6196	0.0156	0.6071	0.6370
7	balanced_accuracy	0.6206	0.0200	0.6014	0.6412
8	fake_f1	0.4495	0.0390	0.4074	0.4843
9	roc_auc	0.6982	0.0174	0.6781	0.7092
10	average_precision	0.4504	0.0505	0.3975	0.4982

Panel S7.2.2.2.1

condition: source_masked. Fields: `pooled`.

CSV row	metric	pooled
6	macro_f1	0.6209
7	balanced_accuracy	0.6204
8	fake_f1	0.4518
9	roc_auc	0.6966
10	average_precision	0.4438

Table S8. Transformer token-budget and seed sensitivity

Direction is FakeHealth to CoAID. Paired differences are 256 minus 512. Budget and paired-delta summaries retain their saved scopes. Sample SDs/ranges describe the tested seeds, not confidence intervals or a new search.

S8.1 — distilroberta_budget_seed_paired_deltas.csv

Exact source CSV.

Panel S8.1.1.1.1

scope: all_eligible. Fields: `value_512`, `value_256`, `delta_256_minus_512`.

CSV row	seed	metric	value 512	value 256	delta 256 minus 512
1	42	macro_f1	0.3226	0.5120	0.1894
2	42	balanced_accuracy	0.3797	0.5366	0.1569
3	42	fake_f1	0.2445	0.3351	0.0905
4	42	roc_auc	0.3495	0.5409	0.1914
5	42	average_precision	0.1688	0.2302	0.0614
6	42	predicted_fake_prevalence	0.6542	0.4068	-0.2474
13	123	macro_f1	0.2543	0.3273	0.0731
14	123	balanced_accuracy	0.4480	0.4670	0.0190
15	123	fake_f1	0.3314	0.3324	0.0010
16	123	roc_auc	0.3958	0.3597	-0.0360

Panel S8.1.1.1.2

scope: all_eligible. Fields: `value_512`, `value_256`, `delta_256_minus_512`.

CSV row	seed	metric	value 512	value 256	delta 256 minus 512
17	123	average_precision	0.1952	0.1696	-0.0256
18	123	predicted_fake_prevalence	0.8728	0.7769	-0.0959
25	2026	macro_f1	0.4389	0.4656	0.0267
26	2026	balanced_accuracy	0.4905	0.4840	-0.0065
27	2026	fake_f1	0.3167	0.2819	-0.0347
28	2026	roc_auc	0.4949	0.4659	-0.0290

CSV row	seed	metric	value 512	value 256	delta 256 minus 512
29	2026	average_precision	0.2358	0.1989	-0.0369
30	2026	predicted_fake_prevalence	0.5516	0.4258	-0.1258

Panel S8.1.2.1.1

scope: same_target_holdout. Fields: `value_512`, `value_256`, `delta_256_minus_512`.

CSV row	seed	metric	value 512	value 256	delta 256 minus 512
7	42	macro_f1	0.3956	0.5773	0.1816
8	42	balanced_accuracy	0.4622	0.5957	0.1334
9	42	fake_f1	0.3113	0.3990	0.0878
10	42	roc_auc	0.4415	0.6460	0.2045
11	42	average_precision	0.2009	0.2935	0.0926
12	42	predicted_fake_prevalence	0.6239	0.3419	-0.2821
19	123	macro_f1	0.2867	0.3846	0.0979
20	123	balanced_accuracy	0.4863	0.5284	0.0421
21	123	fake_f1	0.3629	0.3811	0.0182
22	123	roc_auc	0.4672	0.4375	-0.0297

Panel S8.1.2.1.2

scope: same_target_holdout. Fields: `value_512`, `value_256`, `delta_256_minus_512`.

CSV row	seed	metric	value 512	value 256	delta 256 minus 512
23	123	average_precision	0.2240	0.1948	-0.0293
24	123	predicted_fake_prevalence	0.8704	0.7578	-0.1125
31	2026	macro_f1	0.5101	0.5339	0.0238
32	2026	balanced_accuracy	0.5590	0.5476	-0.0114
33	2026	fake_f1	0.3782	0.3406	-0.0375

CSV row	seed	metric	value 512	value 256	delta 256 minus 512
34	2026	roc_auc	0.6022	0.5739	-0.0283
35	2026	average_precision	0.3094	0.2483	-0.0611
36	2026	predicted_fake_prevalence	0.4943	0.3490	-0.1453

S8.2 — distilroberta_budget_seed_summary.csv

Exact source CSV.

Panel S8.2.1.1.1

kind: budget; scope: all_eligible. Fields: `N_seeds`, `mean`, `sample_sd`, `minimum`.

CSV row	max length	metric	N seeds	mean	sample sd	minimum
1	256	macro_f1	3	0.4350	0.0961	0.3273
2	256	roc_auc	3	0.4555	0.0910	0.3597
3	256	average_precision	3	0.1996	0.0303	0.1696
4	256	fake_f1	3	0.3165	0.0299	0.2819
5	512	macro_f1	3	0.3386	0.0933	0.2543
6	512	roc_auc	3	0.4134	0.0743	0.3495
7	512	average_precision	3	0.1999	0.0338	0.1688
8	512	fake_f1	3	0.2975	0.0465	0.2445

Panel S8.2.1.2.1

kind: budget; scope: all_eligible. Fields: `maximum`, `positive_seeds`.

CSV row	max length	metric	maximum	positive seeds
1	256	macro_f1	0.5120	—
2	256	roc_auc	0.5409	—
3	256	average_precision	0.2302	—
4	256	fake_f1	0.3351	—

CSV row	max length	metric	maximum	positive seeds
5	512	macro_f1	0.4389	—
6	512	roc_auc	0.4949	—
7	512	average_precision	0.2358	—
8	512	fake_f1	0.3314	—

Panel S8.2.2.1.1

kind: budget; scope: same_target_holdout. Fields: `N_seeds`, `mean`, `sample_sd`, `minimum`.

CSV row	max length	metric	N seeds	mean	sample sd	minimum
9	256	macro_f1	3	0.4986	0.1011	0.3846
10	256	roc_auc	3	0.5525	0.1059	0.4375
11	256	average_precision	3	0.2455	0.0494	0.1948
12	256	fake_f1	3	0.3736	0.0299	0.3406
13	512	macro_f1	3	0.3975	0.1117	0.2867
14	512	roc_auc	3	0.5036	0.0863	0.4415
15	512	average_precision	3	0.2448	0.0572	0.2009
16	512	fake_f1	3	0.3508	0.0351	0.3113

Panel S8.2.2.2.1

kind: budget; scope: same_target_holdout. Fields: `maximum`, `positive_seeds`.

CSV row	max length	metric	maximum	positive seeds
9	256	macro_f1	0.5773	—
10	256	roc_auc	0.6460	—
11	256	average_precision	0.2935	—
12	256	fake_f1	0.3990	—
13	512	macro_f1	0.5101	—

CSV row	max length	metric	maximum	positive seeds
14	512	roc_auc	0.6022	—
15	512	average_precision	0.3094	—
16	512	fake_f1	0.3782	—

Panel S8.2.3.1.1

kind: paired_delta; scope: all_eligible. Fields: `N_seeds`, `mean`, `sample_sd`, `minimum`.

CSV row	max length	metric	N seeds	mean	sample sd	minimum
17	—	average_precision	3	-0.0003	0.0538	-0.0369
18	—	balanced_accuracy	3	0.0565	0.0879	-0.0065
19	—	fake_f1	3	0.0189	0.0645	-0.0347
20	—	macro_f1	3	0.0964	0.0838	0.0267
21	—	predicted_fake_prevalence	3	-0.1564	0.0803	-0.2474
22	—	roc_auc	3	0.0421	0.1293	-0.0360

Panel S8.2.3.2.1

kind: paired_delta; scope: all_eligible. Fields: `maximum`, `positive_seeds`.

CSV row	max length	metric	maximum	positive seeds
17	—	average_precision	0.0614	1
18	—	balanced_accuracy	0.1569	2
19	—	fake_f1	0.0905	2
20	—	macro_f1	0.1894	3
21	—	predicted_fake_prevalence	-0.0959	0
22	—	roc_auc	0.1914	1

Panel S8.2.4.1.1

kind: paired_delta; scope: same_target_holdout. Fields: `N_seeds`, `mean`, `sample_sd`, `minimum`.

CSV row	max length	metric	N seeds	mean	sample sd	minimum
23	—	average_precision	3	0.0008	0.0812	-0.0611
24	—	balanced_accuracy	3	0.0547	0.0732	-0.0114
25	—	fake_f1	3	0.0228	0.0628	-0.0375
26	—	macro_f1	3	0.1011	0.0790	0.0238
27	—	predicted_fake_prevalence	3	-0.1800	0.0899	-0.2821
28	—	roc_auc	3	0.0489	0.1348	-0.0297

Panel S8.2.4.2.1

kind: paired_delta; scope: same_target_holdout. Fields: `maximum`, `positive_seeds`.

CSV row	max length	metric	maximum	positive seeds
23	—	average_precision	0.0926	1
24	—	balanced_accuracy	0.1334	2
25	—	fake_f1	0.0878	2
26	—	macro_f1	0.1816	3
27	—	predicted_fake_prevalence	-0.1125	0
28	—	roc_auc	0.2045	1

Table S9. Independent COVID-Rumor target and representation diagnostics

External models are the full-source LR/SVM fits, applied unchanged. Source and clean-primary versus sensitivity scopes remain explicit. Target labels are inherited True/False annotations. Vocabulary coverage counts unigram occurrences without double-counting bigrams.

S9.1 — covid_rumor_external_cohort_summary.csv

Exact source CSV.

Panel S9.1.1.1.1

All saved rows. Fields: `value`.

CSV row	measure	value
1	original_NEWS_N	4129
2	True_N	659
3	False_N	3041
4	Unverified_N	429
5	unusable_or_missing_text_N	0
6	duplicate_ID_excess_N	0
7	otherwise_eligible_binary_N	3700
8	exact_source_overlap_excluded_N	33
9	final_clean_binary_N	3667
10	clean_high_similarity_flagged_N	22

Panel S9.1.1.1.2

All saved rows. Fields: `value`.

CSV row	measure	value
11	all_NEWS_exact_duplicate_text_excess_N	0
12	all_NEWS_exact_duplicate_text_involved_N	0
13	all_eligible_exact_duplicate_text_excess_N	0
14	all_eligible_exact_duplicate_text_involved_N	0
15	clean_primary_exact_duplicate_text_excess_N	0
16	clean_primary_exact_duplicate_text_involved_N	0
17	clean_Fake_N	3015
18	clean_Real_N	652
19	clean_Fake_prevalence	0.8222

S9.2 — covid_rumor_external_transfer_results.csv

Exact source CSV.

Companion-only audit fields: `evaluation_uid_sha256`.

Panel S9.2.1.1.1

scope: clean_primary. Fields: `N`, `fake_N`, `real_N`, `accuracy`.

CSV row	classifier	source dataset	N	fake N	real N	accuracy
1	LR	CoAID	3667	3015	652	0.8064
4	LR	FakeHealth	3667	3015	652	0.7565
7	SVM	CoAID	3667	3015	652	0.8064
10	SVM	FakeHealth	3667	3015	652	0.7737

Panel S9.2.1.2.1

scope: clean_primary. Fields: `balanced_accuracy`, `macro_f1`, `fake_precision`, `fake_recall`.

CSV row	classifier	source dataset	balanced accuracy	macro f1	fake precision	fake recall
1	LR	CoAID	0.5529	0.5580	0.8389	0.9463
4	LR	FakeHealth	0.5021	0.4969	0.8229	0.8968
7	SVM	CoAID	0.5349	0.5314	0.8330	0.9562
10	SVM	FakeHealth	0.5059	0.4973	0.8241	0.9214

Panel S9.2.1.3.1

scope: clean_primary. Fields: `fake_f1`, `real_precision`, `real_recall`, `real_f1`.

CSV row	classifier	source dataset	fake f1	real precision	real recall	real f1
1	LR	CoAID	0.8893	0.3910	0.1595	0.2266
4	LR	FakeHealth	0.8583	0.1837	0.1074	0.1355
7	SVM	CoAID	0.8904	0.3592	0.1135	0.1725
10	SVM	FakeHealth	0.8700	0.1993	0.0905	0.1245

Panel S9.2.1.4.1

scope: clean_primary. Fields: `roc_auc`, `average_precision`, `tn`, `fp`.

CSV row	classifier	source dataset	roc auc	average precision	tn	fp
1	LR	CoAID	0.6995	0.9114	104	548
4	LR	FakeHealth	0.5219	0.8331	70	582
7	SVM	CoAID	0.6816	0.9045	74	578
10	SVM	FakeHealth	0.5248	0.8345	59	593

Panel S9.2.1.5.1

scope: clean_primary. Fields: `fn`, `tp`, `threshold`, `actual_fake_prevalence`.

CSV row	classifier	source dataset	fn	tp	threshold	actual fake prevalence
1	LR	CoAID	162	2853	0.5000	0.8222
4	LR	FakeHealth	311	2704	0.5000	0.8222
7	SVM	CoAID	132	2883	0	0.8222
10	SVM	FakeHealth	237	2778	0	0.8222

Panel S9.2.1.6.1

scope: clean_primary. Fields: `predicted_fake_prevalence`.

CSV row	classifier	source dataset	predicted fake prevalence
1	LR	CoAID	0.9275
4	LR	FakeHealth	0.8961
7	SVM	CoAID	0.9438
10	SVM	FakeHealth	0.9193

Panel S9.2.2.1.1

scope: all_eligible. Fields: `N`, `fake_N`, `real_N`, `accuracy`.

CSV row	classifier	source dataset	N	fake N	real N	accuracy
2	LR	CoAID	3700	3041	659	0.8059
5	LR	FakeHealth	3700	3041	659	0.7551
8	SVM	CoAID	3700	3041	659	0.8059
11	SVM	FakeHealth	3700	3041	659	0.7722

Panel S9.2.2.2.1

scope: all_eligible. Fields: `balanced_accuracy`, `macro_f1`, `fake_precision`, `fake_recall`.

CSV row	classifier	source dataset	balanced accuracy	macro f1	fake precision	fake recall
2	LR	CoAID	0.5527	0.5577	0.8385	0.9461
5	LR	FakeHealth	0.5016	0.4964	0.8224	0.8954
8	SVM	CoAID	0.5349	0.5314	0.8327	0.9559
11	SVM	FakeHealth	0.5054	0.4968	0.8236	0.9198

Panel S9.2.2.3.1

scope: all_eligible. Fields: `fake_f1`, `real_precision`, `real_recall`, `real_f1`.

CSV row	classifier	source dataset	fake f1	real precision	real recall	real f1
2	LR	CoAID	0.8891	0.3903	0.1593	0.2263
5	LR	FakeHealth	0.8574	0.1825	0.1077	0.1355
8	SVM	CoAID	0.8901	0.3589	0.1138	0.1728
11	SVM	FakeHealth	0.8690	0.1974	0.0910	0.1246

Panel S9.2.2.4.1

scope: all_eligible. Fields: `roc_auc`, `average_precision`, `tn`, `fp`.

CSV row	classifier	source dataset	roc auc	average precision	tn	fp
2	LR	CoAID	0.7001	0.9116	105	554
5	LR	FakeHealth	0.5205	0.8320	71	588
8	SVM	CoAID	0.6823	0.9047	75	584
11	SVM	FakeHealth	0.5234	0.8334	60	599

Panel S9.2.2.5.1

scope: all_eligible. Fields: `fn`, `tp`, `threshold`, `actual_fake_prevalence`.

CSV row	classifier	source dataset	fn	tp	threshold	actual fake prevalence
2	LR	CoAID	164	2877	0.5000	0.8219
5	LR	FakeHealth	318	2723	0.5000	0.8219
8	SVM	CoAID	134	2907	0	0.8219
11	SVM	FakeHealth	244	2797	0	0.8219

Panel S9.2.2.6.1

scope: all_eligible. Fields: `predicted_fake_prevalence`.

CSV row	classifier	source dataset	predicted fake prevalence
2	LR	CoAID	0.9273
5	LR	FakeHealth	0.8949
8	SVM	CoAID	0.9435
11	SVM	FakeHealth	0.9178

Panel S9.2.3.1.1

scope: clean_without_high_similarity. Fields: `N`, `fake_N`, `real_N`, `accuracy`.

CSV row	classifier	source dataset	N	fake N	real N	accuracy
3	LR	CoAID	3645	2994	651	0.8055
6	LR	FakeHealth	3645	2994	651	0.7553
9	SVM	CoAID	3645	2994	651	0.8055
12	SVM	FakeHealth	3645	2994	651	0.7726

Panel S9.2.3.2.1

scope: clean_without_high_similarity. Fields: `balanced_accuracy`, `macro_f1`, `fake_precision`, `fake_recall`.

CSV row	classifier	source dataset	balanced accuracy	macro f1	fake precision	fake recall
3	LR	CoAID	0.5528	0.5578	0.8381	0.9459
6	LR	FakeHealth	0.5018	0.4966	0.8220	0.8961
9	SVM	CoAID	0.5348	0.5312	0.8322	0.9559
12	SVM	FakeHealth	0.5057	0.4970	0.8232	0.9208

Panel S9.2.3.3.1

scope: clean_without_high_similarity. Fields: `fake_f1`, `real_precision`, `real_recall`, `real_f1`.

CSV row	classifier	source dataset	fake f1	real precision	real recall	real f1
3	LR	CoAID	0.8887	0.3910	0.1598	0.2268
6	LR	FakeHealth	0.8575	0.1837	0.1075	0.1357
9	SVM	CoAID	0.8898	0.3592	0.1137	0.1727
12	SVM	FakeHealth	0.8693	0.1993	0.0906	0.1246

Panel S9.2.3.4.1

scope: clean_without_high_similarity. Fields: `roc_auc`, `average_precision`, `tn`, `fp`.

CSV row	classifier	source dataset	roc auc	average precision	tn	fp
3	LR	CoAID	0.6985	0.9106	104	547
6	LR	FakeHealth	0.5217	0.8324	70	581
9	SVM	CoAID	0.6808	0.9036	74	577
12	SVM	FakeHealth	0.5246	0.8338	59	592

Panel S9.2.3.5.1

scope: clean_without_high_similarity. Fields: `fn`, `tp`, `threshold`, `actual_fake_prevalence`.

CSV row	classifier	source dataset	fn	tp	threshold	actual fake prevalence
3	LR	CoAID	162	2832	0.5000	0.8214
6	LR	FakeHealth	311	2683	0.5000	0.8214
9	SVM	CoAID	132	2862	0	0.8214
12	SVM	FakeHealth	237	2757	0	0.8214

Panel S9.2.3.6.1

scope: clean_without_high_similarity. Fields: `predicted_fake_prevalence`.

CSV row	classifier	source dataset	predicted fake prevalence
3	LR	CoAID	0.9270
6	LR	FakeHealth	0.8955
9	SVM	CoAID	0.9435
12	SVM	FakeHealth	0.9188

S9.3 — covid_rumor_external_vocabulary_diagnostics.csv

Exact source CSV.

Panel S9.3.1.1.1

scope: clean_primary. Fields: `vocabulary_size`, `source_N`, `target_N`, `target_documents_with_known_token_percent`.

CSV row	classifier	source dataset	vocabulary size	source N	target N	target documents with known token percent
1	LR	FakeHealth	35168	1647	3667	100
3	LR	CoAID	7299	3577	3667	100
5	SVM	FakeHealth	35168	1647	3667	100
7	SVM	CoAID	7299	3577	3667	100

Panel S9.3.1.2.1

scope: clean_primary. Fields: `target_token_occurrence_coverage_percent`, `target_median_known_token_fraction`, `target_empty_token_document_N`, `zero_feature_rows`.

CSV row	classifier	source dataset	target token occurrence coverage percent	target median known token fraction	target empty token document N	zero feature rows
1	LR	FakeHealth	81.8743	0.8182	0	0
3	LR	CoAID	89.4714	0.9032	0	0
5	SVM	FakeHealth	81.8743	0.8182	0	0
7	SVM	CoAID	89.4714	0.9032	0	0

Panel S9.3.1.3.1

scope: clean_primary. Fields: `zero_feature_percent`, `target_nonzero_features`, `target_total_feature_cells`, `target_feature_sparsity`.

CSV row	classifier	source dataset	zero feature percent	target nonzero features	target total feature cells	target feature sparsity
1	LR	FakeHealth	0	66020	128961056	0.9995
3	LR	CoAID	0	52818	26765433	0.9980
5	SVM	FakeHealth	0	66020	128961056	0.9995
7	SVM	CoAID	0	52818	26765433	0.9980

Panel S9.3.1.4.1

scope: clean_primary. Fields: `median_target_nonzero_features`, `total_target_token_occurrences`, `known_target_token_occurrences`, `source_characters_p05`.

CSV row	classifier	source dataset	median target nonzero features	total target token occurrences	known target token occurrences	source characters p05
1	LR	FakeHealth	15	64814	53066	83.1000
3	LR	CoAID	12	64814	57990	57
5	SVM	FakeHealth	15	64814	53066	83.1000
7	SVM	CoAID	12	64814	57990	57

Panel S9.3.1.5.1

scope: clean_primary. Fields: `source_characters_median`, `source_characters_p95`, `source_whitespace_tokens_p05`, `source_whitespace_tokens_median`.

CSV row	classifier	source dataset	source characters median	source characters p95	source whitespace tokens p05	source whitespace tokens median
1	LR	FakeHealth	3665	7850.5000	13.3000	593
3	LR	CoAID	537	573	9	84
5	SVM	FakeHealth	3665	7850.5000	13.3000	593
7	SVM	CoAID	537	573	9	84

Panel S9.3.1.6.1

scope: clean_primary. Fields: `source_whitespace_tokens_p95`, `source_analyzer_unigrams_p05`, `source_analyzer_unigrams_median`, `source_analyzer_unigrams_p95`.

CSV row	classifier	source dataset	source whitespace tokens p95	source analyzer unigrams p05	source analyzer unigrams median	source analyzer unigrams p95
1	LR	FakeHealth	1286.1000	13.3000	583	1250.7000
3	LR	CoAID	97	9	85	96
5	SVM	FakeHealth	1286.1000	13.3000	583	1250.7000
7	SVM	CoAID	97	9	85	96

Panel S9.3.1.7.1

scope: clean_primary. Fields: `target_characters_p05`, `target_characters_median`, `target_characters_p95`, `target_whitespace_tokens_p05`.

CSV row	classifier	source dataset	target characters p05	target characters median	target characters p95	target whitespace tokens p05
1	LR	FakeHealth	40	89	247	6
3	LR	CoAID	40	89	247	6
5	SVM	FakeHealth	40	89	247	6
7	SVM	CoAID	40	89	247	6

Panel S9.3.1.8.1

scope: clean_primary. Fields: `target_whitespace_tokens_median`, `target_whitespace_tokens_p95`, `target_analyzer_unigrams_p05`, `target_analyzer_unigrams_median`.

CSV row	classifier	source dataset	target whitespace tokens median	target whitespace tokens p95	target analyzer unigrams p05	target analyzer unigrams median
1	LR	FakeHealth	14	40	6	14
3	LR	CoAID	14	40	6	14
5	SVM	FakeHealth	14	40	6	14
7	SVM	CoAID	14	40	6	14

Panel S9.3.1.9.1

scope: clean_primary. Fields: `target_analyzer_unigrams_p95`.

CSV row	classifier	source dataset	target analyzer unigrams p95
1	LR	FakeHealth	39
3	LR	CoAID	39
5	SVM	FakeHealth	39
7	SVM	CoAID	39

token definition; CSV rows 1, 3, 5, 7: Occurrences of preprocessed default word-tokenizer unigrams; no bigram double counting

Panel S9.3.2.1.1

scope: all_eligible. Fields: `vocabulary_size`, `source_N`, `target_N`, `target_documents_with_known_token_percent`.

CSV row	classifier	source dataset	vocabulary size	source N	target N	target documents with known token percent
2	LR	FakeHealth	35168	1647	3700	100
4	LR	CoAID	7299	3577	3700	100
6	SVM	FakeHealth	35168	1647	3700	100
8	SVM	CoAID	7299	3577	3700	100

Panel S9.3.2.2.1

scope: all_eligible. Fields: `target_token_occurrence_coverage_percent`, `target_median_known_token_fraction`, `target_empty_token_document_N`, `zero_feature_rows`.

CSV row	classifier	source dataset	target token occurrence coverage percent	target median known token fraction	target empty token document N	zero feature rows
2	LR	FakeHealth	81.8935	0.8182	0	0
4	LR	CoAID	89.5176	0.9048	0	0
6	SVM	FakeHealth	81.8935	0.8182	0	0
8	SVM	CoAID	89.5176	0.9048	0	0

Panel S9.3.2.3.1

scope: all_eligible. Fields: `zero_feature_percent`, `target_nonzero_features`, `target_total_feature_cells`, `target_feature_sparsity`.

CSV row	classifier	source dataset	zero feature percent	target nonzero features	target total feature cells	target feature sparsity
2	LR	FakeHealth	0	66562	130121600	0.9995
4	LR	CoAID	0	53266	27006300	0.9980
6	SVM	FakeHealth	0	66562	130121600	0.9995
8	SVM	CoAID	0	53266	27006300	0.9980

Panel S9.3.2.4.1

scope: all_eligible. Fields: `median_target_nonzero_features`, `total_target_token_occurrences`, `known_target_token_occurrences`, `source_characters_p05`.

CSV row	classifier	source dataset	median target nonzero features	total target token occurrences	known target token occurrences	source characters p05
2	LR	FakeHealth	15	65319	53492	83.1000
4	LR	CoAID	12	65319	58472	57
6	SVM	FakeHealth	15	65319	53492	83.1000
8	SVM	CoAID	12	65319	58472	57

Panel S9.3.2.5.1

scope: all_eligible. Fields: `source_characters_median`, `source_characters_p95`, `source_whitespace_tokens_p05`, `source_whitespace_tokens_median`.

CSV row	classifier	source dataset	source characters median	source characters p95	source whitespace tokens p05	source whitespace tokens median
2	LR	FakeHealth	3665	7850.5000	13.3000	593
4	LR	CoAID	537	573	9	84
6	SVM	FakeHealth	3665	7850.5000	13.3000	593
8	SVM	CoAID	537	573	9	84

Panel S9.3.2.6.1

scope: all_eligible. Fields: `source_whitespace_tokens_p95`, `source_analyzer_unigrams_p05`, `source_analyzer_unigrams_median`, `source_analyzer_unigrams_p95`.

CSV row	classifier	source dataset	source whitespace tokens p95	source analyzer unigrams p05	source analyzer unigrams median	source analyzer unigrams p95
2	LR	FakeHealth	1286.1000	13.3000	583	1250.7000
4	LR	CoAID	97	9	85	96
6	SVM	FakeHealth	1286.1000	13.3000	583	1250.7000
8	SVM	CoAID	97	9	85	96

Panel S9.3.2.7.1

scope: all_eligible. Fields: `target_characters_p05`, `target_characters_median`, `target_characters_p95`, `target_whitespace_tokens_p05`.

CSV row	classifier	source dataset	target characters p05	target characters median	target characters p95	target whitespace tokens p05
2	LR	FakeHealth	40	89	247	6
4	LR	CoAID	40	89	247	6
6	SVM	FakeHealth	40	89	247	6
8	SVM	CoAID	40	89	247	6

Panel S9.3.2.8.1

scope: all_eligible. Fields: `target_whitespace_tokens_median`, `target_whitespace_tokens_p95`, `target_analyzer_unigrams_p05`, `target_analyzer_unigrams_median`.

CSV row	classifier	source dataset	target whitespace tokens median	target whitespace tokens p95	target analyzer unigrams p05	target analyzer unigrams median
2	LR	FakeHealth	14	40	6	14
4	LR	CoAID	14	40	6	14
6	SVM	FakeHealth	14	40	6	14
8	SVM	CoAID	14	40	6	14

Panel S9.3.2.9.1

scope: all_eligible. Fields: `target_analyzer_unigrams_p95`.

CSV row	classifier	source dataset	target analyzer unigrams p95
2	LR	FakeHealth	39
4	LR	CoAID	39
6	SVM	FakeHealth	39
8	SVM	CoAID	39

token definition; CSV rows 2, 4, 6, 8: Occurrences of preprocessed default word-tokenizer unigrams; no bigram double counting

Table S10. Lexical alignment importance subsets

Signed coefficient comparisons use shared unigrams from separately fitted within-context LR models. Raw magnitudes are not comparable causal effect sizes. Importance cutoffs are descriptive.

S10.1 — text_coefficient_alignment_summary.csv

Exact source CSV.

Panel S10.1.1.1.1

All saved rows. Fields: `N`, `pearson`, `spearman`, `absolute_magnitude_spearman`.

CSV row	cutoff	N	pearson	spearman	absolute magnitude spearman
1	all	4174	-0.0148	-0.0034	0.1292
2	above_median_either	3052	-0.0191	-0.0098	-0.1819
3	top1000	1000	-0.0393	0.0094	0.1369
4	top500	500	-0.0569	0.0050	0.1831
5	top100	100	-0.1451	-0.1013	0.2983

Panel S10.1.1.2.1

All saved rows. Fields: `sign_agreement_N`, `sign_agreement_percent`, `sign_reversal_N`, `sign_reversal_percent`.

CSV row	cutoff	sign agreement N	sign agreement percent	sign reversal N	sign reversal percent
1	all	2154	51.6052	2020	48.3948
2	above_median_either	1572	51.5072	1480	48.4928
3	top1000	512	51.2000	488	48.8000
4	top500	265	53	235	47
5	top100	52	52	48	48

Panel S10.1.1.3.1

All saved rows. Fields: `positive_both_N`, `positive_both_percent`, `negative_both_N`, `negative_both_percent`.

CSV row	cutoff	positive both N	positive both percent	negative both N	negative both percent
1	all	496	11.8831	1658	39.7221
2	above_median_either	389	12.7457	1183	38.7615
3	top1000	128	12.8000	384	38.4000
4	top500	56	11.2000	209	41.8000
5	top100	8	8	44	44

Panel S10.1.1.4.1

All saved rows. Fields: `positive_fh_negative_ca_N`, `positive_fh_negative_ca_percent`, `negative_fh_positive_ca_N`, `negative_fh_positive_ca_percent`.

CSV row	cutoff	positive fh negative ca N	positive fh negative ca percent	negative fh positive ca N	negative fh positive ca percent
1	all	1354	32.4389	666	15.9559
2	above_median_either	959	31.4220	521	17.0708
3	top1000	296	29.6000	192	19.2000
4	top500	148	29.6000	87	17.4000
5	top100	31	31	17	17

Panel S10.1.1.5.1

All saved rows. Fields: `either_exact_zero_N`, `either_exact_zero_percent`.

CSV row	cutoff	either exact zero N	either exact zero percent
1	all	0	0
2	above_median_either	0	0
3	top1000	0	0
4	top500	0	0
5	top100	0	0

Table S11. CoAID Specification A hurdle components

Standardized occurrence probability and conditional positive mean by class for the primary CoAID hurdle (logit + zero-truncated NB2, first-release fixed effects), with the full row-level-product estimand for comparison. Point estimates are the saved values; intervals are 2.5th/97.5th linear percentiles of the 1,000 saved successful overlap-cluster bootstrap draws and reproduce the saved intervals exactly. The product of the two component ratios is not the full ratio because the estimand averages row-level products.

S11.1 — coaid_hurdle_components.csv

Source CSV: `outputs/extension_v1/tables/coaid_hurdle_components.csv`; SHA-256 `5d2cd1f9ecef71bbd44d11c9e45ddf2bb36c113670fa874cedb1f92cfaa6bec7`.

Panel S11.1.1

All saved rows. Fields: `measure`, `estimate`, `CI_lower`, `CI_upper`, `note`.

CSV row	measure	estimate	CI lower	CI upper	note
1	P_posts_gt_0_fake	0.1172	0.1008	0.1348	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2
2	P_posts_gt_0_real	0.6168	0.5922	0.6457	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2
3	E_posts_given_positive_fake	51.4005	36.3582	71.2154	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2
4	E_posts_given_positive_real	39.5277	23.5300	64.1466	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2
5	conditional_count_ratio_fake_over_real	1.3004	0.7285	2.5464	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2

CSV row	measure	estimate	CI lower	CI upper	note
6	E_posts_fake_full	8.8278	5.9332	12.4141	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2
7	E_posts_real_full	34.9197	20.3507	57.8835	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2
8	full_ratio_fake_over_real	0.2528	0.1322	0.5220	standardized over the A population covariate mix (first-release fixed effects); hurdle logit + zero-truncated NB2
9	occurrence_probability_ratio_fake_over_real	0.1900	0.1639	0.2182	ratio of marginal standardized probabilities; NOT the conditional logistic odds ratio
10	product_of_component_ratios	0.2471	0.1290	0.5012	product of averaged components; differs from the frozen estimand, which averages row-level P(Y>0|X)*E(Y|Y>0,X) before taking the Fake/Real ratio
11	full_ratio_minus_product	0.0057	0.0012	0.0245	share of draws with product > full ratio = 0.008
12	draw_level_median_product	0.2593	—	—	descriptive only

Table S12. Top Fake- and Real-leaning unigrams of the within-context LR models

The 25 unigrams with the largest positive (Fake-leaning) and largest negative (Real-leaning) coefficients in each saved natural-text within-context LR model (`final_coaid_natural.joblib`, `final_fakehealth_natural.joblib`; Fake = 1). Training document frequency was obtained by transforming the saved training pool with the fitted vectorizer. "In shared vocabulary" is membership in the 4,174-term unigram intersection of Table S10; "other-model sign" is the sign of the same term in the other dataset's model. "Masked as full identifier" marks terms the unchanged source-mask dictionary removes as whole identifiers; "fragment" marks terms that are tokens of a longer dictionary identifier and are removed only inside that phrase. Coefficients are model-specific conditional associations, not effect sizes.

S12.1 — top_fake_real_terms.csv

Source CSV: `outputs/extension_v1/tables/top_fake_real_terms.csv`; SHA-256 `640cc327ee98308e3326b6aa5a3b29a8ae4a3381851cf1f580842307139e29c9`.

Panel S12.1.1

CoAID, Fake-leaning top 25 (training N 2875; other model: FakeHealth).

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
1	1	coronavirus	3.9574	1516	False	—	False	False	True
2	2	trump	3.5883	107	False	—	False	False	False
3	3	facebook	3.5235	137	True	Real	True	True	True
4	4	òthe	3.1391	13	False	—	False	False	False
5	5	you	3.0340	371	True	Real	True	False	False
6	6	says	2.9898	110	True	Real	True	False	False
7	7	the	2.7804	1918	True	Fake	False	False	True
8	8	stimulus	2.7761	14	False	—	False	False	False
9	9	president	2.7028	102	True	Real	True	False	False
10	10	government	2.6324	59	True	Real	True	False	False
11	11	claim	2.5854	25	True	Real	True	False	False
12	12	video	2.5760	48	True	Fake	False	False	True
13	13	florida	2.5659	23	True	Fake	False	False	False
14	14	michigan	2.4294	16	True	Fake	False	False	False
15	15	was	2.4227	322	True	Real	True	False	False
16	16	starting	2.4109	11	True	Real	True	False	False
17	17	came	2.3709	12	True	Real	True	False	False
18	18	died	2.3695	60	True	Real	True	False	False
19	19	gates	2.3631	20	False	—	False	False	False

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
20	20	donõt	2.3353	4	False	—	False	False	False
21	21	election	2.3226	20	False	—	False	False	False
22	22	oil	2.2994	2	True	Real	True	False	False
23	23	media	2.2685	62	True	Fake	False	False	False
24	24	citizens	2.2490	17	False	—	False	False	False
25	25	drinking	2.2345	13	True	Fake	False	False	False

Panel S12.1.2

CoAID, Real-leaning top 25 (training N 2875; other model: FakeHealth).

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
1	1	and	-9.2397	2204	True	Fake	True	False	True
2	2	2020	-8.1817	629	True	Fake	True	False	False
3	3	disease	-6.9610	819	True	Real	False	False	True
4	4	19	-6.5716	1987	True	Real	False	False	False
5	5	april	-5.8565	305	True	Fake	True	False	False
6	6	covid	-5.6598	1996	False	—	False	False	True
7	7	health	-5.3675	719	True	Real	False	False	True
8	8	of	-5.1850	2275	True	Fake	True	False	False
9	9	for	-4.7084	1562	True	Fake	True	False	True
10	10	new	-4.6080	639	True	Real	False	False	True
11	11	2019	-4.3986	578	False	—	False	False	False
12	12	may	-4.3957	728	True	Real	False	False	False
13	13	experts	-4.1899	310	True	Real	False	False	False
14	14	as	-4.0902	858	True	Fake	True	False	False

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
15	15	images	-4.0283	288	True	Fake	True	False	False
16	16	yes	-3.9876	11	True	Real	False	False	False
17	17	getty	-3.9479	288	True	Real	False	False	False
18	18	data	-3.9412	485	True	Real	False	False	False
19	19	in	-3.8493	1775	True	Fake	True	False	False
20	20	march	-3.8296	267	True	Fake	True	False	False
21	21	medscape	-3.5131	218	False	—	False	False	False
22	22	other	-3.4913	349	True	Real	False	False	False
23	23	some	-3.4809	548	True	Real	False	False	False
24	24	at	-3.4439	980	True	Fake	True	False	False
25	25	reading	-3.3935	230	True	Real	False	False	False

Panel S12.1.3

FakeHealth, Fake-leaning top 25 (training N 1317; other model: CoAID).

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
1	1	release	1.0228	127	True	Real	True	False	False
2	2	news	0.8836	433	True	Real	True	False	True
3	3	and	0.8417	1218	True	Real	True	False	True
4	4	the	0.8383	1235	True	Fake	False	False	True
5	5	to	0.7763	1235	True	Real	True	False	False
6	6	artificial	0.6716	27	True	Real	True	False	False
7	7	mri	0.6647	53	False	—	False	False	False
8	8	pfizer	0.6395	34	False	—	False	False	False
9	9	of	0.6238	1228	True	Real	True	False	False

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
10	10	chronic	0.6165	140	True	Real	True	False	False
11	11	london	0.5794	63	True	Fake	False	False	False
12	12	traditional	0.5529	89	True	Real	True	False	False
13	13	cbs	0.5456	22	True	Real	True	False	True
14	14	scans	0.5057	75	True	Real	True	False	False
15	15	able	0.4979	245	True	Fake	False	False	False
16	16	menopause	0.4945	50	False	—	False	False	False
17	17	development	0.4876	159	True	Real	True	False	False
18	18	activity	0.4872	120	True	Real	True	False	False
19	19	imaging	0.4785	74	True	Real	True	False	False
20	20	proton	0.4765	6	False	—	False	False	False
21	21	my	0.4721	196	True	Fake	False	False	False
22	22	hope	0.4550	167	True	Real	True	False	False
23	23	pancreas	0.4519	27	False	—	False	False	False
24	24	2010	0.4518	210	True	Fake	False	False	False
25	25	pump	0.4491	30	False	—	False	False	False

Panel S12.1.4

FakeHealth, Real-leaning top 25 (training N 1317; other model: CoAID).

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
1	1	still	-0.7700	437	True	Fake	True	False	False
2	2	says	-0.7370	455	True	Fake	True	False	False
3	3	there	-0.7016	744	True	Real	False	False	False
4	4	drug	-0.6585	489	True	Real	False	False	False

CSV row	rank	term	coefficient	training doc. frequency	in shared vocabulary	other-model sign	sign reversal	masked as full identifier	fragment of identifier
5	5	costs	-0.6553	146	True	Real	False	False	False
6	6	many	-0.6520	580	True	Real	False	False	False
7	7	vaccine	-0.6427	40	True	Fake	True	False	True
8	8	he	-0.5835	706	True	Fake	True	False	False
9	9	involved	-0.5779	305	True	Fake	True	False	False
10	10	cost	-0.5762	204	True	Real	False	False	False
11	11	some	-0.5414	736	True	Real	False	False	False
12	12	however	-0.5409	339	True	Real	False	False	False
13	13	dr	-0.5364	715	True	Fake	True	False	False
14	14	sleep	-0.5360	87	True	Real	False	False	False
15	15	000	-0.5307	419	True	Real	False	False	False
16	16	therapy	-0.5250	364	True	Real	False	False	False
17	17	last	-0.5240	256	True	Fake	True	False	False
18	18	safe	-0.5209	179	True	Real	False	False	False
19	19	deaths	-0.5173	134	True	Fake	True	False	False
20	20	evidence	-0.5105	308	True	Fake	True	False	False
21	21	vision	-0.5102	58	False	—	False	False	False
22	22	studies	-0.5052	560	True	Real	False	False	False
23	23	intake	-0.4804	64	False	—	False	False	False
24	24	debate	-0.4800	61	True	Real	False	False	False
25	25	experimental	-0.4752	138	True	Real	False	False	False

Table S13. FakeNewsNet title-only external text target: cohort and LR transfer results

Title-only and LR-only. The local FakeNewsNet copy contains titles, URLs and tweet IDs but no article body, so the combined-text models were not applied; the title-only natural LR transfer fits (CoAID source: 3,571 title-eligible rows, cfg_16; FakeHealth source: 1,645 rows, cfg_11) were scored unchanged at threshold 0.5. No title-only SVM fit exists, so the SVM arm of the COVID-Rumor comparison is not replicated. Quality and overlap rules are identical to the COVID-Rumor evaluation; the two PolitiFact ids present in both label files were excluded. Labels are inherited from FakeNewsNet (PolitiFact fact-checker verdicts; GossipCop site scores) and were not verified locally. Results are item-weighted descriptives.

S13.1 — fakenewsnet_external_cohort_summary.csv

Source CSV: `outputs/extension_v1/tables/fakenewsnet_external_cohort_summary.csv`; SHA-256 `c5b9a442fb8df016b7182d87634555190d8121c73141e0dde985ac8d4d093b45`.

Panel S13.1.1

All saved rows; counts are items.

CSV row	subset	label	rows	label conflict excluded ids in subset	body text present locally	news url present	tweet ids present	zero tweet share	median tweet ids	quality excluded	exact duplicate title rows	all eligible	exact source overlap excluded	clean primary	high similarity flagged clean
1	GossipCop	fake	5323	0	0	5067	5135	0.0353	12	8	819	5315	0	5315	0
2	GossipCop	real	16817	0	0	16804	15759	0.0629	45	18	1498	16799	0	16799	0
3	PolitiFact	fake	430	2	0	426	390	0.0930	77.5000	1	9	429	0	429	0
4	PolitiFact	real	622	2	0	565	407	0.3457	7.5000	26	105	596	0	596	0

S13.2 — fakenewsnet_external_transfer_results.csv

Source CSV: `outputs/extension_v1/tables/fakenewsnet_external_transfer_results.csv`; SHA-256 `bebc7cdbef8661c05f886ad25db6d3b4e716158ff3102ee631ac550d0f04d337`.

Panel S13.2.1

Scope `clean_primary`; Fake = 1; LR probability threshold 0.5.

CSV row	classifier	source dataset	subset	scope	representation	N	fake N	real N	balanced accuracy	macro f1	fake f1	roc auc	average precision	threshold	actual fake prevalence	predicted fake prevalence
1	LR	CoAID	GossipCop	clean_primary	title_only	22114	5315	16799	0.5245	0.4624	0.3601	0.5323	0.2609	0.5000	0.2403	0.5693
2	LR	CoAID	PolitiFact	clean_primary	title_only	1025	429	596	0.5748	0.5419	0.5817	0.6084	0.5043	0.5000	0.4185	0.6683
3	LR	FakeHealth	GossipCop	clean_primary	title_only	22114	5315	16799	0.4814	0.4742	0.2689	0.4770	0.2233	0.5000	0.2403	0.3691
4	LR	FakeHealth	PolitiFact	clean_primary	title_only	1025	429	596	0.5395	0.5370	0.4213	0.5034	0.4262	0.5000	0.4185	0.3317

Panel S13.2.2

Scope `all_eligible`; Fake = 1; LR probability threshold 0.5.

CSV row	classifier	source dataset	subset	scope	representation	N	fake N	real N	balanced accuracy	macro f1	fake f1	roc auc	average precision	threshold	actual fake prevalence	predicted fake prevalence
1	LR	CoAID	GossipCop	all_eligible	title_only	22114	5315	16799	0.5245	0.4624	0.3601	0.5323	0.2609	0.5000	0.2403	0.5693
2	LR	CoAID	PolitiFact	all_eligible	title_only	1025	429	596	0.5748	0.5419	0.5817	0.6084	0.5043	0.5000	0.4185	0.6683
3	LR	FakeHealth	GossipCop	all_eligible	title_only	22114	5315	16799	0.4814	0.4742	0.2689	0.4770	0.2233	0.5000	0.2403	0.3691
4	LR	FakeHealth	PolitiFact	all_eligible	title_only	1025	429	596	0.5395	0.5370	0.4213	0.5034	0.4262	0.5000	0.4185	0.3317

Panel S13.2.3

Scope `clean_without_high_similarity`; Fake = 1; LR probability threshold 0.5.

CSV row	classifier	source dataset	subset	scope	representation	N	fake N	real N	balanced accuracy	macro f1	fake f1	roc auc	average precision	threshold	actual fake prevalence	predicted fake prevalence
1	LR	CoAID	GossipCop	clean_without_high_similarity	title_only	22114	5315	16799	0.5245	0.4624	0.3601	0.5323	0.2609	0.5000	0.2403	0.5693
2	LR	CoAID	PolitiFact	clean_without_high_similarity	title_only	1025	429	596	0.5748	0.5419	0.5817	0.6084	0.5043	0.5000	0.4185	0.6683
3	LR	FakeHealth	GossipCop	clean_without_high_similarity	title_only	22114	5315	16799	0.4814	0.4742	0.2689	0.4770	0.2233	0.5000	0.2403	0.3691
4	LR	FakeHealth	PolitiFact	clean_without_high_similarity	title_only	1025	429	596	0.5395	0.5370	0.4213	0.5034	0.4262	0.5000	0.4185	0.3317

Table S14. Source–class association (Cramér's V)

Descriptive association between source identity and the analytical Fake/Real label, for all input rows and for primary-eligible rows, using the publisher-source field and the article host separately. Cramér's V is computed over non-missing sources; missing sources are counted separately and never treated as one publisher. Major sources have at least 10 items and near-exclusive sources draw at least 95% of their items from one class. No classifier or p-value is involved; source is audit metadata, not a predictor.

S14.1 — text_source_label_association.csv

Source CSV: `outputs/tables/text_source_label_association.csv`; SHA-256 `1be6b53c5bbb8d1362ab55cfeda8425c084e3de2849cd6e1feb3a132e614cd64`.

Panel S14.1.1

All saved rows; both row scopes.

CSV row	dataset	scope	source measure	distinct nonmissing sources	missing source rows	cramers V	majority label baseline	major nearly exclusive sources	rows in major nearly exclusive sources
1	FakeHealth	all_input_rows	publisher_source	49	19	0.3532	0.7207	0	0
2	FakeHealth	all_input_rows	article_host	96	1	0.3859	0.7207	0	0
3	FakeHealth	primary_eligible	publisher_source	48	16	0.3519	0.7209	0	0
4	FakeHealth	primary_eligible	article_host	90	1	0.3787	0.7209	0	0
5	CoAID	all_input_rows	publisher_source	198	43	0.9693	0.8272	12	4905
6	CoAID	all_input_rows	article_host	198	43	0.9693	0.8272	12	4905
7	CoAID	primary_eligible	publisher_source	181	43	0.9710	0.7734	9	3298
8	CoAID	primary_eligible	article_host	181	43	0.9710	0.7734	9	3298

Table S15. Engagement comparability support (aggregate rows)

Aggregate support rows from the engagement comparability feasibility audit (`outputs/reports/engagement_comparability_feasibility.md`): cohort totals, publisher-identity coverage, publisher restrictions, the ten shared publisher families, timing-metadata coverage, topic-metadata coverage and the (not established) common-topic support, for CoAID `A_primary_strict` and the FakeHealth HealthStory engagement populations. The 654 native HealthStory review-category and review-tag rows of the source CSV are omitted here; they overlap and must not be summed. `retained_percent` uses the full engagement cohort as denominator; positive-post columns count entities with nonzero stored outcomes. Blank common-topic counts mean support is not established, not zero. In summary, only two publisher families, ABC News and CNN, carry both labels in both datasets, retaining 11 CoAID items (5 Fake, 6 Real) and 97 FakeHealth items (28 Fake, 69 Real); no shared family has five items of each label in each dataset; and restricting CoAID to identified publisher families would discard 66.81% of its Fake items (616 of 922).

S15.1 — engagement_comparability_support.csv

Source CSV: `outputs/tables/engagement_comparability_support.csv`; SHA-256 `37c2987f7b6964d348d72815e8a9d394faf9c62ca2320df156ec68935cc6832a`.

Panel S15.1.1

group_type `cohort` (2 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	CoAID	all	—	5337	922	4415	147	2530	100.0	observed_metadata_support
2	FakeHealth	all	—	1690	472	1218	471	1212	100.0	observed_metadata_support

Panel S15.1.2

group_type `publisher_metadata_coverage` (4 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	CoAID	prepared_source_name_missing	—	43	40	3	3	1	0.8057	observed_metadata_support
2	CoAID	prepared_article_host_missing	—	43	40	3	3	1	0.8057	observed_metadata_support
3	FakeHealth	prepared_source_name_missing	—	19	8	11	8	11	1.1243	observed_metadata_support
4	FakeHealth	prepared_article_host_missing	—	1	1	0	1	0	0.0592	observed_metadata_support

Panel S15.1.3

group_type `publisher_metadata_status` (6 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	CoAID	identified	—	4600	233	4367	96	2522	86.1907	observed_metadata_support
2	CoAID	identified_account_proxy	—	89	73	16	5	2	1.6676	observed_metadata_support
3	CoAID	missing_source_metadata	—	43	40	3	3	1	0.8057	observed_metadata_support
4	CoAID	unresolved_platform_publisher	—	605	576	29	43	5	11.336	observed_metadata_support
5	FakeHealth	identified	—	1689	471	1218	470	1212	99.9408	observed_metadata_support
6	FakeHealth	missing_source_metadata	—	1	1	0	1	0	0.0592	observed_metadata_support

Panel S15.1.4

group_type `publisher_restriction` (20 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	CoAID	identified_families	262	4689	306	4383	101	2524	87.8583	observed_metadata_support
2	CoAID	mixed_label_within_dataset	9	38	24	14	8	2	0.712	observed_metadata_support
3	CoAID	within_min_5_per_class	0	0	0	0	0	0	0.0	observed_metadata_support
4	CoAID	within_min_10_per_class	0	0	0	0	0	0	0.0	observed_metadata_support

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
5	CoAID	within_min_20_per_class	0	0	0	0	0	0	0.0	observed_metadata_support
6	CoAID	shared_families_any_label	10	2190	12	2178	6	1232	41.0343	observed_metadata_support
7	CoAID	shared_families_mixed_in_both	2	11	5	6	2	1	0.2061	observed_metadata_support
8	CoAID	shared_min_5_per_class_per_dataset	0	0	0	0	0	0	0.0	observed_metadata_support
9	CoAID	shared_min_10_per_class_per_dataset	0	0	0	0	0	0	0.0	observed_metadata_support
10	CoAID	shared_min_20_per_class_per_dataset	0	0	0	0	0	0	0.0	observed_metadata_support
11	FakeHealth	identified_families	51	1689	471	1218	470	1212	99.9408	observed_metadata_support
12	FakeHealth	mixed_label_within_dataset	36	1666	461	1205	460	1199	98.5799	observed_metadata_support
13	FakeHealth	within_min_5_per_class	21	1490	429	1061	428	1056	88.1657	observed_metadata_support
14	FakeHealth	within_min_10_per_class	11	1228	337	891	336	887	72.6627	observed_metadata_support
15	FakeHealth	within_min_20_per_class	5	829	235	594	234	592	49.0533	observed_metadata_support
16	FakeHealth	shared_families_any_label	10	382	151	231	150	230	22.6036	observed_metadata_support
17	FakeHealth	shared_families_mixed_in_both	2	97	28	69	28	69	5.7396	observed_metadata_support

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
18	FakeHealth	shared_min_5_per_class_per_dataset	0	0	0	0	0	0	0.0	observed_metadata_support
19	FakeHealth	shared_min_10_per_class_per_dataset	0	0	0	0	0	0	0.0	observed_metadata_support
20	FakeHealth	shared_min_20_per_class_per_dataset	0	0	0	0	0	0	0.0	observed_metadata_support

Panel S15.1.5

group_type `shared_publisher_family` (20 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	CoAID	host:abcnews.go.com	—	4	2	2	2	0	0.0749	observed_metadata_support
2	CoAID	host:azcentral.com	—	1	0	1	0	1	0.0187	observed_metadata_support
3	CoAID	host:cbsnews.com	—	2	2	0	2	0	0.0375	observed_metadata_support
4	CoAID	host:cnn.com	—	7	3	4	0	1	0.1312	observed_metadata_support
5	CoAID	host:foxnews.com	—	4	4	0	2	0	0.0749	observed_metadata_support
6	CoAID	host:medicalnewstoday.com	—	459	0	459	0	289	8.6003	observed_metadata_support
7	CoAID	host:nbcnews.com	—	1	0	1	0	0	0.0187	observed_metadata_support
8	CoAID	host:newsweek.com	—	1	1	0	0	0	0.0187	observed_metadata_support

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
9	CoAID	host:theguardian.com	—	1	0	1	0	0	0.0187	observed_metadata_support
10	CoAID	host:webmd.com	—	1710	0	1710	0	941	32.0405	observed_metadata_support
11	FakeHealth	host:abcnews.go.com	—	29	9	20	9	20	1.716	observed_metadata_support
12	FakeHealth	host:azcentral.com	—	1	1	0	1	0	0.0592	observed_metadata_support
13	FakeHealth	host:cbsnews.com	—	35	19	16	19	16	2.071	observed_metadata_support
14	FakeHealth	host:cnn.com	—	68	19	49	19	49	4.0237	observed_metadata_support
15	FakeHealth	host:foxnews.com	—	29	20	9	20	9	1.716	observed_metadata_support
16	FakeHealth	host:medicalnewstoday.com	—	2	0	2	0	2	0.1183	observed_metadata_support
17	FakeHealth	host:nbcnews.com	—	35	9	26	9	26	2.071	observed_metadata_support
18	FakeHealth	host:newsweek.com	—	26	7	19	7	19	1.5385	observed_metadata_support
19	FakeHealth	host:theguardian.com	—	21	15	6	15	6	1.2426	observed_metadata_support
20	FakeHealth	host:webmd.com	—	136	52	84	51	83	8.0473	observed_metadata_support

Panel S15.1.6

group_type `timing_metadata_coverage` (8 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	CoAID	publication_date_present	—	1081	140	941	73	559	20.2548	observed_metadata_support
2	CoAID	first_observed_post_time_present	—	2677	147	2530	147	2530	50.1593	observed_metadata_support
3	CoAID	last_observed_post_time_present	—	2677	147	2530	147	2530	50.1593	observed_metadata_support
4	CoAID	publication_date_valid	—	1080	139	941	72	559	20.2361	observed_metadata_support
5	FakeHealth	publication_date_present	—	1038	292	746	291	741	61.4201	observed_metadata_support
6	FakeHealth	first_observed_post_time_present	—	1683	471	1212	471	1212	99.5858	observed_metadata_support
7	FakeHealth	last_observed_post_time_present	—	1683	471	1212	471	1212	99.5858	observed_metadata_support
8	FakeHealth	publication_date_valid	—	1038	292	746	291	741	61.4201	observed_metadata_support

Panel S15.1.7

group_type `topic_metadata_coverage` (5 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	FakeHealth	review_tags_nonempty	—	1194	296	898	296	897	70.6509	observed_metadata_support

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
2	FakeHealth	review_tags_empty	—	496	176	320	175	315	29.3491	observed_metadata_support
3	FakeHealth	review_category_nonempty	—	1326	342	984	342	983	78.4615	observed_metadata_support
4	FakeHealth	review_category_missing	—	364	130	234	129	229	21.5385	observed_metadata_support
5	CoAID	no_prepared_topic_annotation	—	5337	922	4415	147	2530	100.0	observed_metadata_support

Panel S15.1.8

group_type `common_topic_support` (2 rows).

CSV row	dataset	group	group count	N	fake N	real N	fake with observed posts N	real with observed posts N	retained percent	status
1	CoAID	harmonized_topic_categories	0	—	—	—	—	—	—	not_established
2	FakeHealth	harmonized_topic_categories	0	—	—	—	—	—	—	not_established

Table S16. CoAID encoding-artifact prevalence and repair sensitivity

Supplementary analysis (Methods S8; protocol `protocols/coaid_encoding_sensitivity.md`). S16.1: holdout metrics of the selected CoAID LR configuration (cfg_05) refitted on the saved 2,875-row training pool after deterministic text repair and scored on the saved 702-row holdout, next to the saved primary result and a natural-text refit check. ftfy is not installed; the repair mapping is hand-written and recorded in `scripts/task_a_encoding_audit.py`. S16.2: share of documents in the combined-text cohorts containing each label-blind pattern, by dataset and class. The CoAID mojibake consists entirely of Mac-Roman curly-quote substitutes (Ò, Ó, Õ read as Latin-1); no Windows-1252/UTF-8 sequences or U+FFFD occur.

S16.1 — coaid_encoding_sensitivity.csv

Source CSV: `outputs/extension_v1/tables/coaid_encoding_sensitivity.csv`; SHA-256 `a062cc7468425eb642f7e2e8edae125e8fc981492bb0073d59540dc6307d82c1`.

Panel S16.1.1

All saved rows; macro F1, ROC-AUC and AP on the saved holdout, Fake = 1, threshold 0.5.

CSV row	variant	text transform	N train	N holdout	macro f1	roc auc	average precision	balanced accuracy	fake f1	vocabulary size	train docs changed	holdout docs changed
1	frozen_saved	none (saved result)	2875	702	0.9260	0.9827	0.9280	0.9313	0.8876	6428	0	0
2	natural_refit_check	none	2875	702	0.9260	0.9827	0.9280	0.9313	0.8876	6428	0	0
3	repaired	mojibake map + NFKC	2875	702	0.9281	0.9824	0.9264	0.9343	0.8909	6412	171	51
4	repaired_strict	mojibake map + NFKC + remove years/months/credits	2875	702	0.9188	0.9793	0.9168	0.9276	0.8772	6379	2651	658

S16.2 — coaid_encoding_prevalence.csv

Source CSV: `outputs/extension_v1/tables/coaid_encoding_prevalence.csv`; SHA-256 `a25d6fa88689dfdcf87be8869e9e71b0bea7a6979997031d1ad806bcd027abde`.

Panel S16.2.1

Cohort rows: share of documents containing at least one match.

CSV row	dataset	label	N	mojibake any share	macroman quote share	cp1252 utf8 share	c1 control share	replacement char share	year token share	month name share	month name excl may share	credit fragment share
1	CoAID	Fake	825	0.2594	0.2594	0	0	0	0.0836	0.0873	0.0570	0.0012
2	CoAID	Real	2752	0	0	0	0	0	0.4898	0.5730	0.3001	0.2482

CSV row	dataset	label	N	mojibake any share	macroman quote share	cp1252 utf8 share	c1 control share	replacement char share	year token share	month name share	month name excl may share	credit fragment share
3	CoAID	All	3577	0.0598	0.0598	0	0	0	0.3961	0.4610	0.2441	0.1912
4	FakeHealth	Fake	459	0.0044	0	0	0.0044	0	0.5882	0.7298	0.3290	0.2309
5	FakeHealth	Real	1188	0.0017	0	0	0.0017	0	0.6246	0.7643	0.3426	0.1785
6	FakeHealth	All	1647	0.0024	0	0	0.0024	0	0.6145	0.7547	0.3388	0.1931

Panel S16.2.2

CoAID publisher-family rows (families with ≥20 eligible rows; unresolved/missing identity reported separately; smaller families pooled).

CSV row	group	label	N	mojibake any share	year token share	month name share	credit fragment share
1	host:cdc.gov	Real	809	0	0.7330	0.3609	0.0012
2	host:healthline.com	Real	562	0	0.0445	0.4858	0.6957
3	host:instagram.com	Fake	21	0.3333	0.0476	0.0476	0
4	host:medicalnewstoday.com	Real	459	0	0.1176	0.7473	0.0196
5	host:nih.gov	Real	74	0	0.2838	0.1216	0
6	host:webmd.com	Real	626	0	0.9569	0.9425	0.4505
7	host:who.int	Real	158	0	0.2532	0.3101	0
8	other (<20 rows)	Fake	197	0.0102	0.2081	0.3046	0.0051
9	other (<20 rows)	Real	28	0	0.2857	0.5714	0
10	unresolved/missing	Fake	607	0.3377	0.0445	0.0181	0
11	unresolved/missing	Real	36	0	0.2222	0.1389	0

Table S17. FakeNewsNet engagement under the primary estimand

Supplementary analysis (Methods S8; protocol `protocols/fakenewsnet_engagement_protocol.md`). Outcome: distinct tweet IDs per item including zeros; design: intercept and Fake indicator (no release structure); family chosen by the primary selection rule (hurdle selected in both subsets); standardized Fake/Real expected-count ratio by the row-level-product rule; dependence via shared-tweet connected components; 1,000 successful component-bootstrap refits, seed 42, linear percentile intervals; ratio-of-ratios against the saved CoAID A and FakeHealth draws by the same Cartesian-product rule (C = subset ratio / reference ratio). Replies and retweets are not available; tweets were collected by title search in 2016–2018 crawls; labels are inherited. Observational dataset-context quantities.

S17.1 — fakenewsnet_overlap_graph_summary.csv

Source CSV: `outputs/extension_v1/tables/fakenewsnet_overlap_graph_summary.csv`; SHA-256 `927370a317e951421b18575082e742e240d9ec5a064225b09cfa566df0027a5c`.

Panel S17.1.1

Dependence graph by subset.

CSV row	subset	N	fake N	real N	zero N	total unique posts	components	singleton components	multi row components	rows in multi row components	largest component rows	largest component fake N	largest component real N	largest component post membership share	largest component row share
1	PolitiFact	1052	430	622	255	578316	948	930	18	122	84	5	79	0.6672	0.0798
2	GossipCop	22140	5323	16817	1246	1479926	20777	20406	371	1734	692	582	110	0.1914	0.0313

S17.2 — fakenewsnet_engagement_model_comparison.csv

Source CSV: `outputs/extension_v1/tables/fakenewsnet_engagement_model_comparison.csv`; SHA-256 `55e4dd922f93d42ffc2b80dfe246e4384d5a50d55e97e6aadbf9ddd817ef8227`.

Panel S17.2.1

Candidate families on the full outcome; Poisson is diagnostic only.

CSV row	subset	model	k	log likelihood	AIC	BIC	converged	stable	boundary	selected
1	PolitiFact	Poisson	2	-1117285.7262	2234575.4523	2234585.3692	True	True	False	False

CSV row	subset	model	k	log likelihood	AIC	BIC	converged	stable	boundary	selected
2	PolitiFact	NB2	3	-5843.4579	11692.9159	11707.7912	True	True	False	False
3	PolitiFact	Hurdle_Logit_ZTNB2	5	-5777.0918	11564.1836	11588.9758	True	True	False	True
4	GossipCop	Poisson	2	-1434907.4013	2869818.8025	2869834.8128	True	True	False	False
5	GossipCop	NB2	3	-111801.6001	223609.2001	223633.2155	True	True	False	False
6	GossipCop	Hurdle_Logit_ZTNB2	5	-111758.9186	223527.8373	223567.8630	True	True	False	True

S17.3 — fakenewsnet_engagement_results.csv

Source CSV: `outputs/extension_v1/tables/fakenewsnet_engagement_results.csv`; SHA-256 `a6ffd970be011b3c4002c1ecbb0895d6bd008d656b107ba9b1ff1e941ec7f3df`.

Panel S17.3.1

PolitiFact: standardized means, ratio and hurdle components with 95% component-bootstrap percentile intervals.

CSV row	subset	model	measure	estimate	CI lower	CI upper	successful draws	attempts	N	fake N	real N
1	PolitiFact	Hurdle_Logit_ZTNB2	standardized_fake_mean	374.9907	212.2860	659.4343	1000	1002	1052	430	622
2	PolitiFact	Hurdle_Logit_ZTNB2	standardized_real_mean	670.5306	134.6226	1406.2999	1000	1002	1052	430	622
3	PolitiFact	Hurdle_Logit_ZTNB2	mean_difference	-295.5398	-775.1475	123.1480	1000	1002	1052	430	622
4	PolitiFact	Hurdle_Logit_ZTNB2	mean_ratio	0.5592	0.4394	1.8677	1000	1002	1052	430	622
5	PolitiFact	Hurdle_Logit_ZTNB2	post_probability_fake	0.9070	0.8798	0.9364	1000	1002	1052	430	622
6	PolitiFact	Hurdle_Logit_ZTNB2	post_probability_real	0.6543	0.5766	0.7427	1000	1002	1052	430	622
7	PolitiFact	Hurdle_Logit_ZTNB2	post_probability_difference	0.2526	0.1621	0.3383	1000	1002	1052	430	622

CSV row	subset	model	measure	estimate	CI lower	CI upper	successful draws	attempts	N	fake N	real N
8	PolitiFact	Hurdle_Logit_ZTNB2	positive_fake_mean	413.4513	234.1129	727.9047	1000	1002	1052	430	622
9	PolitiFact	Hurdle_Logit_ZTNB2	positive_real_mean	1024.7420	223.8414	1944.0982	1000	1002	1052	430	622
10	PolitiFact	Hurdle_Logit_ZTNB2	positive_mean_ratio	0.4035	0.3357	1.2545	1000	1002	1052	430	622

Panel S17.3.2

GossipCop: standardized means, ratio and hurdle components with 95% component-bootstrap percentile intervals.

CSV row	subset	model	measure	estimate	CI lower	CI upper	successful draws	attempts	N	fake N	real N
1	GossipCop	Hurdle_Logit_ZTNB2	standardized_fake_mean	112.3988	62.1222	178.1587	1000	1001	22140	5323	16817
2	GossipCop	Hurdle_Logit_ZTNB2	standardized_real_mean	52.4247	50.6412	54.8465	1000	1001	22140	5323	16817
3	GossipCop	Hurdle_Logit_ZTNB2	mean_difference	59.9741	10.8400	123.5271	1000	1001	22140	5323	16817
4	GossipCop	Hurdle_Logit_ZTNB2	mean_ratio	2.1440	1.2116	3.2904	1000	1001	22140	5323	16817
5	GossipCop	Hurdle_Logit_ZTNB2	post_probability_fake	0.9647	0.9561	0.9723	1000	1001	22140	5323	16817
6	GossipCop	Hurdle_Logit_ZTNB2	post_probability_real	0.9371	0.9334	0.9408	1000	1001	22140	5323	16817
7	GossipCop	Hurdle_Logit_ZTNB2	post_probability_difference	0.0276	0.0183	0.0359	1000	1001	22140	5323	16817
8	GossipCop	Hurdle_Logit_ZTNB2	positive_fake_mean	116.5139	64.8865	183.1834	1000	1001	22140	5323	16817
9	GossipCop	Hurdle_Logit_ZTNB2	positive_real_mean	55.9443	54.1147	58.4590	1000	1001	22140	5323	16817

CSV row	subset	model	measure	estimate	CI lower	CI upper	successful draws	attempts	N	fake N	real N
10	GossipCop	Hurdle_Logit_ZTNB2	positive_mean_ratio	2.0827	1.1849	3.1788	1000	1001	22140	5323	16817

S17.4 — fakenewsnet_engagement_contrast.csv

Source CSV: `outputs/extension_v1/tables/fakenewsnet_engagement_contrast.csv`; SHA-256 `d4944bb55c05de060d83b2de06dfeec579a89daa7b135cc7dd2939613aaed11b`.

Panel S17.4.1

Ratio-of-ratios against the saved references; interval from the 1000 x 1000 Cartesian product of draws.

CSV row	subset	reference	subset ratio	reference ratio	ratio of ratios	CI lower ratio	CI upper ratio	bootstrap median	ratio interval contains 1	cartesian combinations
1	PolitiFact	CoAID_A_primary_strict	0.5592	0.2528	2.2122	0.9866	9.8943	2.5972	True	1000000
2	PolitiFact	FakeHealth_HealthStory	0.5592	0.8403	0.6656	0.4875	2.2906	0.6864	True	1000000
3	GossipCop	CoAID_A_primary_strict	2.1440	0.2528	8.4809	2.8207	18.2246	7.1957	False	1000000
4	GossipCop	FakeHealth_HealthStory	2.1440	0.8403	2.5516	1.3565	4.0868	2.4468	False	1000000

S17.5 — fakenewsnet_duplicate_title_sensitivity.csv

Source CSV: `outputs/extension_v1/tables/fakenewsnet_duplicate_title_sensitivity.csv`; SHA-256 `41e3dd5ad3b8a2c629fcb2a958fb73d44715f8da5809dc84cb042299c5741a74`.

Panel S17.5.1

Rows with repeated normalized titles dropped; fixed family refitted; point estimates only.

CSV row	subset	model	primary N	dropped rows	dropped fake	dropped real	sensitivity N	primary ratio	sensitivity ratio	ratio change	log ratio change	sensitivity fit stable
1	PolitiFact	Hurdle_Logit_ZTNB2	1052	112	7	105	940	0.5592	0.6217	0.0625	0.1059	True

CSV row	subset	model	primary N	dropped rows	dropped fake	dropped real	sensitivity N	primary ratio	sensitivity ratio	ratio change	log ratio change	sensitivity fit stable
2	GossipCop	Hurdle_Logit_ZTNB2	22140	2316	818	1498	19824	2.1440	1.7667	-0.3773	-0.1936	True

Table S18. CoAID A findability: any observed post by publisher resolution and first-recorded release

Supplementary analysis (Methods S8; protocol `protocols/round4_reviewer_checks.md`). Publisher-resolution status is assigned label-blind by the identity helper used for the primary populations (resolved = identified family or account proxy; unresolved = platform host without an account; missing = no source metadata); first-recorded release is the earliest member release; fact-check site is the host of the recorded fact-check URL (Fake entities only). Shares are the fraction of entities with at least one unique observed post. Counts and shares only; title-length summaries are in `round4_title_length.csv`.

S18.1 — round4_findability.csv

Source CSV: `outputs/extension_v1/tables/round4_findability.csv`; SHA-256 `5cff4b1b817d832a89dfebd324018877be93458f87544c66eeff5ca1cc5b9cce`.

Panel S18.1.1

Class × publisher-resolution status.

CSV row	label	group	N	fake N	real N	share any post	any post N
1	Fake	missing	40	40	0	0.0750	3
2	Fake	resolved	306	306	0	0.3301	101
3	Fake	unresolved	576	576	0	0.0747	43
4	Real	missing	3	0	3	0.3333	1
5	Real	resolved	4383	0	4383	0.5759	2524
6	Real	unresolved	29	0	29	0.1724	5

Panel S18.1.2

Class totals.

CSV row	label	group	N	fake N	real N	share any post	any post N
1	Fake	all	922	922	0	0.1594	147
2	Real	all	4415	0	4415	0.5730	2530

Panel S18.1.3

Class × first-recorded release.

CSV row	label	group	N	fake N	real N	share any post	any post N
1	Fake	05-01-2020	572	572	0	0.1276	73
2	Fake	07-01-2020	263	263	0	0.2624	69
3	Fake	09-01-2020	56	56	0	0.0714	4
4	Fake	11-01-2020	31	31	0	0.0323	1
5	Real	05-01-2020	1585	0	1585	0.9129	1447
6	Real	07-01-2020	1076	0	1076	0.9015	970
7	Real	09-01-2020	851	0	851	0.0764	65
8	Real	11-01-2020	903	0	903	0.0532	48

Panel S18.1.4

Fake entities by originating fact-check site (sites with ≥10 entities; smaller sites pooled).

CSV row	label	group	N	fake N	real N	share any post	any post N
1	Fake	politifact.com	352	352	0	0.2216	78
2	Fake	leadstories.com	121	121	0	0.1322	16
3	Fake	hoax-alert.leadstories.com	103	103	0	0.1650	17
4	Fake	healthfeedback.org	100	100	0	0.1500	15
5	Fake	factcheck.org	89	89	0	0.1236	11
6	Fake	factcheck.afp.com	73	73	0	0.0548	4
7	Fake	checkyourfact.com	41	41	0	0.0976	4
8	Fake	washingtonpost.com	20	20	0	0.0500	1
9	Fake	other sites (<10 each; 10 sites)	23	23	0	0.0435	1

Table S19. Document length as a shared cue: score–length correlations and a length-only diagnostic baseline

Supplementary analysis (Methods S8). S19.1: Spearman correlation between log(1 + analyzer tokens) and the model score, with the primary ROC-AUC, per model and target (within-context holdouts and cross-context all-eligible targets re-scored transform-only with the saved pipelines; COVID-Rumor and FakeNewsNet from the saved predictions). S19.2: **a diagnostic model, not a candidate classifier** — logistic regression on the single feature log(1 + tokens), fitted on each source training pool (combined-text pool for combined-text targets, title pool for the FakeNewsNet title targets), scored on the same targets; its ROC-AUC is shown next to the word LR. S19.3: the per-quintile rates from `round4_length_bins.csv` (rank-based quintiles of analyzer token count per target; the source of the 93% figure quoted in Results 3.3), rendered as saved. S19.4 re-scores the saved models within length quintiles without refitting: the ROC-AUC of the saved LR and SVM scores within each length quintile, and the length-stratified ROC-AUC over same-quintile (Fake, Real) pairs.

S19.1 — round4_length_spearman.csv

Source CSV: `outputs/extension_v1/tables/round4_length_spearman.csv`; SHA-256 `8878dd8bbdaf77e37e25064f1d54d1fb6415458544605b439857839c620735f2`.

Panel S19.1.1

All saved rows.

CSV row	target	classifier	source	N	spearman log tokens vs score	frozen roc auc
1	within_CoAID_holdout	LR	CoAID	702	-0.5424	0.9827
2	within_CoAID_holdout	SVM	CoAID	702	-0.5497	0.9795
3	within_FakeHealth_holdout	LR	FakeHealth	330	-0.6280	0.7701
4	within_FakeHealth_holdout	SVM	FakeHealth	330	-0.6355	0.7695
5	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	1647	-0.6872	0.6047
6	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	1647	-0.6741	0.6113
7	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	3577	-0.1185	0.5367
8	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	3577	-0.1383	0.5471
9	external_COVIDRumor_clean	LR	CoAID	3667	-0.3558	0.6995

CSV row	target	classifier	source	N	spearman log tokens vs score	frozen roc auc
10	external_COVIDRumor_clean	SVM	CoAID	3667	-0.3812	0.6816
11	external_COVIDRumor_clean	LR	FakeHealth	3667	0.0398	0.5219
12	external_COVIDRumor_clean	SVM	FakeHealth	3667	0.0262	0.5248
13	external_FakeNewsNet_GossipCop_title_clean	LR	CoAID	22114	0.0091	0.5323
14	external_FakeNewsNet_GossipCop_title_clean	LR	FakeHealth	22114	-0.1253	0.4770
15	external_FakeNewsNet_PolitiFact_title_clean	LR	CoAID	1025	0.0941	0.6084
16	external_FakeNewsNet_PolitiFact_title_clean	LR	FakeHealth	1025	-0.0528	0.5034

S19.2 — round4_length_baseline.csv

Source CSV: `outputs/extension_v1/tables/round4_length_baseline.csv`; SHA-256 `38abbf70c940bbb867be2f38ca898b40d89f870d919f4fd438f7538faf7e1d46`.

Panel S19.2.1

Length-only diagnostic baseline (NEW model; not a candidate classifier).

CSV row	target	source pool	representation	train N	coefficient log tokens	roc auc length only	roc auc frozen LR
1	within_CoAID_holdout	CoAID	combined_text	2875	-2.6543	0.8818	0.9827
2	within_FakeHealth_holdout	FakeHealth	combined_text	1317	-0.1292	0.7289	0.7701
3	transfer_CoAID_to_FakeHealth_all_eligible	CoAID	combined_text	2875	-2.6543	0.6491	0.6047

CSV row	target	source pool	representation	train N	coefficient log tokens	roc auc length only	roc auc frozen LR
4	transfer_FakeHealth_to_CoAID_all_eligible	FakeHealth	combined_text	1317	-0.1292	0.8521	0.5367
5	external_COVIDRumor_clean	CoAID	combined_text	2875	-2.6543	0.7158	0.6995
6	external_COVIDRumor_clean	FakeHealth	combined_text	1317	-0.1292	0.7158	0.5219
7	external_FakeNewsNet_GossipCop_title_clean	CoAID	title_text	2870	0.6540	0.4806	0.5323
8	external_FakeNewsNet_GossipCop_title_clean	FakeHealth	title_text	1315	-0.2341	0.5194	0.4770
9	external_FakeNewsNet_PolitiFact_title_clean	CoAID	title_text	2870	0.6540	0.7388	0.6084
10	external_FakeNewsNet_PolitiFact_title_clean	FakeHealth	title_text	1315	-0.2341	0.2612	0.5034

S19.3 — round4_length_bins.csv

Source CSV: `outputs/extension_v1/tables/round4_length_bins.csv`; SHA-256 `f6f18b3db71670f9658b4c511f6436af7e16e769127ebc2b5cda9cfafdf37485`.

Panel S19.3.1

All saved rows: per length quintile (rank-based quintiles of analyzer token count within each target), N, token range, predicted and actual Fake rates, macro F1 and ROC-AUC of the saved score. Within-context and cross-context rows are transform-only re-scores with the saved pipelines; COVID-Rumor and FakeNewsNet rows use the saved predictions.

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
1	within_CoAID_holdout	LR	CoAID	1	141	4	36	0.9716	0.9291	0.5490	0.5840
2	within_CoAID_holdout	LR	CoAID	2	140	36	84	0.1000	0.0857	0.7456	0.9447

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
3	within_CoAID_holdout	LR	CoAID	3	140	84	88	0.0286	0.0429	0.7926	0.9826
4	within_CoAID_holdout	LR	CoAID	4	140	88	92	0.0286	0.0214	0.9267	0.9951
5	within_CoAID_holdout	LR	CoAID	5	141	92	114	0.0922	0.0993	0.8157	0.9612
6	within_CoAID_holdout	SVM	CoAID	1	141	4	36	1	0.9291	0.4816	0.5260
7	within_CoAID_holdout	SVM	CoAID	2	140	36	84	0.1143	0.0857	0.7222	0.9284
8	within_CoAID_holdout	SVM	CoAID	3	140	84	88	0.0571	0.0429	0.6992	0.9751
9	within_CoAID_holdout	SVM	CoAID	4	140	88	92	0.0429	0.0214	0.8278	0.9951
10	within_CoAID_holdout	SVM	CoAID	5	141	92	114	0.1064	0.0993	0.7886	0.9584
11	within_FakeHealth_holdout	LR	FakeHealth	1	66	5	328	0.6364	0.4697	0.5863	0.6065
12	within_FakeHealth_holdout	LR	FakeHealth	2	66	328	528	0.3333	0.5303	0.6140	0.7005
13	within_FakeHealth_holdout	LR	FakeHealth	3	66	533	706	0.1212	0.2121	0.6182	0.6662
14	within_FakeHealth_holdout	LR	FakeHealth	4	66	715	894	0.0606	0.0909	0.5672	0.7667
15	within_FakeHealth_holdout	LR	FakeHealth	5	66	911	3903	0.0303	0.0909	0.4677	0.7000

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
16	within_FakeHealth_holdout	SVM	FakeHealth	1	66	5	328	0.6667	0.4697	0.5523	0.6046
17	within_FakeHealth_holdout	SVM	FakeHealth	2	66	328	528	0.3333	0.5303	0.6140	0.7023
18	within_FakeHealth_holdout	SVM	FakeHealth	3	66	533	706	0.1212	0.2121	0.6182	0.6662
19	within_FakeHealth_holdout	SVM	FakeHealth	4	66	715	894	0.0606	0.0909	0.5672	0.7694
20	within_FakeHealth_holdout	SVM	FakeHealth	5	66	911	3903	0.0303	0.0909	0.4677	0.6944
21	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	1	330	4	293	0.3273	0.3515	0.3918	0.3460
22	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	2	329	295	494	0	0.4620	0.3498	0.5017
23	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	3	329	496	660	0	0.3040	0.4104	0.5846
24	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	4	329	662	849	0	0.1337	0.4642	0.5624
25	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	5	330	851	3903	0	0.1424	0.4617	0.5717

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
26	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	1	330	4	293	0.3545	0.3515	0.3930	0.3500
27	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	2	329	295	494	0	0.4620	0.3498	0.5135
28	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	3	329	496	660	0	0.3040	0.4104	0.5967
29	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	4	329	662	849	0	0.1337	0.4642	0.5808
30	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	5	330	851	3903	0	0.1424	0.4617	0.5959
31	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	1	716	1	36	0.8087	0.8743	0.4582	0.4196
32	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	2	715	36	82	0.8294	0.0713	0.2005	0.4870
33	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	3	715	82	87	0.7538	0.0322	0.2232	0.5098

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
34	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	4	715	87	91	0.7091	0.0601	0.2880	0.5690
35	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	5	716	91	115	0.6955	0.1145	0.3509	0.6177
36	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	1	716	1	36	0.9022	0.8743	0.4864	0.4213
37	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	2	715	36	82	0.8573	0.0713	0.1790	0.4863
38	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	3	715	82	87	0.7818	0.0322	0.2061	0.5106
39	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	4	715	87	91	0.7343	0.0601	0.2728	0.5691
40	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	5	716	91	115	0.7151	0.1145	0.3355	0.6177
41	external_COVIDRumor_clean	LR	CoAID	1	734	2	9	0.9728	0.9550	0.4813	0.5339
42	external_COVIDRumor_clean	LR	CoAID	2	733	9	12	0.9591	0.8909	0.5086	0.5812

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
43	external_COVIDRumor_clean	LR	CoAID	3	733	12	16	0.9386	0.8649	0.5148	0.6087
44	external_COVIDRumor_clean	LR	CoAID	4	733	16	25	0.9304	0.7790	0.5194	0.6529
45	external_COVIDRumor_clean	LR	CoAID	5	734	25	123	0.8365	0.6213	0.5588	0.7055
46	external_COVIDRumor_clean	SVM	CoAID	1	734	2	9	0.9782	0.9550	0.4827	0.5161
47	external_COVIDRumor_clean	SVM	CoAID	2	733	9	12	0.9632	0.8909	0.5010	0.5574
48	external_COVIDRumor_clean	SVM	CoAID	3	733	12	16	0.9509	0.8649	0.4982	0.5860
49	external_COVIDRumor_clean	SVM	CoAID	4	733	16	25	0.9495	0.7790	0.4971	0.6334
50	external_COVIDRumor_clean	SVM	CoAID	5	734	25	123	0.8774	0.6213	0.5177	0.6702
51	external_COVIDRumor_clean	LR	FakeHealth	1	734	2	9	0.8760	0.9550	0.4715	0.5190
52	external_COVIDRumor_clean	LR	FakeHealth	2	733	9	12	0.9181	0.8909	0.4946	0.5610
53	external_COVIDRumor_clean	LR	FakeHealth	3	733	12	16	0.9072	0.8649	0.5641	0.6046

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
54	external_COVIDRumor_clean	LR	FakeHealth	4	733	16	25	0.8895	0.7790	0.4894	0.5627
55	external_COVIDRumor_clean	LR	FakeHealth	5	734	25	123	0.8896	0.6213	0.4303	0.4662
56	external_COVIDRumor_clean	SVM	FakeHealth	1	734	2	9	0.9005	0.9550	0.4814	0.5180
57	external_COVIDRumor_clean	SVM	FakeHealth	2	733	9	12	0.9454	0.8909	0.4917	0.5594
58	external_COVIDRumor_clean	SVM	FakeHealth	3	733	12	16	0.9277	0.8649	0.5449	0.6039
59	external_COVIDRumor_clean	SVM	FakeHealth	4	733	16	25	0.9154	0.7790	0.5047	0.5632
60	external_COVIDRumor_clean	SVM	FakeHealth	5	734	25	123	0.9074	0.6213	0.4252	0.4664
61	external_FakeNewsNet_GossipCop_title_clean	LR	CoAID	1	4423	2	8	0.6000	0.3070	0.4809	0.5372
62	external_FakeNewsNet_GossipCop_title_clean	LR	CoAID	2	4423	8	10	0.5605	0.2571	0.4733	0.5425
63	external_FakeNewsNet_GossipCop_title_clean	LR	CoAID	3	4422	10	12	0.5595	0.2546	0.4683	0.5320

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
64	external_FakeNewsNet_GossipCop_title_clean	LR	CoAID	4	4423	12	14	0.5625	0.1981	0.4417	0.5251
65	external_FakeNewsNet_GossipCop_title_clean	LR	CoAID	5	4423	14	37	0.5641	0.1849	0.4376	0.5245
66	external_FakeNewsNet_GossipCop_title_clean	LR	FakeHealth	1	4423	2	8	0.3118	0.3070	0.4860	0.4592
67	external_FakeNewsNet_GossipCop_title_clean	LR	FakeHealth	2	4423	8	10	0.3981	0.2571	0.4781	0.4705
68	external_FakeNewsNet_GossipCop_title_clean	LR	FakeHealth	3	4422	10	12	0.3899	0.2546	0.4714	0.4746
69	external_FakeNewsNet_GossipCop_title_clean	LR	FakeHealth	4	4423	12	14	0.3893	0.1981	0.4632	0.4611
70	external_FakeNewsNet_GossipCop_title_clean	LR	FakeHealth	5	4423	14	37	0.3565	0.1849	0.4661	0.4841
71	external_FakeNewsNet_PolitiFact_title_clean	LR	CoAID	1	205	2	5	0.6488	0.1707	0.4353	0.6352

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
72	external_FakeNewsNet_PolitiFact_title_clean	LR	CoAID	2	205	5	8	0.6244	0.2829	0.5178	0.6546
73	external_FakeNewsNet_PolitiFact_title_clean	LR	CoAID	3	205	8	10	0.6976	0.4537	0.5407	0.5876
74	external_FakeNewsNet_PolitiFact_title_clean	LR	CoAID	4	205	10	13	0.6683	0.5366	0.5876	0.5817
75	external_FakeNewsNet_PolitiFact_title_clean	LR	CoAID	5	205	13	53	0.7024	0.6488	0.5382	0.5603
76	external_FakeNewsNet_PolitiFact_title_clean	LR	FakeHealth	1	205	2	5	0.2000	0.1707	0.4508	0.4593
77	external_FakeNewsNet_PolitiFact_title_clean	LR	FakeHealth	2	205	5	8	0.2976	0.2829	0.5915	0.5826
78	external_FakeNewsNet_PolitiFact_title_clean	LR	FakeHealth	3	205	8	10	0.4488	0.4537	0.5223	0.5154
79	external_FakeNewsNet_PolitiFact_title_clean	LR	FakeHealth	4	205	10	13	0.3756	0.5366	0.5232	0.5111

CSV row	target	classifier	source	quintile	N	tokens min	tokens max	predicted fake rate	actual fake rate	macro f1	roc auc
80	external_FakeNewsNet_PolitiFact_title_clean	LR	FakeHealth	5	205	13	53	0.3366	0.6488	0.4438	0.4807

S19.4 — round5_within_length_auc.csv (re-scoring check)

Source CSV: `outputs/extension_v1/round5/round5_within_length_auc.csv`; SHA-256 `837b19a1afffbc23fbaf09102dffbc304f7dcc1fe35dd1e79b2561aff8e1708e`.

Within-length-quintile ROC-AUC of the saved scores. Bins are the same rank-based quintiles of analyzer token count as S19.3, reconstructed by executing the same scoring code as S19.3 and checked cell-for-cell against `round4_length_bins.csv` before any computation; the unstratified ROC-AUC in each pooled row equals Table S19.1. Within-bin ROC-AUC is left blank (`estimable` = False) when a bin has fewer than 5 Fake or 5 Real items. The `pooled_length_stratified` row is the share of same-quintile (Fake, Real) pairs that the score orders correctly, ties counting 0.5, over all such pairs (`same bin pairs`). The small Fake counts in CoAID's longer quintiles (3–14 items) make those bins imprecise; no intervals are reported. No model was refit, retrained or tuned.

Panel S19.4.1

All saved rows.

CSV row	target	classifier	source	bin	tokens min	tokens max	N	fake N	real N	same bin pairs	roc auc	estimable	unstratified frozen roc auc	s19 1 frozen roc auc
1	within_CoAID_holdout	LR	CoAID	1	4	36	141	131	10	1310	0.5840	True	—	—
2	within_CoAID_holdout	LR	CoAID	2	36	84	140	12	128	1536	0.9447	True	—	—
3	within_CoAID_holdout	LR	CoAID	3	84	88	140	6	134	804	0.9826	True	—	—
4	within_CoAID_holdout	LR	CoAID	4	88	92	140	3	137	411	—	False	—	—

CSV row	target	classifier	source	bin	tokens min	tokens max	N	fake N	real N	same bin pairs	roc auc	estimable	unstratified frozen roc auc	s19 1 frozen roc auc
5	within_CoAID_holdout	LR	CoAID	5	92	114	141	14	127	1778	0.9612	True	—	—
6	within_CoAID_holdout	LR	CoAID	pooled_length_stratified	4	114	702	166	536	5839	0.8775	True	0.9827	0.9827
7	within_CoAID_holdout	SVM	CoAID	1	4	36	141	131	10	1310	0.5260	True	—	—
8	within_CoAID_holdout	SVM	CoAID	2	36	84	140	12	128	1536	0.9284	True	—	—
9	within_CoAID_holdout	SVM	CoAID	3	84	88	140	6	134	804	0.9751	True	—	—
10	within_CoAID_holdout	SVM	CoAID	4	88	92	140	3	137	411	—	False	—	—
11	within_CoAID_holdout	SVM	CoAID	5	92	114	141	14	127	1778	0.9584	True	—	—
12	within_CoAID_holdout	SVM	CoAID	pooled_length_stratified	4	114	702	166	536	5839	0.8584	True	0.9795	0.9795
13	within_FakeHealth_holdout	LR	FakeHealth	1	5	328	66	31	35	1085	0.6065	True	—	—
14	within_FakeHealth_holdout	LR	FakeHealth	2	328	528	66	35	31	1085	0.7005	True	—	—

CSV row	target	classifier	source	bin	tokens min	tokens max	N	fake N	real N	same bin pairs	roc auc	estimable	unstratified frozen roc auc	s19 1 frozen roc auc
15	within_FakeHealth_holdout	LR	FakeHealth	3	533	706	66	14	52	728	0.6662	True	—	—
16	within_FakeHealth_holdout	LR	FakeHealth	4	715	894	66	6	60	360	0.7667	True	—	—
17	within_FakeHealth_holdout	LR	FakeHealth	5	911	3903	66	6	60	360	0.7000	True	—	—
18	within_FakeHealth_holdout	LR	FakeHealth	pooled_length_stratified	5	3903	330	92	238	3618	0.6719	True	0.7701	0.7701
19	within_FakeHealth_holdout	SVM	FakeHealth	1	5	328	66	31	35	1085	0.6046	True	—	—
20	within_FakeHealth_holdout	SVM	FakeHealth	2	328	528	66	35	31	1085	0.7023	True	—	—
21	within_FakeHealth_holdout	SVM	FakeHealth	3	533	706	66	14	52	728	0.6662	True	—	—
22	within_FakeHealth_holdout	SVM	FakeHealth	4	715	894	66	6	60	360	0.7694	True	—	—
23	within_FakeHealth_holdout	SVM	FakeHealth	5	911	3903	66	6	60	360	0.6944	True	—	—
24	within_FakeHealth_holdout	SVM	FakeHealth	pooled_length_stratified	5	3903	330	92	238	3618	0.6716	True	0.7695	0.7695

CSV row	target	classifier	source	bin	tokens min	tokens max	N	fake N	real N	same bin pairs	roc auc	estimable	unstratified frozen roc auc	s19 1 frozen roc auc
25	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	1	4	293	330	116	214	24824	0.3460	True	—	—
26	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	2	295	494	329	152	177	26904	0.5017	True	—	—
27	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	3	496	660	329	100	229	22900	0.5846	True	—	—
28	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	4	662	849	329	44	285	12540	0.5624	True	—	—
29	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	5	851	3903	330	47	283	13301	0.5717	True	—	—
30	transfer_CoAID_to_FakeHealth_all_eligible	LR	CoAID	pooled_length_stratified	4	3903	1647	459	1188	100469	0.4990	True	0.6047	0.6047
31	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	1	4	293	330	116	214	24824	0.3500	True	—	—

CSV row	target	classifier	source	bin	tokens min	tokens max	N	fake N	real N	same bin pairs	roc auc	estimable	unstratified frozen roc auc	s19 1 frozen roc auc
32	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	2	295	494	329	152	177	26904	0.5135	True	—	—
33	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	3	496	660	329	100	229	22900	0.5967	True	—	—
34	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	4	662	849	329	44	285	12540	0.5808	True	—	—
35	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	5	851	3903	330	47	283	13301	0.5959	True	—	—
36	transfer_CoAID_to_FakeHealth_all_eligible	SVM	CoAID	pooled_length_stratified	4	3903	1647	459	1188	100469	0.5114	True	0.6113	0.6113
37	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	1	1	36	716	626	90	56340	0.4196	True	—	—
38	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	2	36	82	715	51	664	33864	0.4870	True	—	—

CSV row	target	classifier	source	bin	tokens min	tokens max	N	fake N	real N	same bin pairs	roc auc	estimable	unstratified frozen roc auc	s19 1 frozen roc auc
39	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	3	82	87	715	23	692	15916	0.5098	True	—	—
40	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	4	87	91	715	43	672	28896	0.5690	True	—	—
41	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	5	91	115	716	82	634	51988	0.6177	True	—	—
42	transfer_FakeHealth_to_CoAID_all_eligible	LR	FakeHealth	pooled_length_stratified	1	115	3577	825	2752	187004	0.5176	True	0.5367	0.5367
43	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	1	1	36	716	626	90	56340	0.4213	True	—	—
44	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	2	36	82	715	51	664	33864	0.4863	True	—	—
45	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	3	82	87	715	23	692	15916	0.5106	True	—	—

CSV row	target	classifier	source	bin	tokens min	tokens max	N	fake N	real N	same bin pairs	roc auc	estimable	unstratified frozen roc auc	s19 1 frozen roc auc
46	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	4	87	91	715	43	672	28896	0.5691	True	—	—
47	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	5	91	115	716	82	634	51988	0.6177	True	—	—
48	transfer_FakeHealth_to_CoAID_all_eligible	SVM	FakeHealth	pooled_length_stratified	1	115	3577	825	2752	187004	0.5181	True	0.5471	0.5471

Table S20. Bootstrap shape conditional on dominant-component inclusions, and the PolitiFact dominant-component title audit

Supplementary analysis (Methods S8), from the saved draws only (`version_aware_bootstrap_refits.csv` for CoAID A; `fakenewsnet_engagement_bootstrap_refits.csv` for PolitiFact and GossipCop; 1,000 successful draws each). Inclusions = number of times the dominant component entered a draw; the `kde_local_maxima_log_ratio` row gives the number of local maxima of a Gaussian-kernel density of the log ratio (draws column) and the share of draws containing the component at least once. S20.2 is a label-blind title audit of the 84-row PolitiFact component against the rest of PolitiFact. Histogram bins are in `round4_bootstrap_histogram.csv`; the ten most frequent component titles in `round4_politifact_component_top_titles.csv`. Figure S1 shows the three distributions.

S20.1 — round4_bootstrap_conditional.csv

Source CSV: `outputs/extension_v1/tables/round4_bootstrap_conditional.csv`; SHA-256 `f59af9e3b73ca8d94f7523865d3af7aed99d90b969f7cc04e6d5658b57811e78`.

Panel S20.1.1

All saved rows; ratio quantiles conditional on inclusion class.

CSV row	context	inclusions	draws	share of draws	ratio p025	ratio p25	ratio median	ratio p75	ratio p975	ratio mean
1	CoAID_A	all	1000	1	0.13222351531208054	0.20810261997983176	0.26557963214816327	0.37065545833338406	0.52200670906616609	0.2919
2	CoAID_A	0	363	0.3630	0.26585157260043679	0.35857379588962274	0.40822198975520895	0.45899276520042207	0.56894410124467932	0.4097
3	CoAID_A	1	372	0.3720	0.16928991740070131	0.21875588773132534	0.24864945782718	0.28164102283613807	0.34832405847718142	0.2516
4	CoAID_A	2+	265	0.2650	0.11214577170896536	0.16200433663345893	0.18239080224471961	0.21339629717776772	0.26992668125825636	0.1869
5	CoAID_A	kde_local_maxima_log_ratio	2	0.6370						0.2919
6	PolitiFact	all	1000	1	0.43943590989393948	0.49897558429980093	0.56525632390003866	1.347511511945077	1.867660818484842651	0.8774
7	PolitiFact	0	372	0.3720	1.0547745740899943	1.3025695559932498	1.4652724196145286	1.6594489712512763	1.9843258520289415	1.4818
8	PolitiFact	1	355	0.3550	0.4878796758170835	0.5305864244358055	0.5543327289292894	0.5775860824353447	0.6495722501919685	0.5560
9	PolitiFact	2+	273	0.2730	0.42715961352302	0.45488394676683 1	0.4703186292011441	0.4871826182973249	0.52837115743428 33	0.4719
10	PolitiFact	kde_local_maxima_log_ratio	2	0.6280						0.8774
11	GossipCop	all	1000	1	1.2116206729620254	1.4240569172238643	2.097324069205169	2.617364142008 4763	3.290419801908 16	2.0553
12	GossipCop	0	377	0.3770	1.1500174453632828	1.2919122514 99149	1.3749101635715613	1.4456225895844943	1.6175951986401589	1.3755
13	GossipCop	1	356	0.3560	1.9664864116009517	2.0756308148409452	2.1332278302311285	2.1978103028 06054	2.3385582785 48549	2.1418

CSV row	context	inclusions	draws	share of draws	ratio p025	ratio p25	ratio median	ratio p75	ratio p975	ratio mean
14	GossipCop	2+	267	0.2670	2.557465040458803	2.695415898775128	2.780359557199463 7	3.117402155846714	3.631116395655752	2.8997
15	GossipCop	kde_local_maxima_log_ratio	3	0.6230						2.0553

S20.2 — round4_politifact_component_titles.csv

Source CSV: `outputs/extension_v1/tables/round4_politifact_component_titles.csv`; SHA-256 `5f1b50d49518a4ae6865fffb1a4074ecf1c21e0ab5c6e646bff1c312366edb54`.

Panel S20.2.1

Label-blind title audit; primary near-duplicate rule = bigram Jaccard ≥ 0.95 with ≥ 8 tokens.

CSV row	group	rows	fake N	real N	exact duplicate title share	distinct titles	near duplicate share frozen rule min8	near duplicate share relaxed min4	mean pairwise title bigram jaccard	median title tokens
1	dominant_component_84	84	5	79	0.2024	74	0	0	0.0054	6
2	rest_of_PolitiFact	968	425	543	0.0868	916	0	0	0.0006	10

Figure S1 — bootstrap ratio distributions by dominant-component inclusions

Histograms of the log Fake/Real ratio over the 1,000 saved successful draws, coloured by how many times the dominant component entered the draw (0, 1, 2+). Dashed line: parity. Source: `outputs/extension_v1/figures/round4_bootstrap_shape_<context>.png` (SHA-256 in the caption lines), generated from the saved draws by `scripts/round4_check3_bootstrap.py`.

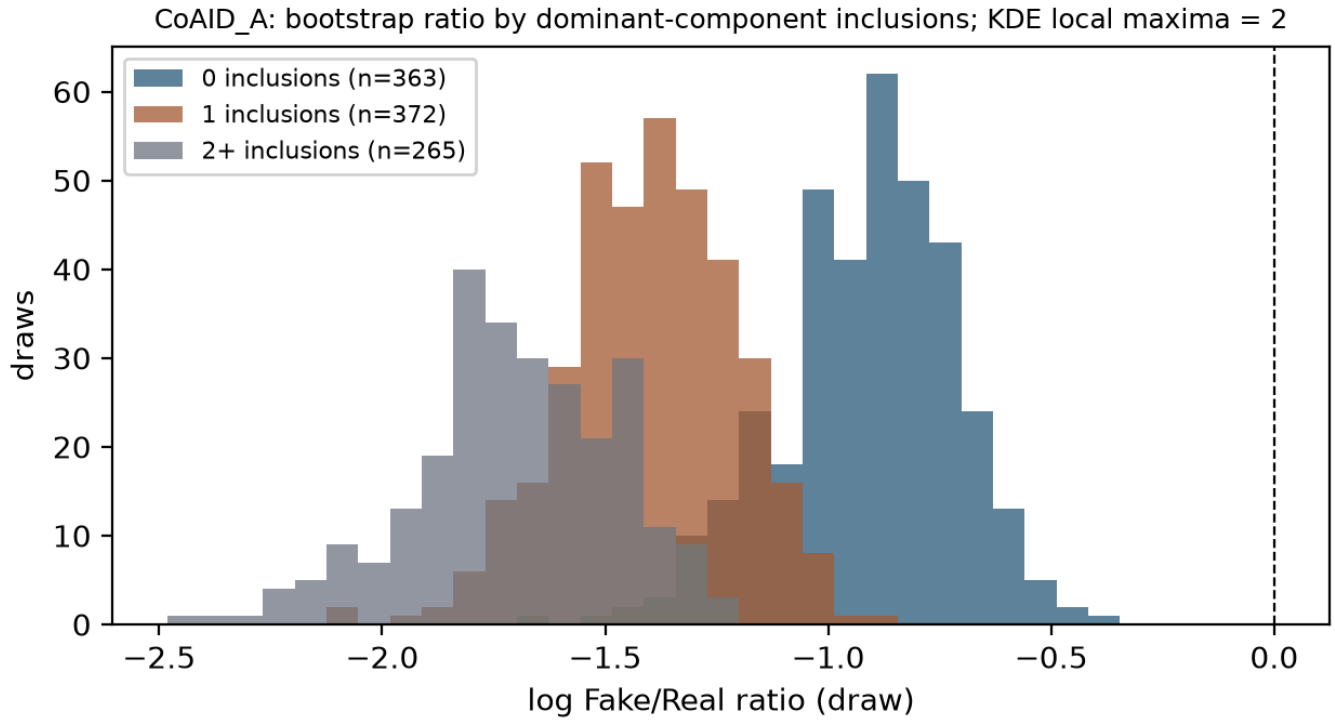


Figure S1A. CoAID A: bootstrap Fake/Real ratio draws by dominant-component inclusions. Source PNG SHA-256 `60bb4ba8a378cf82cad35d6ef858b58e28b4dc7d0651ba06e1f12ecea3cb418b`.

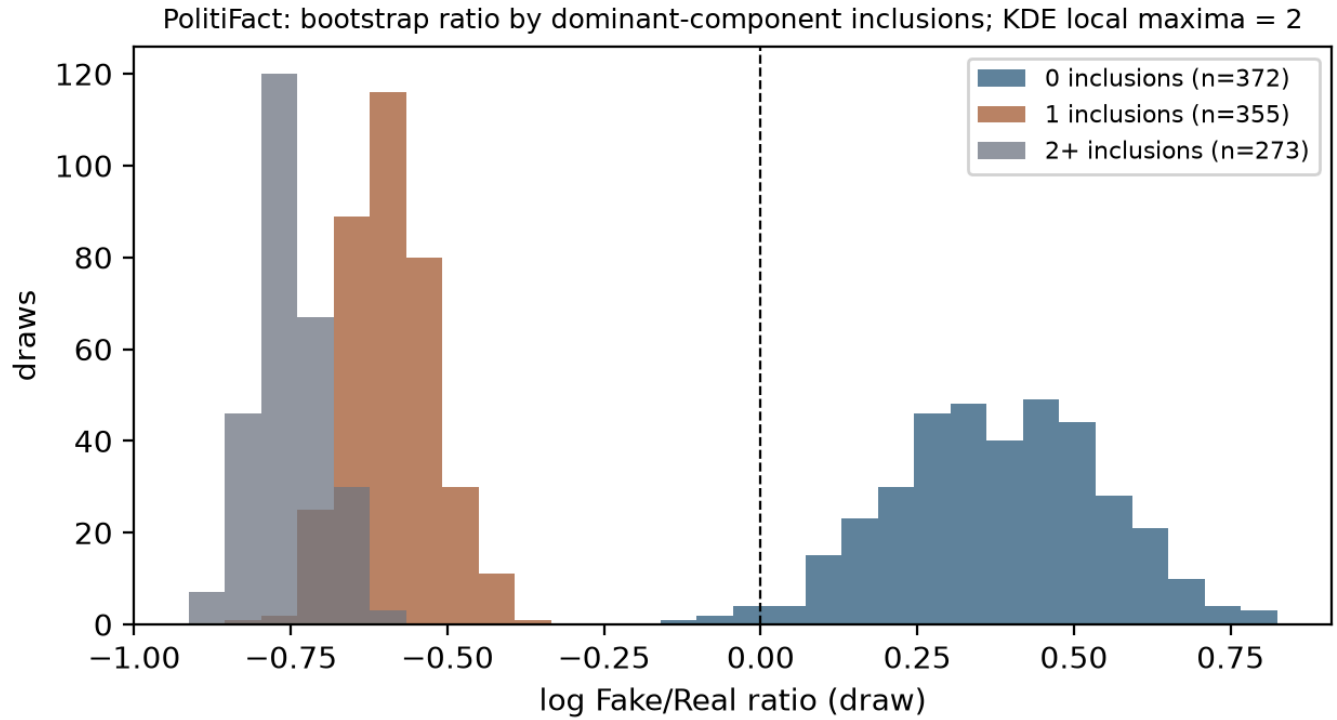


Figure S1B. PolitiFact: bootstrap Fake/Real ratio draws by dominant-component inclusions. Source PNG SHA-256 `dadcd1f34d8164d54a6730e5f64e1a960472cd65787743eb463baa1614bd6de9`.

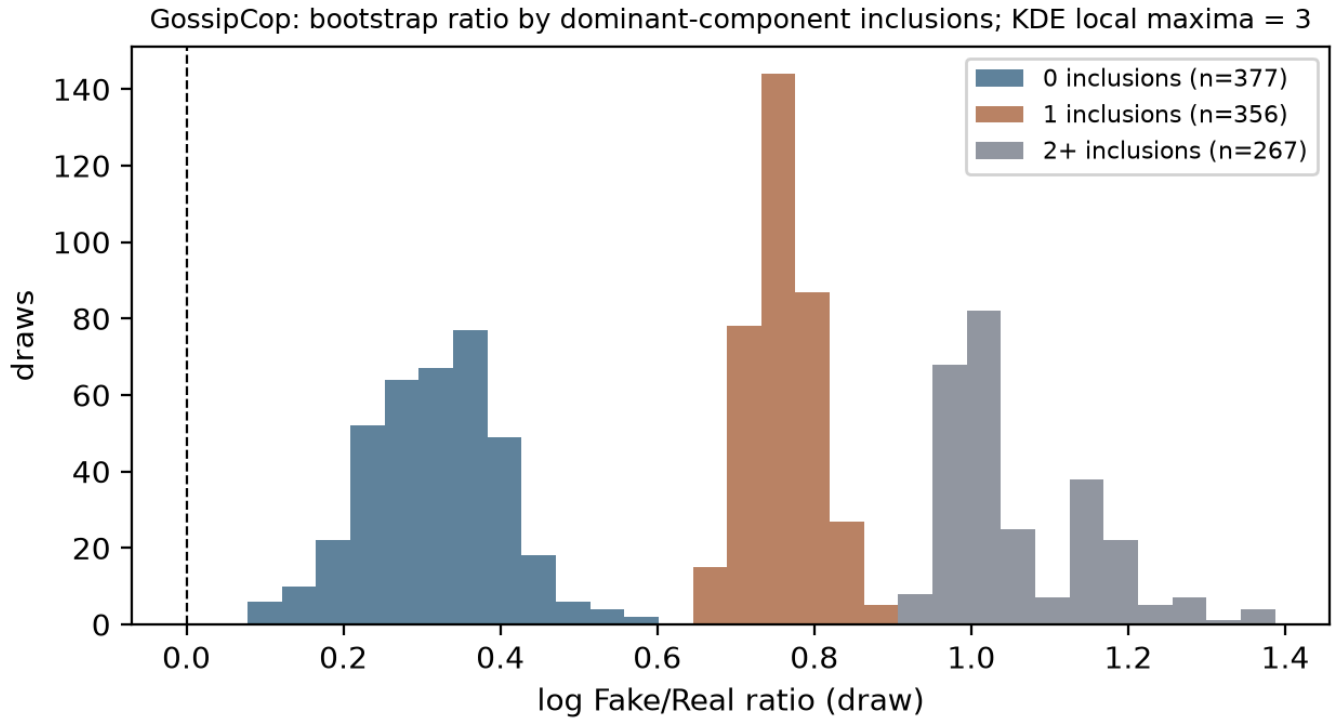


Figure S1C. GossipCop: bootstrap Fake/Real ratio draws by dominant-component inclusions. Source PNG SHA-256 `d009655052ce223515f0af82b4323b9b9edddd72d9e8ef47196a1c38f9b3bd61`.

Table S21. Mean versus typical item: zero-inclusive unique-post distributions by class

Supplementary analysis (Methods S8). Share of items with zero observed posts, median, 75th and 90th percentiles and mean of unique posts, and the probability that a randomly chosen Fake item has more observed posts than a randomly chosen Real item with ties counted one half (Mann–Whitney probability), for CoAID A, FakeHealth HealthStory, PolitiFact and GossipCop (label-conflict ids excluded). Observed values, not model-standardized; descriptive only.

S21.1 — round4_typical_item.csv

Source CSV: `outputs/extension_v1/tables/round4_typical_item.csv`; SHA-256 `4c9596c38e60a634b37757ae5571b0cce2c98bd58bad552fd662939a31a26ac5`.

Panel S21.1.1

All saved rows.

CSV row	context	label	N	share zero	median	p75	p90	mean	prob random fake exceeds random real ties half
1	CoAID_A	Fake	922	0.8406	0	0	3	11.3265	0.2861
2	CoAID_A	Real	4415	0.4270	3	19	54	32.1133	0.2861
3	FakeHealth	Fake	472	0.0021	113	215.2500	400.6000	199.8771	0.4595
4	FakeHealth	Real	1218	0.0049	129	270	505.6000	237.8744	0.4595
5	PolitiFact	Fake	430	0.0930	77.5000	288.7500	752.3000	374.9907	0.6345
6	PolitiFact	Real	622	0.3457	7.5000	154.7500	1541.4000	670.5305	0.6345
7	GossipCop	Fake	5323	0.0353	12	42	303	112.3988	0.3654
8	GossipCop	Real	16817	0.0629	45	66	91	52.4247	0.3654

Table S22. FakeHealth LR refits on the text before and after the 512-token transformer cut

Supplementary analysis (Methods S8). The cut position per item was located with the transformer tokenizer and pair-packing rule (512 budget, four special tokens, full title kept, body prefix fills the remainder; `round4_fakehealth_cut_offsets.csv`, 69.4% of items truncated). The selected FakeHealth LR configuration (cfg_21) was refitted on the saved training pool for each representation and scored on the saved holdout; the natural refit reproduces the primary result. S22.2: the 50 largest-magnitude terms of the primary FakeHealth LR with the share of their training-pool occurrences falling after the cut.

S22.1 — round4_fakehealth_truncation.csv

Source CSV: `outputs/extension_v1/tables/round4_fakehealth_truncation.csv`; SHA-256 `5c14aa88a11439183d3af73518ee18a76ceb51ef712075395dd6be9ffa6773eb`.

Panel S22.1.1

Holdout metrics (Fake = 1, threshold 0.5).

CSV row	variant	macro f1	roc auc	average precision	vocabulary size	empty training texts
1	frozen_saved	0.6752	0.7701	0.5707	28514	0
2	natural_refit_check	0.6752	0.7701	0.5707	28514	0

CSV row	variant	macro f1	roc auc	average precision	vocabulary size	empty training texts
3	before_512_cut	0.6509	0.7182	0.5324	16982	0
4	after_512_cut	0.6525	0.7656	0.4771	14280	392

S22.2 — round4_fakehealth_top50_terms_after_cut.csv

Source CSV: `outputs/extension_v1/tables/round4_fakehealth_top50_terms_after_cut.csv`; SHA-256 `c1e60141af9d68d1ff1ee8797c719403d74269b0bc100d469febe76082f8ac39`.

Panel S22.2.1

Top-50 terms by |coefficient|; occurrences counted with the fitted vectorizer analyzer over the training pool.

CSV row	rank	term	coefficient	leaning	occurrences before cut	occurrences after cut	share after cut
1	1	news release	1.0703	Fake	73	7	0.0875
2	2	release	1.0228	Fake	119	39	0.2468
3	3	news	0.8836	Fake	457	127	0.2175
4	4	and	0.8417	Fake	10187	8379	0.4513
5	5	the	0.8383	Fake	24272	18250	0.4292
6	6	to	0.7763	Fake	10470	9149	0.4663
7	7	still	-0.7700	Real	275	411	0.5991
8	8	was not	-0.7395	Real	109	102	0.4834
9	9	says	-0.7370	Real	937	1465	0.6099
10	10	there	-0.7016	Real	768	971	0.5584
11	11	in news	0.6981	Fake	37	2	0.0513
12	12	artificial	0.6716	Fake	55	29	0.3452
13	13	mri	0.6647	Fake	87	50	0.3650
14	14	drug	-0.6585	Real	1183	607	0.3391
15	15	costs	-0.6553	Real	89	89	0.5000

CSV row	rank	term	coefficient	leaning	occurrences before cut	occurrences after cut	share after cut
16	16	many	-0.6520	Real	595	441	0.4257
17	17	vaccine	-0.6427	Real	163	88	0.3506
18	18	pfizer	0.6395	Fake	31	13	0.2955
19	19	of	0.6238	Fake	13863	9774	0.4135
20	20	reuters health	-0.6168	Real	129	44	0.2543
21	21	chronic	0.6165	Fake	165	80	0.3265
22	22	the new	-0.6062	Real	516	286	0.3566
23	23	involved in	-0.5991	Real	121	114	0.4851
24	24	he	-0.5835	Real	893	1881	0.6781
25	25	london	0.5794	Fake	67	12	0.1519
26	26	involved	-0.5779	Real	221	183	0.4530
27	27	he says	-0.5769	Real	119	246	0.6740
28	28	cost	-0.5762	Real	120	197	0.6215
29	29	said dr	-0.5647	Real	250	201	0.4457
30	30	cbs news	0.5600	Fake	34	8	0.1905
31	31	traditional	0.5529	Fake	92	47	0.3381
32	32	cbs	0.5456	Fake	38	9	0.1915
33	33	some	-0.5414	Real	876	754	0.4626
34	34	however	-0.5409	Real	248	206	0.4537
35	35	dr	-0.5364	Real	912	929	0.5046
36	36	sleep	-0.5360	Real	256	166	0.3934
37	37	said in	0.5359	Fake	163	91	0.3583
38	38	information the	0.5358	Fake	22	72	0.7660
39	39	of medicine	-0.5325	Real	290	195	0.4021

CSV row	rank	term	coefficient	leaning	occurrences before cut	occurrences after cut	share after cut
40	40	000	-0.5307	Real	463	325	0.4124
41	41	therapy	-0.5250	Real	579	382	0.3975
42	42	last	-0.5240	Real	188	207	0.5241
43	43	safe	-0.5209	Real	126	109	0.4638
44	44	deaths	-0.5173	Real	168	87	0.3412
45	45	evidence	-0.5105	Real	291	258	0.4699
46	46	vision	-0.5102	Real	105	49	0.3182
47	47	not involved	-0.5083	Real	63	61	0.4919
48	48	scans	0.5057	Fake	142	85	0.3744
49	49	studies	-0.5052	Real	623	551	0.4693
50	50	the results	-0.4979	Real	251	197	0.4397

Table S23. Lexical signal reliability: LR coefficient agreement across training folds

Supplementary analysis (Methods S8). The selected natural LR configuration was refitted on each of the five saved CV training folds per dataset; coefficients over the 4,174-term shared unigram vocabulary (Table S10) were compared pairwise by Spearman correlation (terms absent from a fold vocabulary excluded pairwise), next to the cross-dataset Spearman of −0.003.

S23.1 — round4_fold_coefficient_summary.csv

Source CSV: `outputs/extension_v1/tables/round4_fold_coefficient_summary.csv`; SHA-256 `fdb84a1d682efed2b779e340d2f3811804f8fa119075992fe4960444c28eaeae`.

Panel S23.1.1

Summary over the ten fold pairs per dataset.

CSV row	dataset	pairs	spearman mean	spearman min	spearman max	cross dataset spearman frozen
1	FakeHealth	10	0.7799	0.7695	0.7941	-0.0034
2	CoAID	10	0.8208	0.8003	0.8355	-0.0034

S23.2 — round4_fold_coefficient_pairs.csv

Source CSV: `outputs/extension_v1/tables/round4_fold_coefficient_pairs.csv`; SHA-256 `b9dfbc68f8d805a012b122bad8663e5907a10ca81def9a3556dc0dbf9204ef82`.

Panel S23.2.1

Fold fits (single-fold rows) and pairwise correlations (pair rows).

CSV row	dataset	fold	train N	vocabulary size	shared terms present	spearman
1	FakeHealth	0	1053.0	23649.0	3955	
2	FakeHealth	1	1055.0	23663.0	3950	
3	FakeHealth	2	1053.0	23968.0	3990	
4	FakeHealth	3	1053.0	23330.0	3947	
5	FakeHealth	4	1054.0	22985.0	3929	
6	FakeHealth	0-1			3824	0.7809531088914924
7	FakeHealth	0-2			3851	0.7812226811441432
8	FakeHealth	0-3			3816	0.7941489452811827
9	FakeHealth	0-4			3813	0.7825096448509486
10	FakeHealth	1-2			3851	0.7752106820196585
11	FakeHealth	1-3			3816	0.7801599176254299
12	FakeHealth	1-4			3821	0.7715457016100478
13	FakeHealth	2-3			3844	0.7853486732013192
14	FakeHealth	2-4			3835	0.7695221986424489
15	FakeHealth	3-4			3812	0.7787774801439918
16	CoAID	0	2143.0	5600.0	3888	
17	CoAID	1	2337.0	5739.0	3850	
18	CoAID	2	2342.0	5674.0	3840	
19	CoAID	3	2339.0	5686.0	3837	
20	CoAID	4	2339.0	5674.0	3860	

CSV row	dataset	fold	train N	vocabulary size	shared terms present	spearman
21	CoAID	0-1			3596	0.8154946342191248
22	CoAID	0-2			3589	0.8320731063029343
23	CoAID	0-3			3579	0.8349442784554193
24	CoAID	0-4			3600	0.821840809115125
25	CoAID	1-2			3572	0.8094056559180103
26	CoAID	1-3			3585	0.8130323115891848
27	CoAID	1-4			3592	0.8003388808102324
28	CoAID	2-3			3565	0.8354537851407671
29	CoAID	2-4			3578	0.8199291063561122
30	CoAID	3-4			3593	0.8257193229380336